

What Cycle Closure Can and Cannot See in a Relative Binding Free Energy Network

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Abstract

Cycle closure is the standard consistency check on a perturbation network, and what such a check can see is fixed by the gradient-cycle decomposition of edge flows (HodgeRank): the cycle part of a network’s error is auditable and biases nothing, while the gradient part biases the affinities and is invisible to any statistic computed from the network’s own values, at any magnitude. Both parts are measured here on 1145 real binding edges. The auditable subspace spans a third of the error space and, in the variance-weighted metric the closure statistic uses, carries under one per cent of the error against experiment. Experimental and annotation error is per-ligand, so it falls in the invisible part by construction and can only have made that share smaller: the measurement is a floor, and it survives correction: in the metric the closure statistic uses on every system measured, at one cell as a tie, and in an isotropic reading on every system of the wider grounded set, failing only on the two weakest of the four the test itself selected. The test built on it behaves as the bound predicts, and its flag set is not stable: the per-edge residuals and the ranking of systems reproduce across independent repeats, and the list of flagged systems does not. Fitting closure against the precisions the estimator already reports gives that test its stated null; it ties the standard fixed 1.0 kcal/mol cutoff, and on every replicate the set of systems it flags lies inside the set that cutoff flags. It flags nothing once the reported errors are widened by half, enough to cover the pose and preparation variance the protocol repeats hold fixed, and it names no removal that recovers accuracy against experiment. Resolving the gradient part needs measurements from outside the network, at worst one per unresolved dimension.

1 Introduction

Relative binding free energy calculations are the most accurate computational tool available for ranking how tightly a set of candidate molecules will bind a target, and structure-based drug discovery programs increasingly build them into decisions about which molecules to synthesize. The calculations come as a network rather than as isolated numbers: each edge is an alchemical transformation between two related candidates, costing hours of molecular dynamics, and a realistic project touches dozens to hundreds of them. Because the affinities a program acts on are read off the network as a whole and not off any single edge, the network’s own internal agreement is what a practitioner has to judge the calculations by before any molecule is made.

That judgement has a standard instrument. A perturbation network must close its thermodynamic cycles, the signed sum of predictions around any loop being zero in expectation, and how far a network fails to close them is routinely reported as a diagnostic and acted on through fixed

hysteresis cutoffs. What has not been asked of that diagnostic is how much of a network’s error it can reach at all: whether a network that closes its cycles is thereby a network whose affinities are right, and if not, what share of what is wrong with it a closure check could ever respond to.

That capacity follows from the algebra of the network rather than from any particular protocol, and it is measurable on public data. A network’s per-edge error splits, uniquely and orthogonally, into a part carried by its cycles and a part that is a difference of per-ligand offsets. The first is the part any closure check sees, and it does not bias the estimated affinities at all; the second is the part that biases them, and no statistic computed from the network’s own edge values responds to it at any magnitude. That fixes the capacity of a closure check; how much of a real network’s error falls inside it is then a measurement. Both are made here, on a public benchmark of real binding edges with independent protocol repeats, and both are small: the auditable subspace spans a third of the error space, and the share of the error against experiment that falls inside it is under one per cent.

The test that runs against this bound is built from an uncertainty the calculation already supplies rather than from one learned alongside the prediction: where the prevailing approach fits a mean–variance head [29] or reads uncertainty off the disagreement of an ensemble [24], the estimator that turns raw simulation output into a free energy for a single edge [7] has a closed-form finite-sampling variance computed from that same output at no label cost. The bar the test carries is the multistate standard error that same calculation reports, of which that closed form is the two-state case, and validating it against independent repeats is what turns cycle closure from a descriptive check into a test with a stated null.

Section 2 places the work against prior free-energy estimators, learned uncertainty and cycle-closure diagnostics. Section 3 gives the decomposition and what it fixes about any closure-based check, and Section 4 measures where a real network’s error lies against that bound. Section 5 characterises the per-edge error bar the test uses as its null and validates it against independent repeats, and Section 6 builds the test and reports what it inherits from the bound. Section 7 gives the data, the estimators, the models and the simulations behind every number reported above, and Section 8 collects the limitations, including an applicability map for the calibrated uncertainty downstream.

2 Related work

BAR [7] and MBAR [33] are the standard alchemical estimators, brought to prospective potency prediction by FEP+ [36], assessed at scale [28, 31], and carrying as their per-edge uncertainty the sandwich variance of M -estimation [18, 37]; the prevailing alternative learns that uncertainty from training examples instead [2, 21, 24, 29]. Cycle closure is long established as a network diagnostic [8, 36], as a weighting and estimation problem [13, 25, 38], and as necessary but not sufficient [28]. The gap addressed here is the quantitative form of that insufficiency: how much of a network’s error a closure statistic can reach, and how much of a real network’s error falls there.

3 What cycle closure can see: the decomposition

A perturbation network must close its cycles: the signed sum of $\Delta\Delta G$ around any loop is zero in expectation. Everything a closure check reports is a function of the residuals of that constraint, so what such a check can see is fixed before any protocol is chosen, by the algebra of the network alone. Fix the fit those residuals come from. Node potentials are estimated by generalized least squares on the signed incidence matrix, whitened by the reported per-edge standard errors, so that

the conductances are $w_e = 1/V_e^{\text{rep}}$ with V_e^{rep} the per-edge variance the free-energy estimator reports (Section 5; Theorem S1 in the Supporting Information identifies that conductance with the sandwich precision I_e^2/B_e whenever the work samples are available, which on the benchmark used here they are not, so V_e^{rep} is the reported variance composed from the complex and the solvent leg, Section 7). The weighted sum of squared residuals of that fit is the closure statistic whose null Section 6 turns into a test; only its geometry is needed here. Two results follow, one per edge and one about the whole error space.

Theorem 1 (Estimation–detection conservation law). *In the whitened GLS fit let $\Pi = \mathbb{I} - H$ be the residual-maker projector onto the cycle space (Π , to keep M for M -estimation and R for a Pearson correlation). Its diagonal, the curl-leverage $h_e = \Pi_{ee}$, satisfies $h_e + w_e \Omega_e = 1$ with $w_e = 1/V_e^{\text{rep}}$ and Ω_e the effective resistance of edge e [22]; hence $\sum_e h_e = \text{dof}$, and the closure- χ^2 noncentrality against a systematic shift μ on edge e is $h_e \mu^2 / V_e^{\text{rep}}$. A bridge edge (in no cycle) has $h_e = 0$ and is undetectable at any μ ; a node-consistent (gradient) bias lies in $\text{range}(H)$ and is annihilated by Π . A closure check’s blindness to node-consistent force-field bias is therefore a derived property, not an assumption, and h_e is a per-edge observability certificate with resolution $\delta_e^* = \sqrt{V_e^{\text{rep}}/h_e}$, the shift at which the closure statistic carries unit noncentrality.*

The algebra is standard: $h_e = \Pi_{ee}$ is the hat-matrix leverage of the whitened design [17], its complement $w_e \Omega_e$ is the correspondence between leverage and effective resistance that graph sparsification uses [34], and $\sum_e h_e = \text{tr}(I - H) = \text{dof}$ is the trace identity for a projector. What the theorem adds is the reading of h_e as a per-edge observability certificate, which turns the test’s blindness to node-consistent bias into a consequence rather than an assumption and names, before any simulation is run, the edges that can carry no evidence at all. The resulting map lists the structurally un-auditable (bridge) edges per system; on this benchmark the curl-leverage does not additionally predict the *location* of reproducible systematic error (studentized Spearman $\rho = -0.073$, permutation $p = 0.9571$, not significant; Supplementary Figure S9), so only the structural certificate is claimed.

Theorem 1 states the diagonal of a result about the whole error space, and that full result is what bounds the method. Write the per-edge systematic error as a vector μ over the E edges. Whitening and projecting splits it, orthogonally and uniquely, into a part that lies in the cycle space and a part that is a difference of per-ligand offsets.

Theorem 2 (Auditable and unauditable error). *In the model $y = A\varphi + \mu + \varepsilon$ with A the signed incidence matrix of a network of N ligands, E edges and c components, and φ unknown: (i) for every δ , the parameter pairs (φ, μ) and $(\varphi + \delta, \mu - A\delta)$ induce the same distribution of y , so μ is identifiable only through $\Pi\tilde{\mu}$. The auditable subspace has dimension $\text{dof} = E - N + c$ and its orthogonal complement, the gradient fields $\mu_e = b_j - b_i$, has dimension $N - c$ and is unidentifiable: no statistic computed from a network’s own edge values distinguishes a per-ligand bias from no bias, at any magnitude. (ii) The bias such an error induces in the fitted potentials, $(\tilde{A}^\top \tilde{A})^+ \tilde{A}^\top \tilde{\mu}$, depends only on the gradient part, while the closure noncentrality $\|\Pi\tilde{\mu}\|^2$ depends only on the cycle part. The detectable part and the biasing part are exact orthogonal complements. (iii) For an error field with no preferential alignment, the expected visible fraction $\|\Pi\tilde{\mu}\|^2/\|\tilde{\mu}\|^2$ is dof/E . (iv) Network estimation of $\Delta\Delta G$ under loop closure, with and without experimental constraints, is established [13, 38]; what the rank counting adds is the cost of the constraints. Given unbiased absolute measurements on a set S of ligands, the unidentifiable dimension is $(N - |S|) - c_{\text{free}}(S)$, with c_{free} the number of components holding none of them: the first anchor in a component removes only that component’s arbitrary offset and no unidentifiable direction, each further anchor removes exactly one dimension, and resolving a per-ligand bias in full requires an absolute measurement on every ligand.*

The decomposition is not new. Splitting an edge field on a graph into a gradient part and a divergence-free part, and reading the size of the second as whether the first is meaningful at all, is HodgeRank [19], where the gradient component is the global ranking being estimated and the cyclic component certifies it; a third, harmonic term appears there under a choice of two-cells and is not needed here. The hat-matrix and leverage machinery of part (ii) is likewise standard on networks of relative comparisons, where it already locates inconsistency in network meta-analysis [23]. That the closure check sees only cycles is in this sense a restatement, and it matches the long-standing reading of cycle closure as necessary and not sufficient [28].

The count in part (iii) is older still and closer to this setting than HodgeRank is. A network of relative comparisons under a consistency assumption has, in the language of mixed treatment comparisons, inconsistency degrees of freedom equal to the number of independent loops [26], which is dof here; the modern treatment of that model and its multi-arm complications is Higgins et al. [16].

Part (i) states that a bias common to every comparison involving one node is absorbed into that node’s effect and is invisible to any inconsistency test. That literature does not state it in those terms, but it is immediate in that framework and may well be folklore there, as the corresponding fact is here, so the absence of an explicit statement does not establish that the fact is new.

The bias-adjustment strand of the same literature weakens the substance further. Bias-adjustment models in mixed treatment comparisons [9] treat study-level bias terms as quantities the network cannot identify on its own and inform them with external evidence or informative priors, which is part (i) for a different index and the same conclusion: a bias living where the design cannot separate it is resolved only from outside. The orthogonality of part (ii) and the anchor counting of part (iv) are likewise not located in that strand, and not finding them there does not establish that they are absent.

Part (ii) follows from part (i) in two lines. The bias in the fitted potentials is $(\tilde{A}^\top \tilde{A})^+ \tilde{A}^\top \tilde{\mu}$ and $\tilde{A}^\top \Pi = 0$, so the gradient part alone determines it while $\|\Pi \tilde{\mu}\|^2$ alone determines the noncentrality; it is the defining property of an orthogonal projector, read on the error rather than on the signal. What is left to this article is that reading for alchemical networks and the measurement of where a real network’s error falls between the two halves, which is Section 4 and what the rest of this article rests on. Part (iv) is a dimension count and is not asserted as new. Network estimation of $\Delta\Delta G$ under loop closure is solved in Xu [38] by Fisher information and in Giese and York [13] by a constrained variational solution of the whole network; neither is the object part (iv) counts, which is the number of absolute measurements needed to remove the unidentifiable directions. That counting may well be present in either, and it is used here to state that cost.

The map as a design rule. The observability certificate is prospective: alone among the quantities measured here it can be acted on before a simulation is run. It prices two distinct purchases on the replicate benchmark (1143 edges over the 48 systems that carry a cycle; Section 7). Coverage costs 29 added ligand pairs, 2.5% more perturbations than were already run, and leaves no bridge anywhere; connecting every system as well costs 36. Both counts are certified minimum against the Eswaran–Tarjan augmentation bound and assume nothing about the edges they add (Supplementary Table S5; `make figDesign`).

Resolution is the second purchase, and the certificate prices it together with the point at which it stops being purchasable. An edge freshly placed on one long cycle has small h_e and correspondingly poor δ_e^* , so coverage does not buy resolution and the two are costed apart: where a target is reachable and not already met, a greedy design reaches a median δ^* of 1.0 kcal/mol on 39 of the 48 systems for a median of two further edges, on an assumed variance for edges not yet run. Past that

point the graph is no longer the binding constraint. Because $h_e \leq 1$ pointwise, no design pushes an edge’s resolution below its own reported standard error, and measuring every remaining ligand pair still leaves nine systems with a median δ^* above 1.0 kcal/mol and nineteen above 0.5. There the sampling budget is what has to change, which is the bound of Section 3 presenting itself as a design constraint rather than as a measurement.

Two properties of the certificate carry into the rule. Its targets are on a resolution scale and not a detection threshold, at the ratio Section 4 measures; and it says where the test could see, not where reproducible error lands, which the same curl-leverage does not predict.

4 Where the error actually lies

Theorem 2 makes two quantities measurable that were previously matters of assumption: how much of a network is auditable at all, and how much of its real error the audit can reach. Both are answered on this benchmark in Figure 7.

Of the 1143 edge dimensions carried by the 48 systems with a cycle, 372 lie in the cycle space, so two-thirds of the error space is node-consistent and invisible to every closure-based diagnostic, this one included. Forty-eight edges are bridges, carrying no evidence at any magnitude. Half the remaining edges cannot resolve a shift below 0.79 kcal/mol, and only 27% of all 1143 can resolve one of 0.5 (Figure 7A).

Both numbers overstate the test’s reach, for two reasons that compose. δ^* is the shift at unit noncentrality rather than a detection threshold: power there runs from 0.170 to 0.070 across the degrees of freedom present, and 80% at $\alpha = 0.05$ needs 2.8 to 4.7 times it, moving the median resolution to roughly 2.2–3.7 kcal/mol. δ^* is also carried on the raw reported standard errors, where the calibrated scale is 0.92 of them (Section 6).

Composing the two gives operational numbers worse than δ^* alone suggests. Only 6.6% of auditable edges reach 80% power against a 1.0 kcal/mol shift on that edge alone, at $\alpha = 0.05$ and before any multiplicity correction. At the level the test actually fires, a system’s most observable edge needs a median 1.35 kcal/mol for 80% power (quartiles 0.81 and 2.06) and its median edge needs 3.33; a third of systems could detect a 1.0 kcal/mol error somewhere, and 58% could detect 1.5. The regime the test operates in therefore begins at about the top of the band that relative-binding-free-energy decisions turn on, which is a sharper statement of the same limitation the rest of this section measures.

That is the capacity of the audit; its reach on real error is smaller, and measured on different graphs. Figure 7B carries two grounded readings of that reach on one shared axis: B1 the four sub-networks the closure test itself selected, on a ChEMBL join, and B2 the eight the benchmark’s own curated per-edge labels reach, six of which the test does not flag. They run on different edges and different labels, neither corrects the other, and the differences between them are set out with the second reading below. Affinities cover part of each ligand set, so Figure 7B1 runs on each system’s grounded sub-network rather than on the full networks of Supplementary Table S4. As (N, E, c, dof) these are (24, 31, 2, 9), (41, 59, 1, 19), (34, 51, 1, 18) and (23, 26, 2, 5) for **cdk8**, **hif2a**, **p38** and **bace**, with auditable fractions 0.290, 0.322, 0.353 and 0.192 rather than the tabulated ones (**cdk8** is 0.46 on its full network); two are disconnected, which the fit absorbs into one offset per component.

All four sub-networks were selected by the calibrated closure test of Section 6, so what follows characterises the networks that test flags and not the benchmark. One of the four, **cdk8**, carries its flag from a naming collision between two releases rather than from either release’s simulations.

The quantity measured on them is the visible fraction $f = \|\Pi\tilde{\mu}\|^2/\|\tilde{\mu}\|^2$ of Theorem 2(iii), here

the share of the measured error against experiment that lies in the auditable subspace. Pooled it is 0.6%, against those chance levels (Figure 7B1); per system 0.0065, 0.0017, 0.0086 and 0.0460. Dropping **cdk8** leaves the pooled figure at 0.6% (0.0060 against 0.0060) and moves it from 5.45% to 6.10% in the unweighted metric, so the pooled figure does not rest on it.

The per-system figures are less stable than the pooled one. Leaving out one edge at a time moves them by 1.9, 1.7 and 2.6 times on the first three and by 17 on **bace**, whose range spans a factor of 23.

Both sides of the ratio are measured in the variance-weighted metric the closure statistic lives in, and the margin over chance depends on that choice. It narrows to a factor of two on **bace** under isotropy in kcal/mol, and the pooled fraction rises to 5.45% in the unweighted metric, which recomputes the projector without variance weighting. Supplementary Table S2 gives all three conventions for all four systems with their own chance levels and their single-edge-deletion ranges; its weakest cell is **bace** isotropic under deletion, where f is smaller than its own chance level by a factor of 1.36.

Against the frozen H1 criterion, a pooled degrees-of-freedom-adjusted gradient-model R^2 of at least 0.50 and positive on three of the four systems, the pooled value is 0.980 and the per-system values are 0.978, 0.995, 0.976 and 0.761, so the criterion is met. Pooling is over the four sub-networks jointly, f as a ratio of summed squared norms and the adjustment carried on the summed $E = 167$ edges and rank 116, that is on 51 residual degrees of freedom, and not as an average of per-system adjustments.

The criterion evaluated is the degrees-of-freedom-adjusted rule the pre-registration froze and not the weaker unadjusted form $1 - f_{\text{pooled}} \geq 0.50$. These figures support a bound and not a point estimate: the visible fraction is small, and at its weakest it sits a factor of two below its chance level, not the two orders of magnitude the pooled figure alone suggests.

The measurement is not an artefact of the fit, because it is a near-identity: per-ligand affinities make the experimental $\Delta\Delta G$ vector a gradient field, so $\Pi\varepsilon$ reduces to the closure residual and f is the closure statistic over the total squared standardized error. It survives the obvious contaminants, holding on medians rather than sums and holding on **bace**, whose ligands carry no enantiomer pair collapsed by the stereo-blind join (Supporting Information). Changing the label source to the curated per-edge affinities the OpenFF protein–ligand benchmark of Section 7 carries, available for three of the four systems, agrees where it can be run (Supporting Information). Per-edge error against experiment runs 1.06–1.45 kcal/mol at the median against reported standard errors of 0.14–0.40: these networks are wrong by several standard errors while closing their cycles, in the one direction closure cannot look.

The design does not license attribution. Experimental measurement error and the annotation join are per-ligand, hence gradient fields too, so what is established is that the dominant part of the error is node-consistent, not that it is force-field bias.

The curated cross-check named above is not confined to the four systems the test selected, and it is the second block of the same panel. That source shares fourteen system names with the replicate set, and eight of them keep a cycle once the two edge sets are matched by structure rather than by name; pooled over those eight, on 205 edges and 31 residual degrees of freedom, the visible fraction is 0.00052 against chance levels of 0.151 whitened and 0.0137 isotropic (Figure 7B2; Supplementary Figure S6 and Table S3; **make figGround**). The systems the closure test does not flag show the effect at least as strongly as the ones it does, 0.00042 over six against 0.00071 over two, which turns the measurement from a statement about test-selected networks into a statement about the benchmark.

The expanded measurement neither replaces the four-system one nor corrects it, which is why Figure 7B keeps the two as separate blocks rather than as one series. **bace** is absent from the curated set entirely, **p38**’s matched sub-network is acyclic and contributes to neither side, the two run on

different edge sets, and the expanded set carries 31 residual degrees of freedom against the original’s 51: the two are measurements of different things on different labels, and both are reported. On genuinely identical edges the cleaner labels give the *higher* f , by $4.4\times$ on **cdk8** and $2.9\times$ on **hif2a**. That is the direction a per-ligand label error predicts and not its reverse, because such an error is a gradient field: the whitened projector annihilates it, so it lands wholly in the denominator of f and pushes the noisier source’s reading down.

One degree of freedom in the protocol was undeclared; the naming finding of Section 6 settles it. A ligand name is not an identifier here, so the figures quoted throughout key nodes by resolved structure, which is the less favourable reading: merging two names for one molecule knits components together and adds cycles, and cycles can only raise the auditable share. **ptp1b** is the only system of the eight it touches, going from $f = 0.0023$ to 0.0159 , its worst single-edge-deletion isotropic margin from $10.3\times$ to $2.0\times$, and the pooled figure from 0.00046 to the 0.00052 above. Neither key is safe in general, and the Supporting Information shows where this one fails: a 2D structure merges poses a release names apart. Both readings are reported.

The label source can move f in one direction only, which is what makes it a bound. Those per-ligand errors reach an edge as a difference of node offsets, exactly the gradient fields part (i) describes, and the whitened projector annihilates them. Injecting a per-ligand field at the mixed-laboratory reproducibility public affinity data show, $0.68 \log_{10}$ units per ligand [20], moves the numerator of f by at most 1.96×10^{-11} of itself over 800 seeded draws, against a norm of order one, while the denominator grows by 1.6 to 2.2 times at the median, which is the inflation the injection produces (Supplementary Figure S7; **make figNoise**). Label noise therefore contributes nothing to the numerator and all of it to the denominator, so a correction for it can only raise f , and the measured 0.6% is a floor on the share attributable to the calculation’s own error and not an estimate of it.

The room that leaves is not uniform across systems, and the pooled crossing is not the reading to quote alone. Pooled, f would first reach its chance level at a denominator shrinkage of $50.6\times$ whitened and $14.1\times$ isotropic. What a correction could deliver is the reverse accounting: removing the share a literature-scale per-ligand noise implies would shrink the denominator by $4.13\times$ pooled, once each system is held to its own arithmetic. **bace** needs only $4.2\times$ whitened and $2.0\times$ isotropic, and under the least favourable single-edge deletion **p38** needs $1.46\times$ and **bace** $1.36\times$ isotropic, which a plausible correction does supply.

The two metrics accordingly give different verdicts, and both are reported. The whitened reading, the metric f is defined in, survives every level tried on every system with one cell excepted. That cell is **bace** under the least favourable single-edge deletion, where the $2.55\times$ needed and the $2.55\times$ supplied by the level carrying half the mixed-laboratory per-edge variance coincide to within one per cent, which is a tie rather than a margin. The isotropic reading does not survive on **p38** and **bace**, the two weakest cells of the isotropic convention already tabulated.

How far a correction can be pushed follows from arithmetic. At the full mixed-laboratory level the implied label-noise share exceeds one on **cdk8**, **p38** and **bace**, asserting more label variance than those networks’ entire measured error against experiment contains, so that level is an upper bound and not an estimate for a join that takes one assay per system and never mixes IC_{50} with K_i .

On **eg5**, whose ligands carry ChEMBL identifiers, the network $\Delta\Delta G$ from the same generalized-least-squares fit agrees with real KIF11 experimental affinity [40] at a mean unsigned error (MUE) of 0.79 kcal/mol (95% bootstrap CI $[0.59, 0.97]$, conditional on the fitted network; Pearson $R = 0.65$ $[0.30, 0.83]$, $p \approx 2 \times 10^{-4}$, $n = 28$), and its clean cycle-closure reduced $\chi^2 = 0.64$ is consistent. That is not evidence that a clean system is accurate, and the five grounded systems cannot settle the question in either direction.

The ordering by mean unsigned error is **eg5** $0.79 = \text{p38}$ $0.79 < \text{bace}$ $0.88 < \text{cdk8}$ $0.91 < \text{hif2a}$

1.47, so the one unflagged system ties the best flagged one and is below the other three. That direction favours the flag, but it rests on a single unflagged system against four, and the grounding is not matched: **eg5** comes from a single named ChEMBL assay while the four flagged systems come from a target-wide connectivity search under the coverage rule. No separation claim follows from it in either direction.

Acting on the flag does not improve accuracy on **eg5** (internal $|z|$ against error-against-experiment, $r = +0.13$), which is what the scope predicts: on a clean system the residual error against experiment is dominated by node-consistent force-field bias, and cycle closure cannot see that. The complementary test, that repairing a flagged system improves its accuracy, needs experimental affinities for the flagged systems; these are not directly tabulated in the replicate set but are recoverable by per-paper ChEMBL structure-search. Of the six flagged systems, four clear a 60% public-ChEMBL coverage gate (**cdk8**, **hif2a**, **p38**, **bace**; connectivity-based structure matching, one assay per system, IC₅₀ before K_i , never mixed); their baseline network MUE against this grounded experimental affinity is 0.79–1.47 kcal/mol, typical-magnitude FEP error, which confirms the grounding itself is sound.

On these four, guided removal of the flagged high- $|z|$ edges (the same repair order as Figure 6A, fixing a removal count $K = 1$ –3 per system) does not reduce MUE-against-experiment more than an equal-size random removal does (Supplementary Figure S10; per-system one-sided permutation $p = 0.905, 0.136, 0.880, 0.071$; combined Stouffer $p = 0.48$, sign test $p = 0.69$). Two systems improve slightly over their random-null mean and two worsen slightly, with no system individually significant. The frozen criterion is conjunctive, requiring Stouffer $p < 0.05$ and a guided Δ MUE, defined throughout as MUE(full) – MUE(after) so that an improvement is positive, above the 95th percentile of its own random null in a majority of grounded systems; neither conjunct is met. The remaining two flagged systems (**brd4**, **faah**) did not clear the coverage gate and are not included.

That null is what Theorem 2 predicts and what the paragraph above measures. A removal rule reads the visible part of the error; the visible part carries under one per cent of what is wrong with these networks, and by part (ii) it is the part that biases the fitted potentials least. A second pre-registered rule tests whether the ranking was simply the wrong one, and fails to find that it was; a second failure to reject is weaker evidence than a rejection would be. Part (ii) suggests ranking edges by influence rather than by residual. The statistic used is $|z_e|\sqrt{(1-h_e)/h_e}$ with $z_e = r_e/\sqrt{V_e^{\text{rep}}}$ the raw standardized residual, which is the classical influence form scaled by $\sqrt{h_e}$ and not that form itself; edges with h_e below 10^{-9} are treated as bridges and given influence zero. Writing it in the unscaled form changes the removal set on two of the four systems, and that variant is reported with the outcome below.

At the same removal counts the influence rule performs worse than the rule it was meant to replace, against the same conjunctive criterion: better on one system of four (per-system Δ MUE $-0.055, -0.099, -0.612, +0.140$ kcal/mol at one-sided permutation $p = 0.937, 0.819, 1.000, 0.023$; Figure 7C), at a pre-registered sign test of $p = 0.94$ against the residual rule’s 0.69. On four systems that test attains only $\{1.0000, 0.9375, 0.6875, 0.3125, 0.0625\}$, so neither value could reach its own level and the two do not form an ordering, one system flipping separating them.

The pre-registration’s other combination, a one-sided Stouffer, returns no usable number here. One system’s permutation p saturates at the resolution of its own grid and Stouffer maps that to an infinite z ; clipping to the grid’s range gives 0.96, and the residual rule, which has no saturated input, stands at 0.48.

Neither rule beats random removal, and on one system the influence rule does measurable harm. On **p38** its guided Δ MUE of -0.612 kcal/mol falls below every one of the 1000 random draws (minimum -0.416 , no ties), so the opposite tail is at most 1/1001. That tail is post hoc, the pre-registration having asked whether the rule helps; it is one system of four, about 0.004 under

a Bonferroni allowance; and the magnitude is convention-dependent, -0.612 under greedy removal that refits after each edge and -0.232 when the statistic is applied once to the original fit. The sign survives both conventions, and survives the strict influence form, which changes the removal set on two of the four systems but not here. The two edges removed carry $h_e = 0.099$ and 0.388 , so the harm is not the degenerate branch where a near-bridge is divided by its own vanishing leverage.

The other tail points the other way and is reported alongside it. On `bace` the same rule improves the error against experiment by 0.140 kcal/mol at a one-sided $p = 0.023$, which the conjunctive criterion does not let the summary report, and which is exactly as post hoc and as single-system as the harm. Neither tail clears the pre-registered bar, and both are reported.

The rule fails overall for the reason measured above: the influence form divides a residual by a small curl-leverage to recover the error of a single anomalous edge, and a smooth per-ligand field is not concentrated on any edge, so the division returns amplified noise.

A flag remains a system-level signal that a protocol has produced results inconsistent beyond sampling, and that signal reproduces across replicates in its per-edge residuals and in its ranking of systems, though not in the list of names it returns. It is not a repair target: deleting the edges it names cannot recover accuracy against experiment, which is set by a component the audit is blind to by construction.

Reaching that component needs information from outside the network, and it has to be information the network does not already contain: an absolute binding free energy calculation carries exactly the force-field component of the per-ligand systematic error part (i) is about, so it does not resolve that component at all; the mapping, soft-core, dummy-atom, schedule and restraint contributions are protocol-specific and would not transfer. A measured anchor resolves its dimension only to its own precision. By part (iv) the requirement does not soften when the anchors are unbiased: one anchor per component fixes only an arbitrary offset, and every dimension after that costs one more measured ligand.

5 The per-edge error bar as a null hypothesis

The test of Section 6 needs a per-edge variance it can treat as known. Two quantities are involved, and they are kept apart throughout: the closed form for a single BAR window, which the replicate benchmark cannot exercise because it releases no work samples, and the bar the test actually carries, which is the multistate standard error that benchmark ships, composed in quadrature from the two legs and validated against independent repeats. Everything below consumes the variance as a number or as a fixed graph weight.

Theorem 3 (Sampling variance via the sandwich). *For one alchemical edge with n_f forward and n_r reverse works, let p be the logistic BAR weights at the root $\widehat{\Delta F}$ of Bennett’s estimating equation $S(\Delta F) = 0$ and $I = \sum p(1 - p)$ its curvature, the work-distribution overlap. Then $\widehat{\Delta F}$ is an M -estimator, so its asymptotic sampling variance is the sandwich $\text{Var}(\widehat{\Delta F}) = A^{-1}BA^{-1} = B/I^2$, with $A = I$ and $B = n_f \text{Var}_f[p] + n_r \text{Var}_r[p]$ [18]. Here $A \neq B$ arises from BAR’s stratified two-sample design and not from model misspecification: the information equality $A = B$, which would give $1/I$, holds for a single-sample MLE and fails for a two-sample one. Evaluated at the root, with works drawn from the correct pair of ensembles, $B/I^2 = 1/I - (1/n_f + 1/n_r)$ identically: the sandwich is not an approximation to the Bennett/Shirts–Chodera (MBAR) uncertainty [7, 33] that `pymbar` reports but the same quantity, and $-(1/n_f + 1/n_r)$ is exactly the $O(1/n)$ gap between it and the naive plug-in $1/I$. The two sample plug-ins still differ through the moments they substitute, by a paired discrepancy falling monotonically and by more than an order of magnitude from $n=40$ to $n=320$, the range over which it exceeds Monte-Carlo noise at a modest replicate budget.*

Remark (scope of “calibrated”). B/I^2 is the variance of the *sampling* error of $\widehat{\Delta F}$, conditional on the works being drawn from the correct alchemical ensembles. It says nothing about systematic force-field bias, which is outside B/I^2 ; a separate epistemic correction term is attached where the downstream uses of the uncertainty are audited (Section 8). It is also non-negative by construction, where the plug-in expression can return a negative value at very high overlap with tiny samples.

Figure 2 checks the closed form on real work samples, the 19 adjacent- λ BAR windows of the complex leg of the `alchemtest` BACE1 relative-binding calculation. The sandwich over decorrelated-bootstrap truth is 1.00 [0.96, 1.07], against $2\text{--}12\times$ for naive $1/I$ on the same reference, every window sitting at overlap ≥ 0.84 (per-window values, Supplementary Table S1). That reference reuses the same work samples, so the figure establishes agreement with a decorrelated bootstrap on real molecular dynamics in the high-overlap regime this schedule occupies, and not a ligand-to-ligand binding quantity; and since the replicate benchmark releases no work samples, the work-sample route to B/I^2 is exercised on these 19 windows alone. Both the sandwich inputs and the bootstrap truth are subsampled to the statistical inefficiency g [33] before B is formed, so Figure 2 uses effective n (Supplementary Figure S1).

The low-overlap behaviour rests instead on a controlled Monte-Carlo overlap sweep, in which the closed form tracks MBAR and Monte-Carlo truth across the range at a reported/true se of ≈ 1 and at no Monte-Carlo cost, bare $1/I$ overstates by $1.08\text{--}2.65\times$, worst at high overlap, entering as the information-equality plug-in the sandwich corrects and not as a baseline anyone reports, and a learned mean-variance head given the same overlap information is overconfident at both label budgets tested; at extreme low overlap ($n=20$) the bias-corrected plug-in itself runs $\sim 10\%$ below MBAR, both ≈ 1 . That last result is a property of the head’s per-edge parameterisation and not a limit on learning the quantity: pooling the same single labels by overlap recovers the sandwich to $\sim 6\%$, and a null miscalibrated by that much leaves the test of Section 6 intact. The sweep, and the two restrictions that travel with the learned head, are Supplementary Figure S2.

The reported bar against independent replicates. Figure 3 validates the bar the test actually uses, on samples it does not reuse. Each edge’s reported bar is composed in quadrature from the complex-leg and solvent-leg MBAR uncertainties shipped with the benchmark, $\text{se} = \sqrt{\text{se}_{\text{complex}}^2 + \text{se}_{\text{solvent}}^2}$. Each leg term is a multistate MBAR uncertainty over that leg’s λ windows and not the two-state BAR quantity of Theorem 3, so what is validated here is the bar as a practitioner receives it and not the two-state identity itself.

Across 1145 real binding edges from the OpenFE IndustryBenchmarks2024 [4] (3 independent replicates each), the reported MBAR se is calibrated-to-conservative against the across-replicate truth: the replicate ratio is 1.41 [1.26, 1.62], and the bar is conservative on 72% of edges and on 28 of the 34 systems carrying at least eight edges. Both sides of that ratio pool in quadrature, so the $n=3$ replicate term is unbiased at the variance level and needs no small-sample correction. The three replicates are independent stochastic repeats of the sampling process from identical starting coordinates (OpenFE `protocol_repeats` [4]), so the across-replicate SD reflects sampling and convergence noise alone and is a *lower bound* on full reproducibility, excluding pose and system-preparation variance. The reported se does not systematically under-state real reproducibility.

That conclusion disagrees with the closest measurement in the literature. Wade et al. [35], comparing the analytic MBAR uncertainty against the spread over five independent replicas on 54 protein-ligand transformations across five targets in two engines, report that MBAR consistently under-estimates the error, by up to 0.9 kcal/mol. Two differences bear on the disagreement, in opposite directions. Their replicas resample the full ensemble, whereas the OpenFE repeats share starting coordinates, so the denominator used here is a lower bound on run-to-run variability and the

ratio an upper bound on conservatism. Their own functional, applied to these edges, sharpens the disagreement instead of resolving it: the analytic standard error falls below the replicate spread on 0.279 [0.224, 0.332] of edges, against the 0.368 correctly scaled bars produce at three replicates, and the interval excludes that value. Settling the question needs replicates that resample preparation and pose, which this benchmark does not provide, and its eight multi-release systems supply no usable substitute (Section 6).

6 The test, and what it inherits from the bound

The calibrated closure test, called the quality-control (QC) test in this article’s figures and the calibrated test below, makes the bound operational: the residuals of the same projector whose diagonal Theorem 1 reads as an observability certificate, weighted by the bar of Section 5, are a statistic with a stated null. It separates the reproducible part of a perturbation network’s edge-level inconsistency from its sampling noise. Candidate physical sources of a reproducible part include non-convergence, hysteresis, and charge, water or protonation artifacts, but this design does not identify which of them is present, and it does not separate them from a fixed consequence of the one starting structure every repeat begins from.

The calibration underwriting the null is validated against seed and velocity replicate spread, a lower bound on full run-to-run reproducibility, since it excludes pose and system-preparation variance. That restriction bounds what the detection claims below can mean: an inconsistency that survives repeats sharing their starting coordinates is established as more than sampling noise, and not as physical error.

The test performs detection rather than weighting of a fixed estimate. The Gauss–Markov ceiling that makes the calibrated variance second-order for ranking, in a connected graph the GLS contrast mean being unbiased for any positive weights, does not bind a goodness-of-fit test, whose statistic depends on the weights directly.

A calibrated cycle-closure test. The fit of Section 3 gives residuals r_e and the goodness-of-fit statistic $X^2 = \sum_e r_e^2 / V_e^{\text{rep}} \sim \chi^2(\text{dof})$, with dof the number of independent cycles (E minus the incidence rank). Under correctly-sampled physics $\mathbb{E}[X^2/\text{dof}] \approx 1$; here it runs below one, at a median reduced χ^2 of 0.34, qualitatively consistent with the conservative bars of Figure 3. A reduced χ^2 well above one signals edge-level inconsistency in excess of sampling noise, and the test is correctly blind to node-consistent force-field bias, which cancels around a loop.

Two variances have to be told apart from here on. The variance a practitioner is handed, V_e^{rep} , is what the fit uses as its weight; the variance the edge actually has is V_e^{true} . They are related by $V_e^{\text{rep}} = c_e^2 V_e^{\text{true}}$, so the per-edge calibration ratio c_e is the reported standard error over the true one: $c_e > 1$ is a reported bar that is too wide (conservative) and $c_e < 1$ one that is too tight (overconfident). On this benchmark c_e is measured as the reported standard error over the across-replicate standard deviation.

Four summaries of that one distribution appear in this article. Each is a different functional of it, each is named here, and each is called by that name from here on. The *replicate ratio* is the pooled reported standard error over replicate spread, 1.41 (Figure 3). The *closure-weighted mean* is $\mathbb{E}[\chi_\nu^2]$, measured here at 0.85: whitening by the reported errors gives $\mathbb{E}[X^2] = \text{tr}(\Pi D)$ with $D = \text{diag}(c_e^{-2})$, so by Theorem 1 $\mathbb{E}[\chi_\nu^2]$ is the curl-leverage-weighted *mean* of c_e^{-2} . The *calibrated scale* is its square root, 0.92, the factor a reported bar has to be multiplied by to be right. The *inverted median* is one over the square root of the median reduced χ^2 , and it recovers a scalar conservatism only when c_e is constant across edges, which Figure 3B shows it is not.

No single scalar conservatism factor is therefore identified by these data, and the apparent disagreement between the replicate ratio and the closure-weighted mean is an artefact of comparing different functionals of a heavy-tailed distribution, not a physical discrepancy. That distribution is drawn with the published summaries marked on it (Figure 4; `make figCal`): the per-edge ratio spans a factor of 128 between its first and ninety-ninth percentiles while the summaries occupy a $1.57\times$ window. Part of that spread is guaranteed rather than measured, since a spread estimated from three replicates would put the same family of functionals between 0.71 and 2.88 even if every reported bar were exactly right.

Residual correlation among edges sharing a ligand endpoint, which the diagonal- V GLS does not model, is separately negligible (excess correlation over the projector-induced null -0.0079 $[-0.0314, +0.0164]$), and the Benjamini–Hochberg flag set is unchanged by it.

Network maximum-likelihood estimation of $\Delta\Delta G$ networks [38], cycle-closure hysteresis as a diagnostic [8, 36], weighted cycle closure [25], and the constrained variational solution of a whole network under loop closure [13] are established; what is new is the *calibrated* null, a per-edge aleatoric variance measured against seed and velocity repeats instead of assumed, which gives the test per-edge adaptive thresholds and complements the fixed hysteresis cutoffs in standard use. Those repeats share one pose and one system preparation, so the variance they validate excludes the component the alternative hypothesis is built from, no controlled false-positive rate follows from them, and nothing on this benchmark flags once the bars are widened to cover that component.

The calibrated null comes together with the cross-replicate reproduction check of Figure 6B and the estimation–detection conservation law of Theorem 1, which fixes which edges are auditable at all. What is shared with the works above is the weighted least-squares treatment of the cycle graph itself.

On the same 1143 edges a pre-registered head-to-head scored the calibrated test against the standard fixed 1.0 kcal/mol hysteresis cutoff and against a fixed-se χ^2 test, by area under the receiver-operating-characteristic curve; the criterion required beating both, and the verdict was a tie (Supplementary Figure S11). No further reading attaches to the ordering inside that tie, because the only label these data supply divides each residual by the same per-edge variance the calibrated rule uses as its weight: the ordering is monotone in proximity to that variance and not in detection skill, and the comparison ranks nothing.

The two rules are also not independent. On replicate 0 the three flag sets are strictly nested, the calibrated test flagging 6 systems, the fixed-se test 11 and the 1.0 kcal/mol cutoff 26, so the calibrated test catches nothing the cutoff misses and instead removes 20 of its flags.

The three-way chain does not survive the other repeats intact: on replicate 1 the calibrated test flags 6, the fixed-se test 13 and the cutoff 24, and two systems break it, `ciordia_retro` flagged by the calibrated test but not by the fixed-se test and `mup1` flagged by that test but not by the cutoff. On replicate 2 the chain holds again at 3, 11 and 24. The containment any claim here rests on is the two-set one, between the calibrated test and the cutoff, and it holds on all three replicates.

Whether removing those 20 flags is an improvement is what the contaminated label cannot say, and adjudicating them needs new sampling or new experiment, which public retrospective data cannot supply. The calibrated test’s advantage over a fixed cutoff is accordingly structural and is not demonstrated here: per-edge adaptive thresholds, a null measured against seed and velocity repeats instead of assumed, and the observability map of Theorem 1. That measurement is not a controlled error rate, since the repeats hold pose and preparation fixed.

The null is conservative in the bulk, at the median reduced χ^2 above, so on a typical system the test under-flags. The median of the statistic does not bound the realized false-positive rate: Benjamini–Hochberg controls the false discovery rate through the behaviour of the null p -values near the rejection threshold and not through the median of the statistic, and the per-system calibration of

Figure 3 is heterogeneous enough that a conservative bulk and an anti-conservative tail can coexist.

The correction is applied across systems, and the guarantee is carried by Benjamini–Yekutieli [6], which assumes no dependence structure at all, so the independence Benjamini–Hochberg needs is not required. It costs a factor $H_{48} = 4.46$ in the threshold and returns the identical six systems on replicate 0 and the identical three on replicate 2, dropping one, **brd4**, on replicate 1 (Supporting Information). Positive regression dependence is a plausible remark, and nothing here rests on it as an assumption: the systems are fitted separately and share no edges, though they share a force field and, within a target, a preparation.

The level quoted is nominal and not verified. The case for the flags therefore rests on the distribution-free cross-replicate reproduction check below rather than on the χ^2 tail; that check is independent of the χ^2 scaling and assumes no Gaussian null.

A subtler concern is that four of the six flagged systems are themselves locally overconfident against their own across-replicate spread (Figure 3: **brd4** 0.76 $\times$, **faah** 0.90 $\times$, **bace** and **hif2a** 0.96 $\times$), so a flag could partly reflect that system’s own too-tight bars. Retesting each against the width its own replicates measure keeps all six at the per-system granularity, and two, **bace** and **cdk8**, at the per-edge granularity this work pre-specified. The per-edge arm is the only self-calibrated arm holding its level (1/400, that is 0.003 with an exact binomial interval [0.000, 0.014], against a nominal 0.05), and so the conservative answer (Supplementary Table S8).

The test on the replicate benchmark. On the OpenFE IndustryBenchmarks2024 [4] (1145 edges over 49 systems, of which the 48 with ≥ 1 independent cycle, holding 1143 edges, admit the test; 3 replicates), the median reduced χ^2 is 0.34 and a Benjamini–Hochberg false-discovery-rate (FDR) test at 0.05 applied to the first replicate flags six chemically-sensible systems (**brd4**, **bace**, **faah**, **cdk8**, **hif2a**, **p38**; Figure 5A; **brd4**’s data set is a buried-water set, **bace** is protonation-sensitive). That particular set of six is a single-replicate draw and should not be read as a census; Section 6.1 measures how much of it survives the other two replicates.

One of the six has an identifiable cause. The benchmark labels a system by target, and eight of its names union edges from more than one separately released data set. Where those releases share ligands the union carries cycles present in neither release alone, and the cycle spaces sit orthogonally inside the union’s, so the closure statistic splits exactly. On **cdk8** the split is decisive: the union gives $\chi^2 = 77.7$ on 29 degrees of freedom, of which the two releases carry 31.1 on 23 between them, leaving 46.6 on the six cycles that span them, a reduced χ^2 of 7.8 at $p = 2.3 \times 10^{-8}$. Neither release is inconsistent on its own: **merck_cdk8** (51 edges, 20 degrees of freedom) gives $p = 0.218, 0.865$ and 0.189 on the three replicates, and **misc_cdk8** (12 edges, 3 degrees of freedom) gives $0.091, 0.992$ and 0.985 , the first being the smallest of the six and still well short of any threshold used here. The spanning cycles exist because the two releases share seven ligand *names*, and all seven denote chemically different molecules in the two releases, name 13 being a benzyl-indazole carboxamide in one and a chlorophenoxy-thienopyridine carboxamide in the other. Every spanning cycle therefore asserts that two distinct ligands have the same free energy, and resolving each edge within its own release removes all six on every replicate, taking the degrees of freedom from 29 to the within-release total of 23. The 46.6 is a bookkeeping collision in the benchmark’s ligand naming and not a disagreement between two preparations. **cdk8** is flagged for that collision, and the flag is retained in the published set with its cause named, because a system whose network asserts a false identity is what an inconsistency test should reject.

The hazard belongs to the analysis and not to the released data. Each release is internally consistent and each names its own ligands unambiguously; what fails is the join, and it fails for anyone who unions two releases of a target on ligand names without checking that the names denote

the same molecules. That is a common enough way to assemble a benchmark network that it is worth saying plainly: match on structure, not on name.

A closure statistic reads the network it is given, and an inconsistency produced by the book-keeping that assembled that network enters as the same cyclic residual as one produced by the simulations. No other system carries across-release cycles that matter. Two shared ligands are needed before a union carries a cycle that spans its releases, and in six of the eight the releases share at most one, so the union is disconnected or joined by a tree and no such cycle exists at any magnitude; **tyk2**, whose two shared names do denote the same molecules, gives one spanning cycle that is unremarkable ($p = 0.67$). The single cross-release comparison that joins matching ligands shows no excess.

What the benchmark supplies on between-preparation variance is therefore a negative result rather than a number. With **cdk8**’s six spanning cycles resolved away as a naming collision, the only genuine cross-release join anywhere in the benchmark is **tyk2**’s single cycle, which never departs from its null ($p = 0.67$, 0.40 and 0.84 on the three replicates) and whose across-release block never rises above 0.64 of the within-release one (Supplementary Figure S8; **make figRelease**). The public data contain no usable estimate of what a second preparation of the same target adds, which bounds what can be said about the pose and preparation component throughout this article rather than supplying it.

The grouping is therefore a choice the analysis makes, and the flag set is not invariant to it. Re-running the same replicate-0 test on the release’s own blocks gives 56 testable systems and again six flags, but a different six: **cdk8** leaves and **jacs_mcl1** enters. The grouping by target name was not pre-registered, and the Benjamini–Hochberg family size depends on it. The published list is the by-target one throughout. Supplementary Tables S6 and S7 give the per-replicate degrees of freedom, reduced χ^2 and adjusted q -value behind every flag.

The test’s selectivity is entirely a function of calibration (Figure 5B). A uniform $\times 0.15$ shrink standing in for the learned head’s central overconfidence (Supplementary Figure S2) flags 88% of systems (42 of 48), which leaves almost nothing unflagged and so no longer separates anything; the reported bar as shipped flags 12% (6 of 48), a $\times 1.3$ inflation one system of 48, and a $\times 2$ inflation none. These are flagged fractions and not false-positive rates: which of the flags are false is not known here (Section 8).

The count is not separable from the level, and the shipped operating point is the conservative one. The scale at which the reported bars are correct is the calibrated scale, which follows from the closure-weighted mean, the functional that governs the statistic, and not from a pooled ratio; at that mean the bars are wide by $1.09\times$. Section 6.2 puts the closure-weighted mean in $[0.63, 1.08]$, and the calibrated scale accordingly in $[0.79, 1.04]$. Correcting every reported bar to a scale in that interval and re-running the test on the real data gives six flags at 1.04 and at 1.00, seven at 0.92 and nine at 0.79 (**make figSelf**).

Two scales have to be kept apart here, and they are reciprocal. The calibrated scale is $se^{\text{true}}/se^{\text{rep}}$, the reciprocal of the per-edge ratio c_e , measured above as 0.92; a simulation that keeps the test on the reported bars must therefore draw its truth at that many times them, which is a null $1/0.92$ times the truth. Quoted on the calibrated scale, the realized probability that the test flags anything when nothing is wrong is 0/400 at 0.79, 1/400 at 0.92, 18/400 at 1.00 and 40/400 at 1.04. As levels with exact binomial intervals those are 0.000 $[0, 0.009]$, 0.003 $[0.000, 0.014]$, 0.045 $[0.027, 0.070]$ and 0.100 $[0.072, 0.134]$. Four hundred draws do not separate the two lower points from each other or from zero, so nothing is read from their ordering. What the grid supports is the upper end, where the interval clears the nominal level, and the direction between the ends.

That simulation verifies the Benjamini–Hochberg implementation and not the real-data operating point, which is why the level quoted above is nominal and not verified. What it does establish is

a direction: the level sits below nominal at the measured calibration and above it at the other end. Because the bars are measured wide, the level across the calibration interval runs from essentially zero at 0.79 to 0.100 at 1.04, so the exposure that matters is the anti-conservative end, where a bar slightly tighter than the truth doubles the nominal rate.

Sensitivity is exposed in the opposite direction. At $1.3\times$ one system of 48 flags and at $1.5\times$ and $2\times$ none do, so nothing on this benchmark survives a bar widened to cover pose and preparation variance. Widening changes the null being tested instead of moving the same test along it, and what is fragile on that side is sensitivity and not validity.

The uniform shrink is a stress test and not a realistic head, and the comparison is a counterfactual in a further sense: no deployed or published pipeline is known to feed a learned per-edge σ into a network consistency check. The head’s own measured, overlap-dependent profile applied per edge flags 42 of 48, so the degradation is not an artifact of uniformity, and a shuffled control with a swept magnitude shows it is the magnitude that destroys the test while over-wide bars cost power the other way (Supporting Information). The demonstration is accordingly scoped to heads as overconfident as the measured one (reported se 5–11 $\times$ too small): it establishes that such heads saturate the test, and not the shrink magnitude at which selectivity is first lost.

The three independent replicates confirm the flagged inconsistency is reproducible and not sampling. Pooling the replicates shrinks the bars (se $\rightarrow$ se/ $\sqrt{3}$) yet increases the flagged systems’ reduced χ^2 toward the $3\times$ variance-components ceiling (Figure 5C), which occurs only for a component that repeats (single-replicate $\chi^2_\nu = 1 + \sigma_{\text{sys}}^2/V_{\text{samp}}$ vs pooled $1 + 3\sigma_{\text{sys}}^2/V_{\text{samp}}$, with σ_{sys}^2 the reproducible variance component and V_{samp} the sampling one; the ratio $\rightarrow 3$ iff reproducible-dominated). The argument fixes the component’s variance and not its origin: it is invariant to whether σ_{sys}^2 is physical error or a fixed consequence of the one starting structure the three repeats share, both being constant across repeats.

Cross-replicate reproduction of the flagged inconsistency. Greedily removing the flagged (highest- $|z|$) edges restores consistency (reduced $\chi^2 \leq 1$) in 3–7 edges, versus 12–37 removed at random (Figure 6A). That contrast illustrates the repair procedure and is not evidence, because it cannot fail: the statistic being driven down is $X^2 = \sum_e r_e^2/V_e^{\text{rep}}$ and the removal order is descending z_e^2 , that sum’s own per-edge summand, so a network with no localized systematic error at all reproduces the same contrast.

The evidential check is instead out-of-sample. Fitting each replicate’s network separately, the per-edge standardized residuals correlate across independent replicates ($r = +0.30$ to $+0.42$, $n = 1143$, each interval excluding zero under a bootstrap over systems; Figure 6B), where sampling noise alone would give zero. What the flags identify is therefore reproducible across runs and not a property of one draw. The correlation carries the same restriction as the pooling argument: any per-edge quantity held fixed across the three repeats produces it, the shared starting structure is such a quantity, and what is established is reproducibility and not a physical origin.

Summary. The calibrated bar supplies, at no label cost, the null that turns cycle closure into a test separating reproducible from sampling inconsistency, and a bar as overconfident as the learned σ measured here destroys it. Theorem 2 bounds what that test, or any other of its kind, can be asked for, which is why its flags are not a repair list.

6.1 What survives a second replicate

The benchmark ships three independent repeats, so the flag set can be recomputed instead of assumed, and recomputing it shows that the set does not reproduce.

Two conventions bound how the union below should be read. It is descriptive and carries no controlled error rate. Three tests each holding a nominal 0.05 would put the chance of at least one false flag among them at $1 - 0.95^3 = 0.143$ if they were independent, and these three are not; Benjamini–Hochberg in any case controls the false discovery rate rather than the probability of any false flag. The counts are also per system and not per target, the benchmark’s 49 systems carrying 41 protein labels, with **bace** contributing four and **hsp90** four; the six flagged systems are six different proteins, so that count at least is not inflated by system splitting.

The same pipeline at the same FDR level returns {**bace**, **brd4**, **cdk8**, **faah**, **hif2a**, **p38**} on the first replicate, {**bace**, **brd4**, **ciordia_retro**, **hif2a**, **p38**, **renin**} on the second and {**bace**, **cdk8**, **renin**} on the third: eight systems are flagged at least once, one (**bace**) is flagged every time, and pairwise Jaccard overlap is 0.50, 0.29, 0.29 against 0.056 for size-matched random draws, averaged over the three pairs. That instability is not a threshold artefact and persists from $\alpha = 0.01$ to $\alpha = 0.20$ (Supplementary Figure S12). Pooling the three replicates flags seven systems, the union minus **faah**.

The instability is not explained by how cyclic the networks are. Cycle counts on this benchmark are low: nine of the 48 systems carry exactly one independent cycle, twenty-three carry three or fewer, and the quartiles are 2, 5 and 12. A reduced χ^2 on one or two degrees of freedom is an unstable statistic, and those systems do vary more across repeats. But they are not the systems that move in and out of the flag set: seven of the eight ever flagged are well connected, and restricting the comparison to the best-connected third of the benchmark leaves the pairwise overlap no better than it is over all 48. The instability sits where the test has power, which makes it a property of the measurement rather than of thin networks.

Two properties do reproduce. The ranking of systems by closure inconsistency reproduces across replicates (Spearman 0.649, 0.644, 0.723 over all 48 systems), and the dependence of selectivity on σ -calibration, which is the claim this section makes, is essentially unmoved: the $\times 0.15$ shrink flags 42, 41, 41 systems, the reported bar 6, 6, 3, a $\times 1.3$ inflation 1, 1, 1, and a $\times 2$ inflation none, on the three replicates respectively. A single replicate therefore supports a ranking and a calibrated operating point, and not a list of names.

Among the published six, **faah** is the weakest on every measure. Its reduced χ^2 crosses one in both directions across replicates (3.71, 0.45, 1.47), its standardized residuals barely reproduce between replicates ($r = -0.085$), and it is the one member the pooled fit drops: a flag a second replicate does not support, and not a member of the set on the footing of **bace**.

6.2 Out-of-sample behaviour

Two checks use one replicate to predict another and so cannot be circular. First, the per-edge miscalibration measured from replicates two and three predicts the closure statistic of the replicate they exclude. With $V_e^{\text{rep}} = c_e^2 V_e^{\text{true}}$ as above, Theorem 1 makes the closure-weighted mean predictable from c_e alone, and that prediction correlates with the observed per-system χ_v^2 at Spearman 0.648 ($n = 48$). Both sides divide by the same reported variance, so the reference is not zero: under a simulation with no systematic error anywhere the same statistic returns 0.059 with a 95% upper limit of 0.364, and 0.648 exceeds every one of 200 draws (Supplementary Figure S13A).

The aggregate level agrees less well. The predicted closure-weighted mean is 0.85, with interval [0.63, 1.08]. The observed degree-of-freedom-weighted value is 1.32, with interval [0.81, 1.80]. The two intervals overlap, and the prediction does not cover the observation.

The scope of that agreement is narrow. The prediction is built entirely from measured miscalibration, so a rank agreement of this size says that the closure statistic largely reads out how well each system’s bars are scaled. It is a statement about what the test responds to, and not evidence

that the flagged systems are in error. The flagged systems do sit above the line, which is what any reproducible offset does: it inflates a closure residual without inflating the spread between repeats. Being reproducible across repeats that share a starting structure is what places a system above the line, and that is weaker than being in error.

Second, edges selected on one replicate are more inconsistent on the two held out than equal-size random selections are (mean gain in held-out reduced χ^2 , Stouffer $p = 1.3 \times 10^{-3}$, 1.3×10^{-3} , 4.4×10^{-3} over the three rotations), while a topology-only control built from curl-leverage is indistinguishable from random ($p = 0.21$ – 0.91). Each candidate statistic was calibrated against a no-error null before use and the realized level of every one is reported; two fired far above nominal and are not used. Only the consistency-gain statistic is specific to the flagged systems: an $|z|$ -rank-matched control drawn from unflagged systems reproduces the leverage-normalised residual signal but not the consistency gain. The effect is an aggregate one, significant individually in two of the five testable systems.

7 Computational details

Data. Every closure fit consumes the release’s combined per-replicate table directly, taking the per-leg complex and solvent ΔG and $d\Delta G$ columns and composing the edge value and its standard error from them; no network-level or cycle-closure correction is applied to an edge before it enters the fit. Figure 2 uses the `alchemtest` BACE1 relative-binding and benzene sets, from which forward and reverse work samples are available per λ window. The replicate validation, the cycle-closure test and every experiment on real binding data use the OpenFE IndustryBenchmarks2024 release [4], which ships three independent protocol repeats per transformation: 1145 edges across 49 benchmark systems. A system here is one of the benchmark’s named edge networks and not one protein, `jak2_set1` and `jak2_set2` being separate systems built on the same protein, so the system count exceeds the number of distinct proteins (Section 6.1). The 48 systems with at least one independent cycle, which between them hold 1143 of the 1145 edges, admit the closure test, and each counts as one hypothesis in the Benjamini–Hochberg correction; the single remaining system is acyclic and contributes the other two edges. The per-system calibration ratios of Figure 3B are pooled instead over the 34 systems that carry at least eight edges, eight being the smallest edge count at which a ratio pooled against an $n = 3$ replicate spread is stable enough to plot on its own row; the other fifteen systems enter the aggregate ratio and the per-edge panel but are not shown individually. Grounding against measured affinity uses public ChEMBL IC_{50} values. The downstream uses audited in Section 8 run instead on the public OpenFF protein–ligand benchmark [15], which aggregates the FEP+ and Merck data sets [31, 36] and carries curated per-edge experimental $\Delta\Delta G$ labels, under both-endpoint scaffold-disjoint out-of-distribution splits; interval sharpness, commit-to-synthesis selection, reward re-ranking and the commit-trust decision run on it, while edge ranking and the stopping decision run in controlled simulations. No free-energy simulation was run for this work; all alchemical output is taken as published.

Per-edge variance. $\widehat{\Delta F}$ is obtained by solving Bennett’s estimating equation $S(\Delta F) = 0$ by scalar root-find, and its variance from the closed form B/I^2 of Theorem 3 evaluated at the root. The variance is therefore computed from the same simulation output as the estimate, at no label cost.

Cycle-closure test. Node potentials are fitted by generalized least squares on the signed incidence matrix of the perturbation network, whitened by the per-edge standard errors, so that the fit is a

weighted least-squares problem with conductances $1/V_e^{\text{rep}}$. Writing N for that incidence matrix, to keep B for the sandwich meat of Theorem 3, the statistic is the reduced χ^2 of the residuals with $\text{dof} = E - \text{rank}(N)$; per-system tail probabilities use the exact regularized upper incomplete gamma function rather than a Wilson–Hilferty approximation, and systems are flagged by Benjamini–Hochberg control of the false discovery rate at $\alpha = 0.05$ across systems. The population for the closure test is the 48 systems that carry at least three replicate-0 edges and at least one independent cycle, which between them hold 1143 of the benchmark’s 1145 edges.

The two baseline detectors. The fixed hysteresis cutoff flags a system when the largest absolute closure sum over a fundamental cycle basis exceeds 1.0 kcal/mol. That basis is the cotree of a spanning tree of the perturbation network, so it holds one cycle per degree of freedom of the closure fit, and the aggregation over them is a maximum. The rule consumes no per-edge uncertainty and applies no multiplicity correction. The fixed-se test is the closure fit above with one standard error per system, the median of that system’s per-edge reported values, substituted for every edge; its per-system tail probabilities pass through the same Benjamini–Hochberg control at $\alpha = 0.05$. The three detectors accordingly differ only in what they do with the per-edge bar: the cutoff ignores it, the fixed-se test replaces it by one number per system, and the calibrated test uses all of them.

Simulation protocols and model architectures. Two protocols underwrite numbers quoted above: the pure-null simulation that supplies every realized false-flag level, and the reproducible-systematic magnitude used as the label in detection comparisons. The amortized predictor used in the acquisition, stopping and reward experiments, whose predictive uncertainty $\sigma_{\text{total}} = \gamma \sqrt{\sigma_{\text{ens}}^2 + \sigma_{\text{ale}}^2}$ composes deep-ensemble epistemic spread with a benchmark aleatoric floor and is conformally scaled at miscoverage $\alpha = 0.1$, and the learned-variance baseline of Supplementary Figure S2 are separate and much smaller models. Both protocols, and the features, layer sizes, objectives and training budgets of both models, are specified in the Supporting Information.

Software. Python ≥ 3.11 with numpy 2.5.0, scipy 1.18.0, pymbar 4.0.3, torch 2.12.1, RDKit 2026.3.3 and matplotlib 3.11.0. The pipeline is deterministic: in the environment recorded here, re-running a figure’s command reproduces that figure’s rendered raster byte for byte, and this was checked on the figures that depend on ensemble training rather than assumed. Ensemble training runs full batch on CPU for a fixed number of epochs, each member is seeded from its own index, and the ligand ordering reaching the featurizer and the sampler is sorted rather than read out of a set, so no unrecorded source of variation enters the training path.

Use of generative artificial intelligence. The tool named in the acknowledgement wrote and ran the code and drafted the text, so what constrains a reported number is the pipeline and not the drafting. Every quantity here and in the Supporting Information is emitted by committed code and regenerated by one documented command, with the record beside the figure it explains; the pre-registered criteria are frozen in files committed before the commit that computes what they govern, and anchored by a digest the test suite asserts on every run. Neither constraint says the tool did not err. They are what makes an error findable by re-running the pipeline rather than by re-reading the prose.

8 Limitations

The sandwich is aleatoric-only and conditional on correctly sampled ensembles. Systematic force-field bias lies outside it, and the out-of-distribution behaviour audited here rests on an ensemble plus a scalar aleatoric floor: there is no learned force-field-correction head, so the three-way decomposition of the synthetic illustration is not reproduced on real data.

The binding-edge se is validated against three independent replicates (Figure 3). In aggregate the replicate ratio is $\sim 1.41\times$ and the bar is conservative, but calibration is heterogeneous: of the 34 systems carrying at least eight edges, 28 are conservative and six overconfident, namely **t4_lysozyme** $0.99\times$, **hif2a** and **bace** $0.96\times$, **faah** $0.90\times$, **brd4** $0.76\times$, and the protonation variant **bace_p3_arg368_in** $0.41\times$, the only one of the six that is not marginal. Those replicates are seed and velocity repeats from identical starting coordinates, so the across-replicate SD is a lower bound on full reproducibility, excluding pose and preparation variance, and the measured conservatism may be closer to neutral against real reproducibility. The same restriction limits what the quality-control test detects: a pose or a protonation assignment carried unchanged into all three repeats reproduces exactly as physical error does, so the flags are reported as reproducible inconsistency throughout.

The test detects only *edge-level* inconsistency, and neither removal rule improves accuracy against experiment: guided removal ties random on the four grounded systems (Stouffer $p = 0.48$; Figure S10) and the influence rule does no better overall, doing measurable harm on **p38** and reaching a nominal improvement on **bace**, neither clearing the conjunctive criterion (Section 4; Figure 7C). Theorem 2 offers a structural reading of both nulls, bounded by the margin Section 4 measures and not established independently of it. The pre-registered falsifier for that reading is a guided removal beating its random null in a majority of grounded systems at a combined p below 0.05, which would show the visible part actionable; it is not met.

The measurement rests on four systems selected by the test, so it characterises those systems rather than the benchmark, and for **cdk8** it characterises less even than that. Its grounded sub-network is all 31 edges of **merck_cdk8**, the block whose own closure statistic is not significant on any replicate, so the evidence that flagged it lies outside the sub-network measured. **p38**’s grounded sub-network is the **jacs_p38** block, which is individually significant ($\chi^2_r = 2.63$, $p = 1.9 \times 10^{-4}$), and **hif2a** and **bace** are single-block systems.

The measurement also runs on a connectivity-based ChEMBL join that is stereo-blind, so enantiomer pairs sharing a 2D graph collapse to one experimental value. That inflates absolute MUE while cancelling in the guided-versus-random contrast, since both arms read the identical cached affinity. The experimental and annotation errors the join carries are per-ligand quantities entering the same unauditable subspace as force-field bias, so the measurement locates the error without attributing it.

The influence rule is a second pre-registered hypothesis on the data that returned the first null, so it is a second look and not independent confirmation. The theorem bounds what a network’s own edge values can support and says nothing against methods bringing information from outside the network. A prospective flag, repair and remeasure test with new molecular dynamics lies outside what public data can support.

The cross-replicate check shows that what is flagged reproduces across independent runs; it is not a false-positive-rate guarantee and does not certify the nominal Benjamini–Hochberg rate. Recall cannot be measured at all, there being no ground-truth error labels. Replacing each system’s reported standard error by its own replicate-measured calibration changes the flag set in both directions, by an amount depending on the granularity and the direction, from retaining two of the six to reaching sixteen. The arms, their realized false-flag probabilities and the single one that

holds its level are tabulated in the Supporting Information, and none is reported here as a flag set. The flag set is therefore sensitive to calibration error of the size actually present, in both directions, which limits how far any single flag list should be trusted.

Together with the median reduced χ^2 , which implies limited single-replicate power, that makes the flagged systems a floor rather than a prevalence estimate. `renin` carries the reproducible signature by the pooled-replicate ratio, its reduced χ^2 rising to 4.7 pooled from a per-replicate mean of 2.5, and clears the false-discovery threshold on two of the three replicates though not on the first, which is the one the published list is drawn from; it is a system that list misses rather than one the test misses, and `ciordia_retro` is a second such case.

A calibrated uncertainty was audited on six downstream uses: interval sharpness, edge ranking in active learning, commit-to-synthesis selection, reward re-ranking over a measured congeneric series, and two decision tasks, commit trust and calibrated stopping. The quantity under test there is not the per-edge sandwich bar of Section 5 but the predictive uncertainty of an amortized $\Delta\Delta G$ predictor, a conformally scaled combination of ensemble spread and an aleatoric floor; the Supporting Information defines it. It leads a fair baseline in none of the six, and on one it does harm: the calibrated reward lowers the true hit-rate of a generator on the single evaluable congeneric series, because the uncertainty there tracks chemical novelty rather than quality. On commit trust the physics σ matches post-hoc recalibration by temperature scaling [14] or isotonic regression [39] rather than beating it, and the separation in the stopping simulation is absent at the six per cent miscalibration a same-budget pooled estimator reaches. Calibration is a precondition for decisions of this kind rather than a multiplier on their quality. Supplementary Table S11 collects every effect against its baseline.

The stopping experiment is a controlled simulation, and the reward re-ranking audit rests on a single σ -varying target, so its conclusion that σ tracks novelty rather than quality is directional and should be tested across more chemotypes.

9 Conclusion

A perturbation network’s own edge values determine only the cycle part of its error. Fitting cycle closure against the precisions the free-energy estimator already reports turns a descriptive consistency check into a test with a stated null, orders networks by how far their own calculations disagree, and names in advance the edges that lie in no cycle and can carry no evidence. The test consumes only quantities the estimator computes anyway, and its selectivity is set by how well those quantities are calibrated. On these data the test catches nothing the fixed hysteresis cutoff misses: the two flag sets are nested and the head-to-head between them is a tie, so the advantage is structural, being per-edge adaptive thresholds, a null measured against seed and velocity repeats, and the observability map. The flags are also not a repair list: removing the edges they name does not recover accuracy against experiment, and a rule built from the theorem does no better and on one system does harm, because the component that sets accuracy against experiment lies in the gradient part, which no closure statistic reaches.

Those limits compound in the flag set this article publishes, and the sum is worth stating once. Of the six systems flagged on the first replicate, one carries its flag from a naming collision created by the grouping the analysis chose, one is the weakest on every measure applied to it and drops when the replicates are pooled, and the membership of the set does not reproduce across the three repeats. What survives all three is the ordering of systems and the per-edge residuals behind it, which is what a reader should take the test to deliver.

Closing that gap needs information the network does not contain, and in absolute measurements

the worst case costs one ligand per dimension. That is a worst case: a bias smooth in chemical space is a far smaller object, and a noisy anchor resolves only to its own precision.

A second ceiling comes from the measurement rather than from the decomposition. The bar the test uses as its null is measured against repeats that resample seeds and velocities while holding one pose and one system preparation fixed, so it prices sampling noise and not the preparation an entire network inherits, and widening it by half, which is what covering that would take, leaves nothing on this benchmark flagged at all. The test is therefore bounded twice: by the geometry of the network, and by a variance component its own null never sees. The practical use of cycle closure is accordingly as a calibrated alarm, with the error that moves a ranking to be sought in absolute measurements or independent preparations instead.

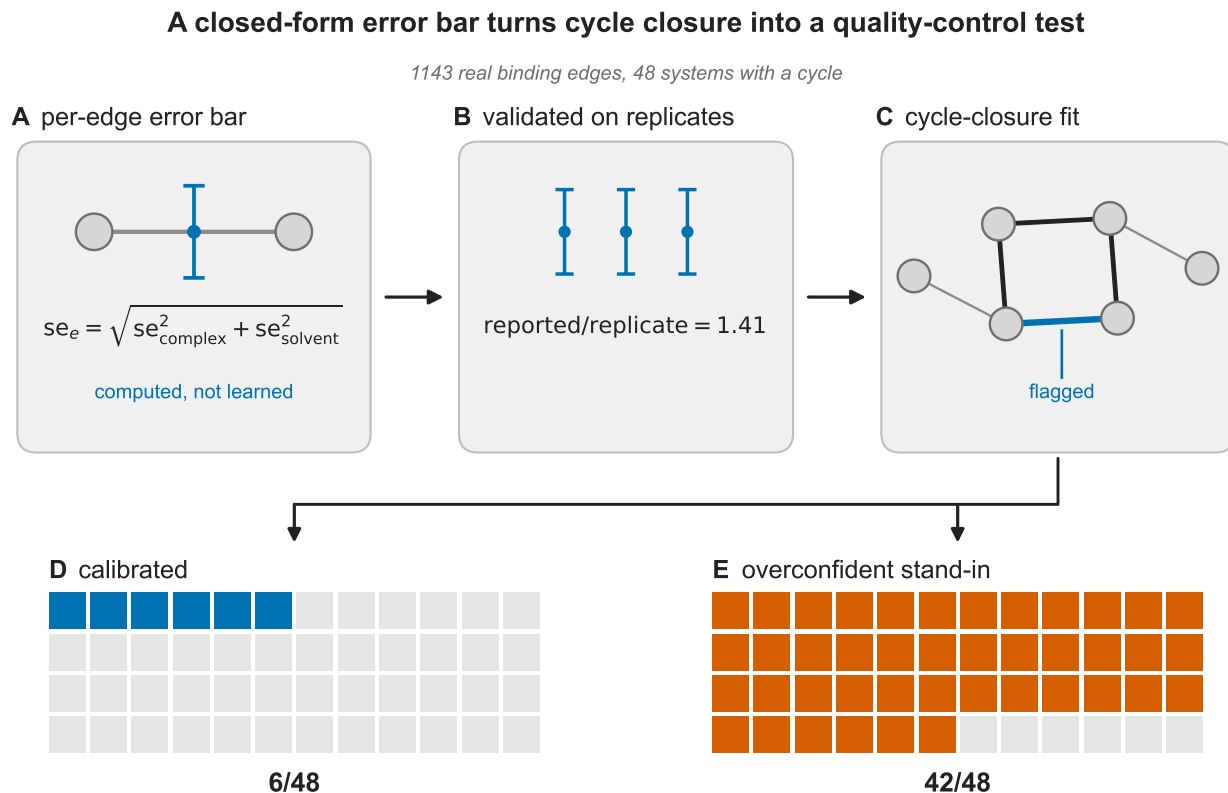


Figure 1: **Overview.** The pipeline this article runs on the benchmark, left to right. Every edge arrives with a per-edge standard error the estimator computes rather than learns, composed in quadrature from the complex-leg and solvent-leg uncertainties; that bar is validated against the benchmark’s three independent protocol repeats (reported over replicate spread 1.41); and it is then the null of a generalized-least-squares cycle-closure fit, drawn as a small perturbation network with one cycle highlighted and one edge marked as flagged. Across the 1143 real binding edges that lie in the 48 benchmark systems carrying a cycle (of 1145 edges over 49 systems), that calibrated bar flags 6, 6 and 3 systems on the three repeats, one of them on all three, and a comparison bar built edge by edge from a learned head’s measured miscalibration profile, not that head run here, flags 42.

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Generative artificial intelligence was used in this work, throughout and substantially. Anthropic’s Claude, driven through the Claude Code command-line interface, was used from the start of the project to submission: to write the analysis library, its unit tests and the figure-generation scripts; to run those analyses and produce the figures, tables and per-figure records; to draft and to revise the text of this article and of its Supporting Information; and to run the adversarial review passes whose surviving objections are reported in the text. Section 7 states what that does and does not leave open. The author directed the work, verified its results, arguments and references, and takes full responsibility for the content.

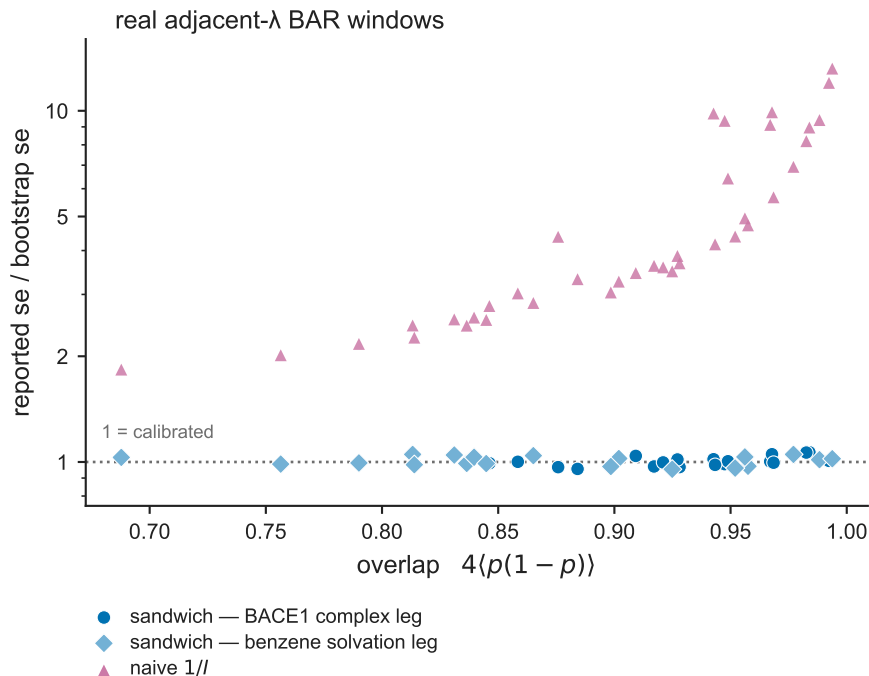


Figure 2: **Sandwich variance against a decorrelated bootstrap on real BAR windows.** Reported standard error divided by the decorrelated-bootstrap standard error, versus work-distribution overlap, for the sandwich B/I^2 on the 19 adjacent- λ BAR windows of the complex leg of the `alchemtest` BACE1 relative-binding calculation (filled circles) and on the 19 windows of the benzene solvation legs (filled diamonds), and for the naive information-equality plug-in $1/I$ on the same windows (triangles). Each point is one two-state BAR estimate between neighbouring λ values, and every window sits at overlap ≥ 0.84 . The controlled low-overlap sweep and the learned-variance baselines are Supplementary Figure S2.

Author Contributions

N.L.P. is the sole author and is responsible for the conception and design of the study, the theory, the software, the analysis, the preparation of the figures, and the writing of the manuscript.

Conflicts of Interest

The author declares no competing financial interest.

Data and Software Availability

All data used in this work are public. The `alchemtest` BACE1 and benzene alchemical sets supply the work samples of Figure 2; the OpenFE IndustryBenchmarks2024 replicate set supplies the 1145 binding edges and their three protocol repeats; the OpenFF protein-ligand benchmark, which aggregates the FEP+ and Merck data sets, supplies the downstream audit tasks; and ChEMBL supplies the experimental affinities used for grounding. No free-energy simulation was run for this work; all alchemical output is taken as published. Two findings reported here were sent to the maintainers of that benchmark as public issues on its repository, <https://github.com/OpenFreeEnergy/>

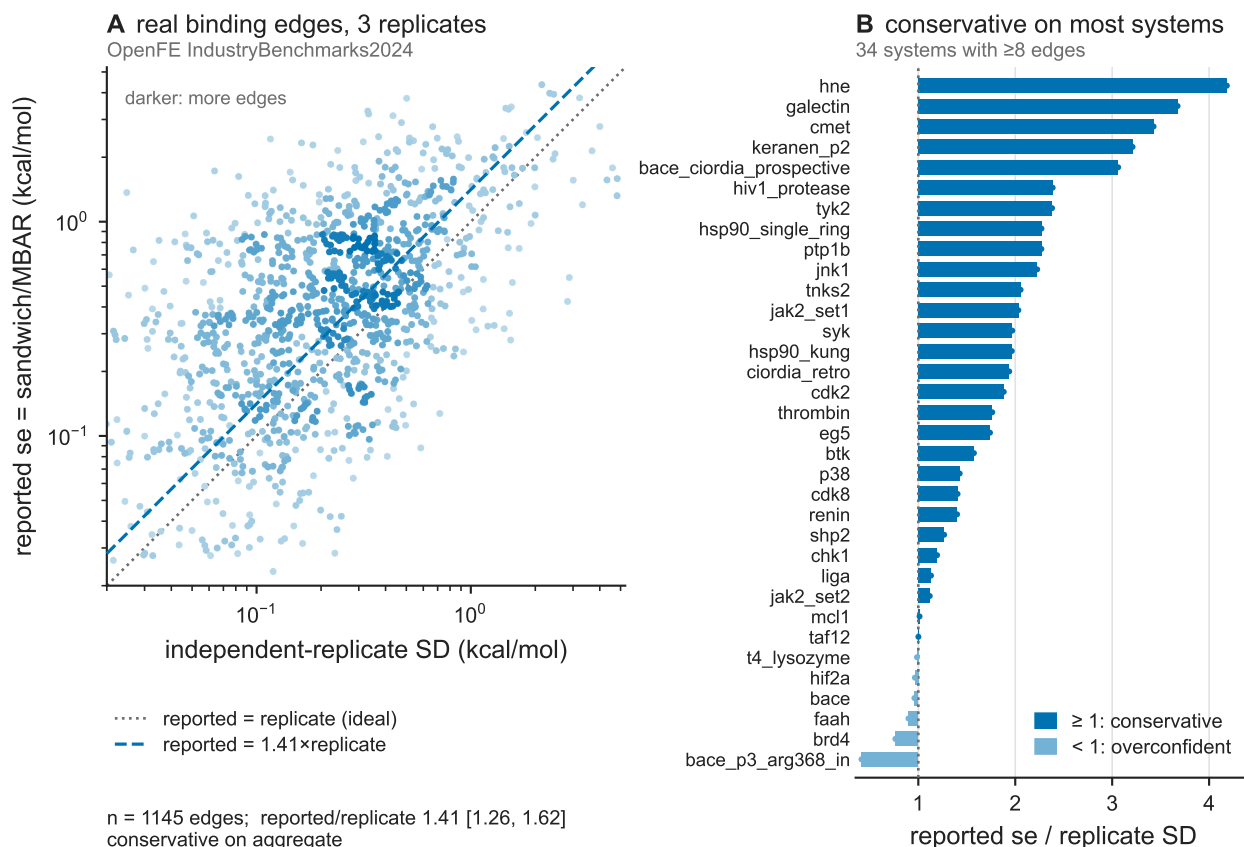


Figure 3: **Independent-replicate validation on real binding edges.** Reported per-edge standard error versus the across-replicate empirical standard deviation, on 1145 OpenFE IndustryBenchmarks2024 [4] edges (3 independent replicates each). (A) Reported standard error versus across-replicate standard deviation, with the $y=x$ line. (B) Per-system pooled ratio of reported standard error to across-replicate standard deviation, for the 34 benchmark systems carrying at least eight edges.

IndustryBenchmarks2024: the per-edge auditability and detectable-shift tables of Section 3 as issue 331 on 9 August 2026, and the `cdk8` ligand-name collision of Section 6 as issue 332 on 22 August 2026. Both were open and unanswered at the time of submission. The BAR layer, the cycle-closure quality-control module, the theorem-verification and quality-control unit tests, and every figure-regeneration script are released as open source under a permissive licence, with each figure in this article regenerated by a single documented command. The code is available at <https://github.com/doctawho42/fep-surrogate-pipeline> under the MIT licence; the version from which every figure in this article was generated is tagged v1.0.0. The archived release is deposited at <https://doi.org/10.5281/zenodo.22046693>.

References

- [1] Matteo Aldeghi, Alexander Heifetz, Michael J. Bodkin, Stefan Knapp, and Philip C. Biggin. Accurate calculation of the absolute free energy of binding for drug molecules. *Chemical Science*, 7(1):207–218, 2016. doi: 10.1039/c5sc02678d.

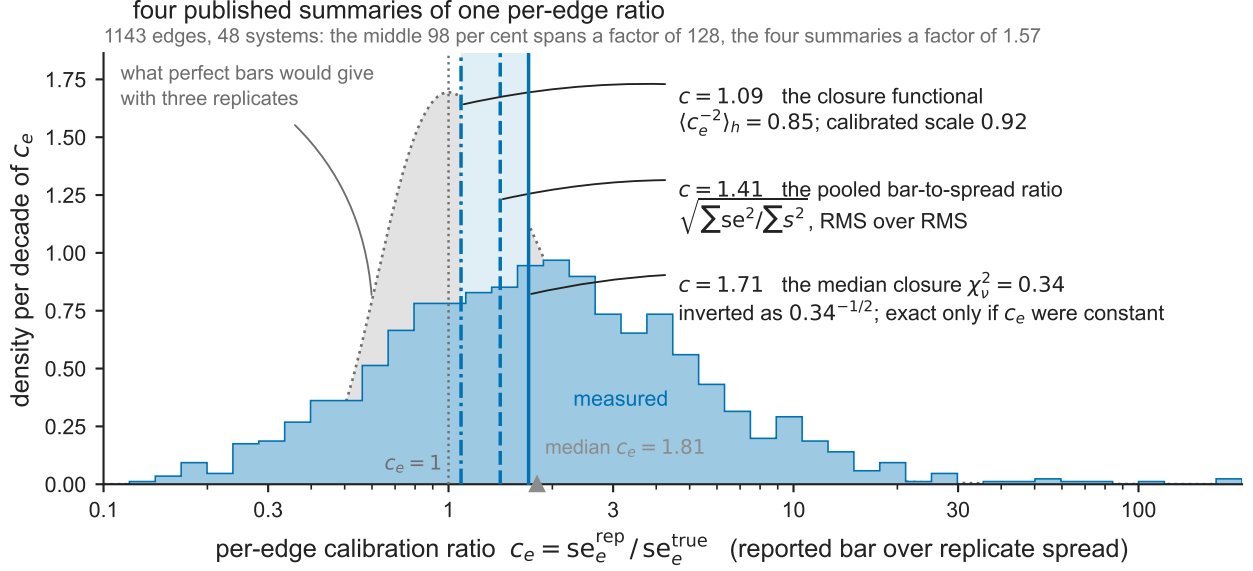


Figure 4: **One heavy-tailed per-edge ratio, and the summaries this article quotes from it.** The calibration ratio $c_e = \text{se}_e^{\text{rep}} / \text{se}_e^{\text{true}}$ over the 1143 edges of the 48 systems that carry a cycle, on a logarithmic axis because c_e is a ratio, so a bar twice too wide and one twice too tight are equally miscalibrated, and because the tail is the reason the summaries disagree rather than a nuisance to be clipped. Quartiles 0.91, 1.81 and 3.60. Each mark is a different functional of this one distribution: the curl-leverage-weighted mean of c_e^{-2} that Theorem 1 makes $\mathbb{E}[\chi_\nu^2]$, together with its root the calibrated scale, which are one mark and not two; the replicate ratio of Figure 3, the pooled reported standard error over replicate spread, 1.41; and the inverted median, the median reduced χ^2 of 0.34 inverted, which recovers a scalar conservatism only if c_e were constant across edges and carries that caveat on the figure. The grey silhouette is the distribution perfect bars would give at three replicates, where $s^2/\sigma^2 \sim \chi_2^2/2$ makes $c_e = 1/\sqrt{\text{Exp}(1)}$ a parameter-free null with median 1.20: part of the measured spread is the denominator’s own small-sample noise and only the excess is per-edge heterogeneity. The panel adds no measurement, and no claim beyond the text’s: no single scalar conservatism is identified by these data. Reproduce with `make figCal`.

- [2] Anastasios N. Angelopoulos and Stephen Bates. A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. arXiv preprint, 2021.
- [3] Maximilian Balandat, Brian Karrer, Daniel R. Jiang, Samuel Daulton, Benjamin Letham, Andrew Gordon Wilson, and Eytan Bakshy. BoTorch: A Framework for Efficient Monte-Carlo Bayesian Optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [4] Hannah M. Baumann, Joshua T. Horton, Michael M. Henry, Alyssa Travitz, Benjamin Ries, Richard J. Gowers, David W. H. Swenson, Iván Pulido, Dominic Rufa, David L. Dotson, Nupur Bansal, Joseph P. Bluck, Howard Broughton, Kira Campbell, Lili Cao, Benedikt Frieg, Vytautas Gapsys, Hendrik Göddeke, Marco Klähn, Sirish Kaushik Lakkaraju, Stephanie M. Linker, Thomas Löhr, Aniket Magarkar, Sergio Pérez-Conesa, Hans E. Purkey, Hayk Saribekyan, Jenke Scheen, Christina E. M. Schindler, Thomas Steinbrecher, Chaya D. Stern, Patricia Suriana, William C. Swope, Gary Tresadern, Lev Tsidilkovski, Binqing Wei, Alexander H. Williams, Yao Wu, Ivy Zhang, John D. Chodera, James R. B. Eastwood, David L. Mobley, and Irfan

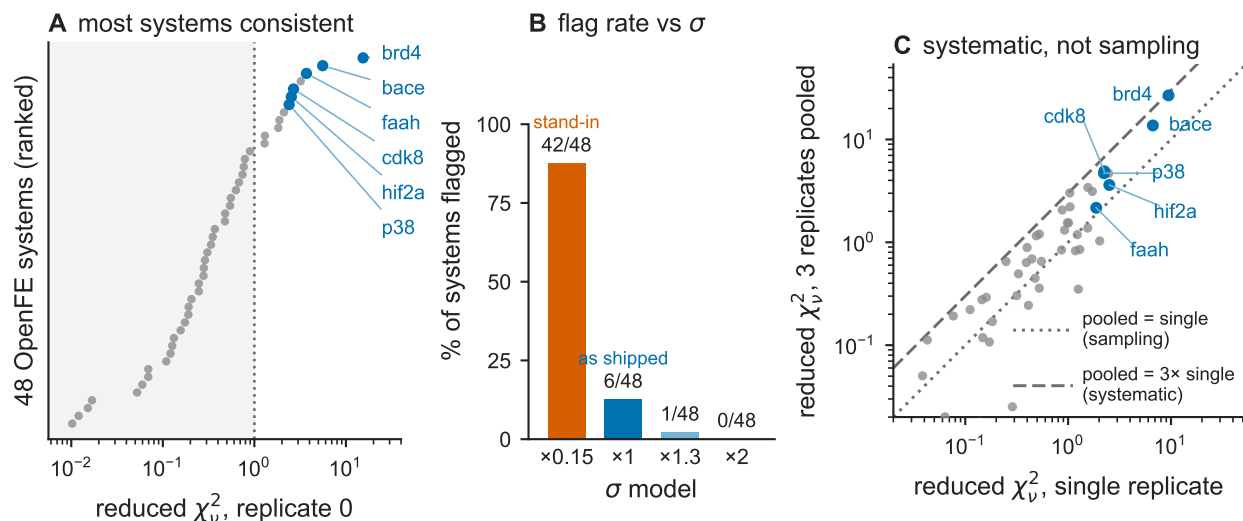


Figure 5: **Calibrated cycle-closure quality control.** (A) Per-system reduced χ^2 from a generalized-least-squares cycle-closure fit, ranked, with shading below $\chi^2_v = 1$ and the flagged systems marked. (B) Percentage of systems flagged under four σ models, in plotted order: a uniform $\times 0.15$ shrink, the reported bar as shipped, a $\times 1.3$ inflation and a $\times 2$ inflation. The shipped bar is the tick marked $\times 1$; the scale at which those bars are correct is measured as 0.92 of them, which is a different quantity from the tick. (C) Reduced χ^2 from a single replicate against the value from the three replicates pooled, with reference lines at pooled/single = 1 and = 3.

Alibay. Large-Scale Collaborative Assessment of Binding Free Energy Calculations for Drug Discovery Using OpenFE. *Journal of Chemical Information and Modeling*, 66(11):6429–6452, 2026. doi: 10.1021/acs.jcim.6c00089.

- [5] Emmanuel Bengio, Moksh Jain, Maksym Korablyov, Doina Precup, and Yoshua Bengio. Flow Network based Generative Models for Non-Iterative Diverse Candidate Generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [6] Yoav Benjamini and Daniel Yekutieli. The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*, 29(4), 2001. doi: 10.1214/aos/1013699998.
- [7] Charles H Bennett. Efficient estimation of free energy differences from Monte Carlo data. *Journal of Computational Physics*, 22:245–268, 1976. doi: 10.1016/0021-9991(76)90078-4.
- [8] Di Cui, Bin W. Zhang, Zhiqiang Tan, and Ronald M. Levy. Ligand Binding Thermodynamic Cycles: Hysteresis, the Locally Weighted Histogram Analysis Method, and the Overlapping States Matrix. *Journal of Chemical Theory and Computation*, 16(1):67–79, 2020. doi: 10.1021/acs.jctc.9b00740.
- [9] Sofia Dias, Nicky J. Welton, Valéria C. C. Marinho, Georgia Salanti, Julian P. T. Higgins, and A. E. Ades. Estimation and Adjustment of Bias in Randomized Evidence by Using Mixed Treatment Comparison Meta-Analysis. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 173(3):613–629, 2010. doi: 10.1111/j.1467-985x.2010.00639.x.
- [10] Panagis Filippakopoulos, Jun Qi, Sarah Picaud, Yao Shen, William B. Smith, Oleg Fedorov, Elizabeth M. Morse, Tracey Keates, Tyler T. Hickman, Ildiko Felletar, Martin Philpott,

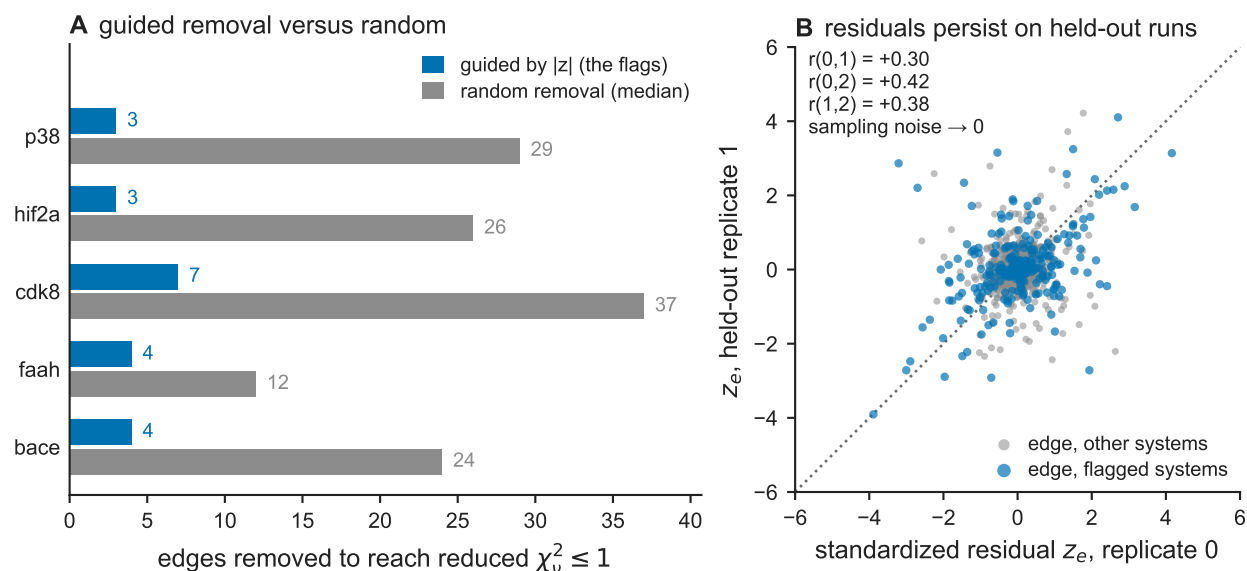


Figure 6: **Edge-removal repair test and cross-replicate residual reproduction.** (A) Number of edges removed, in descending order of $|z|$, before reduced $\chi^2 \leq 1$ is restored: guided removal of the flagged high- $|z|$ edges versus random removal, per system. (B) Standardized residual z_e from a replicate-0 network fit versus z_e from a replicate-1 fit, for edges in flagged and in non-flagged systems, with the $y = x$ line and the Pearson correlation between each pair of replicates.

Shonagh Munro, Michael R. McKeown, Yuchuan Wang, Amanda L. Christie, Nathan West, Michael J. Cameron, Brian Schwartz, Tom D. Heightman, Nicholas La Thangue, Christopher A. French, Olaf Wiest, Andrew L. Kung, Stefan Knapp, and James E. Bradner. Selective inhibition of BET bromodomains. *Nature*, 468(7327):1067–1073, 2010. doi: 10.1038/nature09504.

- [11] Peter I. Frazier, Warren B. Powell, and Savas Dayanik. A Knowledge-Gradient Policy for Sequential Information Collection. *SIAM Journal on Control and Optimization*, 47:2410–2439, 2008. doi: 10.1137/070693424.
- [12] Jacob R. Gardner, Geoff Pleiss, David Bindel, Kilian Q. Weinberger, and Andrew Gordon Wilson. GPyTorch: Blackbox Matrix-Matrix Gaussian Process Inference with GPU Acceleration. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [13] Timothy J. Giese and Darrin M. York. Variational Method for Networkwide Analysis of Relative Ligand Binding Free Energies with Loop Closure and Experimental Constraints. *Journal of Chemical Theory and Computation*, 17(3):1326–1336, 2021. doi: 10.1021/acs.jctc.0c01219.
- [14] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On Calibration of Modern Neural Networks. In *International Conference on Machine Learning (ICML)*, 2017.
- [15] David Hahn, Christopher Bayly, Melissa L. Boby, Hannah Bruce Macdonald, John Chodera, Vytautas Gapsys, Antonia Mey, David Mobley, Laura Perez Benito, Christina Schindler, Gary Tresadern, and Gregory Warren. Best Practices for Constructing, Preparing, and Evaluating Protein-Ligand Binding Affinity Benchmarks [Article v1.0]. *Living Journal of Computational Molecular Science*, 4:1497, 2022. doi: 10.33011/livecoms.4.1.1497.

- [16] J. P. T. Higgins, D. Jackson, J. K. Barrett, G. Lu, A. E. Ades, and I. R. White. Consistency and inconsistency in network meta-analysis: concepts and models for multi-arm studies. *Research Synthesis Methods*, 3(2):98–110, 2012. doi: 10.1002/jrsm.1044.
- [17] David C. Hoaglin and Roy E. Welsch. The Hat Matrix in Regression and ANOVA. *The American Statistician*, 32(1):17–22, 1978. doi: 10.1080/00031305.1978.10479237.
- [18] Peter J. Huber. The behavior of maximum likelihood estimates under nonstandard conditions. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability, Volume 1*, pages 221–233. University of California Press, 1967.
- [19] Xiaoye Jiang, Lek-Heng Lim, Yuan Yao, and Yinyu Ye. Statistical ranking and combinatorial Hodge theory. *Mathematical Programming*, 127(1):203–244, 2011. doi: 10.1007/s10107-010-0419-x.
- [20] Tuomo Kalliokoski, Christian Kramer, Anna Vulpetti, and Peter Gedeck. Comparability of mixed IC₅₀ data – a statistical analysis. *PLoS ONE*, 8(4):e61007, 2013. doi: 10.1371/journal.pone.0061007.
- [21] Alex Kendall and Yarin Gal. What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [22] D. J. Klein and M. Randić. Resistance distance. *Journal of Mathematical Chemistry*, 12:81–95, 1993. doi: 10.1007/BF01164627.
- [23] Ulrike Krahn, Harald Binder, and Jochem König. A graphical tool for locating inconsistency in network meta-analyses. *BMC Medical Research Methodology*, 13:35, 2013. doi: 10.1186/1471-2288-13-35.
- [24] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [25] Yishui Li, Runduo Liu, Jie Liu, Haibin Luo, Chengkun Wu, and Zhe Li. An Open Source Graph-Based Weighted Cycle Closure Method for Relative Binding Free Energy Calculations. *Journal of Chemical Information and Modeling*, 63(2):561–570, 2023. doi: 10.1021/acs.jcim.2c01076.
- [26] Guobing Lu and A. E. Ades. Assessing Evidence Inconsistency in Mixed Treatment Comparisons. *Journal of the American Statistical Association*, 101(474):447–459, 2006. doi: 10.1198/016214505000001302.
- [27] Nikolay Malkin, Moksh Jain, Emmanuel Bengio, Chen Sun, and Yoshua Bengio. Trajectory balance: Improved credit assignment in GFlowNets. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [28] Antonia S.J.S. Mey, Bryce K. Allen, Hannah E. Bruce Macdonald, John D. Chodera, David F. Hahn, Maximilian Kuhn, Julien Michel, David L. Mobley, Levi N. Naden, Samarjeet Prasad, Andrea Rizzi, Jenke Scheen, Michael R. Shirts, Gary Tresadern, and Haufeng Xu. Best Practices for Alchemical Free Energy Calculations [Article v1.0]. *Living Journal of Computational Molecular Science*, 2(1):18378, 2020. doi: 10.33011/livecoms.2.1.18378.

- [29] D.A. Nix and A.S. Weigend. Estimating the mean and variance of the target probability distribution. In *Proceedings of 1994 IEEE International Conference on Neural Networks (ICNN'94)*, pages 55–60 vol.1, 1994. doi: 10.1109/ICNN.1994.374138.
- [30] Yaniv Romano, Evan Patterson, and Emmanuel J. Candès. Conformalized Quantile Regression. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [31] Christina E. M. Schindler, Hannah Baumann, Andreas Blum, Dietrich Böse, Hans-Peter Buchstaller, Lars Burgdorf, Daniel Cappel, Eugene Chekler, Paul Czodrowski, Dieter Dorsch, Merveille K. I. Eguida, Bruce Follows, Thomas Fuchß, Ulrich Grädler, Jakub Gunera, Theresa Johnson, Catherine Jorand Lebrun, Srinivasa Karra, Markus Klein, Tim Knehans, Lisa Koetzner, Mireille Krier, Matthias Leiendecker, Birgitta Leuthner, Liwei Li, Igor Mochalkin, Djordje Musil, Constantin Neagu, Friedrich Rippmann, Kai Schiemann, Robert Schulz, Thomas Steinbrecher, Eva-Maria Tanzer, Andrea Unzue Lopez, Ariele Viacava Follis, Ansgar Wegener, and Daniel Kuhn. Large-Scale Assessment of Binding Free Energy Calculations in Active Drug Discovery Projects. *Journal of Chemical Information and Modeling*, 60:5457–5474, 2020. doi: 10.1021/acs.jcim.0c00900.
- [32] Maximilian Seitzer, Arash Tavakoli, Dimitrije Antic, and Georg Martius. On the pitfalls of heteroscedastic uncertainty estimation with probabilistic neural networks. In *International Conference on Learning Representations (ICLR)*, 2022.
- [33] Michael R. Shirts and John D. Chodera. Statistically optimal analysis of samples from multiple equilibrium states. *The Journal of Chemical Physics*, 129(12):124105, 2008. doi: 10.1063/1.2978177.
- [34] Daniel A. Spielman and Nikhil Srivastava. Graph Sparsification by Effective Resistances. *SIAM Journal on Computing*, 40(6):1913–1926, 2011. doi: 10.1137/080734029.
- [35] Alexander D. Wade, Agastya P. Bhati, Shunzhou Wan, and Peter V. Coveney. Alchemical Free Energy Estimators and Molecular Dynamics Engines: Accuracy, Precision, and Reproducibility. *Journal of Chemical Theory and Computation*, 18(6):3972–3987, 2022. doi: 10.1021/acs.jctc.2c00114.
- [36] Lingle Wang, Yujie Wu, Yuqing Deng, Byungchan Kim, Levi Pierce, Goran Krilov, Dmitry Lupyan, Shaughnessy Robinson, Markus K. Dahlgren, Jeremy Greenwood, Donna L. Romero, Craig Masse, Jennifer L. Knight, Thomas Steinbrecher, Thijs Beuming, Wolfgang Damm, Ed Harder, Woody Sherman, Mark Brewer, Ron Wester, Mark Murcko, Leah Frye, Ramy Farid, Teng Lin, David L. Mobley, William L. Jorgensen, Bruce J. Berne, Richard A. Friesner, and Robert Abel. Accurate and Reliable Prediction of Relative Ligand Binding Potency in Prospective Drug Discovery by Way of a Modern Free-Energy Calculation Protocol and Force Field. *Journal of the American Chemical Society*, 137:2695–2703, 2015. doi: 10.1021/ja512751q.
- [37] Halbert White. Maximum Likelihood Estimation of Misspecified Models. *Econometrica*, 50:1, 1982. doi: 10.2307/1912526.
- [38] Huafeng Xu. Optimal Measurement Network of Pairwise Differences. *Journal of Chemical Information and Modeling*, 59(11):4720–4728, 2019. doi: 10.1021/acs.jcim.9b00528.
- [39] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002. doi: 10.1145/775047.775151.

- [40] Barbara Zdrazil, Eloy Felix, Fiona Hunter, Emma J. Manners, James Blackshaw, Sybilla Corbett, Marleen de Veij, Harris Ioannidis, David Mendez Lopez, Juan F. Mosquera, Maria Paula Magarinos, Nicolas Bosc, Ricardo Arcila, Tevfik Kiziloren, Anna Gaulton, A. Patricia Bento, Melissa F. Adasme, Peter Monecke, Gregory A. Landrum, and Andrew R. Leach. The ChEMBL Database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*, 52(D1):D1180–D1192, 2024. doi: 10.1093/nar/gkad1004.

Supporting Information

Reproducibility. The analysis code released with this work regenerates every figure in this article, each from a single command, and it emits the per-system tables reported here from the same run that draws the corresponding figure, so the two cannot drift apart. This covers the quality-control results (Figures 5 and 6 of the main text and Figures S12, S13 and S14 here), the observability map of Figure S9, the fixed-cutoff comparison of Figure S11, the σ -profile sweeps of Figures S15 and S16, the autocorrelation panel of Figure S1, the curated-label grounding of Figure S6 and Table S3, the label-noise floor of Figure S7, and the cross-release split of Figure S8. The constants appearing in the theorems and the invariants of the cycle-closure test are additionally asserted as unit tests that run in the same environment. Every input is public: the alchemtest BACE1 and benzene sets, the OpenFF protein–ligand benchmark, ChEMBL [40], and the OpenFE IndustryBenchmarks2024 replicate set [4]. Recalibration baselines use temperature scaling [14] and isotonic regression [39].

BACE1 per-window overlap values (Figure 2). Figure 2 reports that naive $1/I$ overstates the bootstrap se by $2\text{--}12\times$ on the 19 real BACE1 adjacent- λ BAR windows (mean 6.07, $[2.78, 12.01]$), the worst overstatement occurring at high overlap, consistent with the controlled panel (invariant #1, main text). These 19 rows are the windows of one λ schedule, taken from the complex leg of the alchemtest AMBER `load_bace_example` relative-binding calculation across its decharge, recharge and van der Waals legs; each is a two-state BAR estimate between neighbouring λ values, so the set tests the closed-form variance against a decorrelated bootstrap on real molecular-dynamics works, and it does not test 19 separate binding free energies or reach beyond the high-overlap band this schedule occupies. Table S1 lists the (λ_i, λ_j) window labels, per-window overlap $4\langle p(1-p) \rangle$, and both ratios for all 19, emitted verbatim by the released analysis code, λ labels included, so the λ pairing is re-derivable from it (seed 20260629; both sandwich inputs and the bootstrap truth subsampled to the statistical inefficiency g before B is formed, Section 5; numerically identical to the binding rows of the corresponding panel). Every window sits at overlap ≥ 0.84 (mean 0.93); the worst naive/bootstrap ratio ($12.01\times$) occurs at the single highest-overlap window ($4\langle p(1-p) \rangle = 0.992$), and the best (least overstated, $2.78\times$) occurs at the lowest-overlap window in the set (0.846), the same high-overlap-is-worst regime as the controlled sweep of Figure S2 rather than a property of the particular windows sampled. The trend across the full 19-window set is not confined to the extremes: the Spearman rank correlation between overlap and the naive/bootstrap ratio is $\rho = 0.84$ ($n = 19$, computed from Table S1), so the relationship holds within the narrow $[0.846, 0.992]$ overlap band, not just at its endpoints.

Autocorrelation robustness: raw n vs $n_{\text{eff}} = n/g$ (Figure S1). The sandwich $B = n_f \text{Var}_f[p] + n_r \text{Var}_r[p]$ (Theorem 3) assumes independent samples, whereas real molecular-dynamics works are autocorrelated. On the 19 real BACE1 adjacent- λ BAR windows of Figure 2 (no new molecular dynamics), the sandwich se is recomputed at the raw sample count and again after decorrelating

Table S1: **Per-window overlap on the 19 real BACE1 adjacent- λ BAR windows.** λ_i, λ_j are the BAR window endpoints; $4\langle p(1-p) \rangle$ is the normalized BAR overlap; sandwich/boot and naive/boot are the reported-se/bootstrap-se ratios for the sandwich B/I^2 and the naive $1/I$, respectively, against a decorrelated-bootstrap reference standard error. Rows are sorted by overlap, ascending; repeated (λ_i, λ_j) pairs are distinct windows from BACE1’s decharge, recharge, and van der Waals alchemical legs, which share a common λ schedule.

λ_i	λ_j	$4\langle p(1-p) \rangle$	sandwich/boot	naive/boot
0.437	0.563	0.846	0.992	2.780
0.316	0.437	0.858	1.001	3.020
0.206	0.316	0.876	0.967	4.377
0.750	1.000	0.884	0.956	3.315
0.563	0.684	0.909	1.041	3.451
0.500	0.750	0.917	0.971	3.618
0.684	0.794	0.921	0.997	3.581
0.250	0.500	0.927	1.017	3.860
0.000	0.250	0.928	0.967	3.678
0.500	0.750	0.943	1.018	9.823
0.794	0.885	0.943	0.981	4.165
0.000	0.250	0.948	0.986	9.360
0.115	0.206	0.949	1.005	6.414
0.250	0.500	0.967	1.004	9.116
0.750	1.000	0.968	1.052	9.901
0.885	0.952	0.969	0.995	5.670
0.952	1.000	0.983	1.064	8.193
0.048	0.115	0.984	1.066	8.960
0.000	0.048	0.992	1.007	12.007

each leg to its statistical inefficiency g [33] (geometric-mean $\bar{g} = 1.48$, range $[1.15, 2.16]$; median count $n=500 \rightarrow 349$). Against the correlation-aware truth (the decorrelated bootstrap standard deviation, the same reference as Figure 2), the n_{eff} sandwich is calibrated, at a reported/truth ratio of 1.00 $[0.93, 1.07]$, reproducing Figure 2. The raw- n sandwich is overconfident by $0.83 \times [0.67, 0.97]$, matching the $1/\sqrt{\bar{g}} = 0.82$ autocorrelation prediction. The sandwich also tracks its own matched (raw) bootstrap (1.00, $[0.92, 1.05]$): the closed form is internally consistent, and the only quantity that must be right is the effective sample count, which the paper decorrelates to g throughout (Section 5, main text). The calibration verdict is therefore robust to autocorrelation once n_{eff} is used, as it is in every real-data panel.

The learned-variance comparison (Figure S2). Figure S2 is the controlled overlap sweep behind the comparison the main text states: the sandwich B/I^2 coincides with pymbar-MBAR and with Monte-Carlo truth across the range, while a fair learned mean-variance head, given the same overlap information the closed form uses, is overconfident at a realistic label budget and remains so at a twenty-fold larger one. Two restrictions travel with the figure. The head is trained and measured here, but its use as a comparison bar downstream is a counterfactual: no deployed or published pipeline is known to feed a learned per-edge σ into a network consistency check, and the arms built from its measured miscalibration in the quality-control and stopping experiments are comparison models rather than estimators run in those experiments. And the overconfidence is a property of the per-edge parameterisation rather than of learning from labels, since on the identical labels pooling single-label edges by overlap recovers the sandwich to $\sim 6\%$ at 4000 edges (below).

Corrected learned-variance foils and the data-efficiency floor (Figure S2). Neither correction restores the head’s calibration at a realistic budget. A head trained on the β -weighted negative log-likelihood (β -NLL) [32] ($\beta=0.5$, the standard objective correction) does not recover calibration (se $\approx 0.11\times$, slightly worse than plain NLL’s $0.15\times$), and at this training seed a 5-member deep ensemble [24] reaches only the large-budget floor (se $\approx 0.20\times$, the same $\approx 5\times$ overconfidence a $20\times$ -larger-budget plain head reaches). The residual $\sim 5\times$ overconfidence of the plain, oracle and β -NLL heads is therefore a property of the per-edge parameterisation rather than of the label budget or of the Gaussian-NLL objective. The conditional variance $se(\text{overlap})$ is identifiable from single labels, and pooling single-label edges by overlap recovers the sandwich to $\sim 6\%$ at 4000 edges ($\sim 4\%$ at 40000) on the same labels the heads were given. This bears on the scope of the quality-control comparison. Placing a 6% miscalibration on the frozen scale grid of Figure S16 gives seven flagged systems of 48 against the published six, and a 6% over-estimate gives six; the test is unchanged. The collapse to 42 of 48 requires the $5\text{--}11\times$ under-estimate the trained per-edge heads produce, so the contrast is scoped to heads of that kind and is not evidence that a same-budget learned estimator must break the test.

Seed stability of the learned-variance foils (Figure S2). These ratios are stable across independent training seeds for the plain and β -NLL heads: over 5 seeds the plain head spans $0.1229\text{--}0.2103\times$ (mean $0.1537\times$) and the β -NLL head $0.1024\text{--}0.1256\times$ (mean $0.1147\times$), so their overconfidence is a property of the estimator at this budget rather than of a single initialization. The 5-member ensemble is not stable: across the same 5 seeds, each with a disjoint 5-network member block, it spans $0.1984\text{--}1.2200\times$ (mean $0.4198\times$, sd $0.4481\times$), and one seed overestimates the true se by 22%, crossing perfect calibration. Unlike the plain, oracle ($0.1941\text{--}0.2571\times$, mean $0.2101\times$), and β -NLL heads, whose overconfidence is robust across independent retrains, the ensemble is unreliable in both directions at this budget.

Synthetic decomposition (illustration). In a controlled one-dimensional problem whose generative structure is known, the decomposed interval is up to $2.2\times$ sharper than a split-conformal interval ($2.0\times$ sharper than conformalized quantile regression [30]) at equal coverage, with better conditional calibration. The problem illustrates the mechanism; the evidence on real data is Figure S18.

Edge weighting (Figure S3). The acquisition policy outperforms random selection by a factor of 2.2. Changing the edge weighting from naive to sandwich changes calls-to-top- k by a factor of 0.98 [0.93, 1.03] (24-seed bootstrap confidence interval), which is the Gauss–Markov ceiling on the choice of weights.

The Fisher–resistance correspondence. The theorem below establishes what the per-edge bar buys the decision layer: with the inverse sandwich variances as conductances, the Gaussian posterior precision of the node potentials is a weighted Laplacian, contrast variance is effective resistance, acquisition updates are $O(1)$, and the knowledge gradient is exactly zero on gauge directions. It is stated here because what the article measures downstream of it is that the weight *value* is second-order for ranking: in a connected graph the generalized-least-squares contrast mean is unbiased for any positive conductances, so by Gauss–Markov the choice of weights cannot improve a ranking (Table S11 below). The theorem’s content is accordingly the gauge-invariance, which is structural and holds for any positive conductances, and which the simulation of Figure S4 exercises.

Theorem S1 (Fisher–resistance correspondence). *For relative measurements $y_e = (\varphi_j - \varphi_i) + \varepsilon_e$, $\varepsilon_e \sim \mathcal{N}(0, V_e^{\text{true}})$ with $V_e^{\text{true}} = B_e/I_e^2$, the Gaussian posterior precision of the node potentials is the weighted Laplacian $L = \sum_e w_e(e_i - e_j)(e_i - e_j)^\top$ with conductances $w_e = 1/V_e^{\text{true}} = I_e^2/B_e$. Contrast variance equals effective resistance [22], a single edge recovers V_e^{true} , Sherman–Morrison gives $O(1)$ acquisition updates, and $\ker L \supseteq \text{span}\{\mathbf{1}\}$, so the knowledge gradient (KG) [11] is exactly zero on gauge/redundant directions: the acquisition is automatically gauge-invariant.*

The edge weights are the inverse sandwich variances, the readout of Section 5 entering the decision layer; the acquisition model identifies the reported variance with the true one ($c_e = 1$), an assumption the replicate validation of Section 5 tests rather than one this section adds. The gauge-zero property is structural: $\text{KG} = 0$ on gauge directions follows from the kernel inclusion alone, so it does not depend on the weight value, and by Gauss–Markov the value does not improve ranking (Table S11 below). For a *connected* network the inclusion is an equality, $\ker L = \text{span}\{\mathbf{1}\}$, for any positive conductances, and the overall additive offset is then the only unidentifiable direction; a network with k components has a k -dimensional kernel, one offset per component. The Laplacian-precision and resistance-distance machinery is classical generalized least squares (GLS) on graphs and the resistance distance of Klein and Randić [22]; what is specific here is the sandwich precision I_e^2/B_e as the conductance, in place of raw overlap, and the automatic gauge-invariance of the acquisition.

Gauge-aware acquisition (Figure S4). The Fisher–resistance Laplacian (Theorem S1) makes the knowledge gradient exactly zero on gauge (all-ones) contrasts. A controlled simulation gives $\text{KG} = 10^{-21}$ on the all-ones gauge contrast, an exact zero, and the identifiability statement behind it transfers structurally to multi-target networks. A gauge-unaware acquisition spends budget resolving a decision-irrelevant cross-cluster offset, reaching the within-target top- k at regret 0.27 where the gauge-aware one reaches 0.01.

Structural context of the flagged systems (Figure S5). Two of the systems the closure test flags (Figure 5) carry an independently-documented hard structural feature, read here directly from public co-crystal structures. In BACE1 (PDB 4DJW) the catalytic aspartic-acid dyad hydrogen-bonds the inhibitor (Asp93 and Asp289, both 2.7 Å); the protonation state of an aspartic-protease dyad is a recognized source of free-energy error. In BRD4 BD1 (PDB 3MXF, in complex with JQ1 [10]) the acetyl-lysine pocket packs 16 crystallographic waters within 4.5 Å of the ligand (8 within 3.5 Å) around the recognition asparagine Asn140 (3.1 Å); this conserved water network is the known BRD4 free-energy challenge [1]. The other flags carry analogous a-priori difficulties: **faah** (covalent serine hydrolase, large acyl channel), **cdk8** and **p38** (large scaffold and R-group changes), and **hif2a** (a buried polar cavity). The context is retrospective and structural rather than a per-edge causal claim: cycle closure is edge-level and blind to node-consistent bias (main text). It establishes only that the flagged set coincides with the features a domain expert would identify a priori. In Figure S5 the colours annotate structural objects rather than methods.

Per-edge observability map and predictive falsifier (Figure S9). The estimation–detection conservation law (Theorem 1) certifies $\sum_e h_e = \text{dof}$ to machine precision (maximum deviation 1.78×10^{-14}) across the 48 cyclic systems and identifies 48 structurally un-auditable bridge edges ($h_e = 0$, one per system on average); a pre-registered test of whether the curl-leverage also predicts the location of reproducible systematic error, run on 934 pooled edges over 45 systems against the rule confirmed if and only if $\rho > 0$ and $p < 0.05$ (10^4 within-system block permutations), is negative (studentized Spearman $\rho = -0.073$, $p = 0.9571$), so only the structural certificate is claimed.

Per-system auditability (Table S4). Theorem 2 makes a network’s auditable fraction a property of its topology and its reported precisions alone, so it can be tabulated for every benchmark system before any question of what that system’s error turns out to be. Table S4 gives, per system, the ligand and edge counts, the number of independent cycles ν , the auditable fraction ν/E , the bridge count, and the median detectable shift over that system’s auditable edges. The auditable fraction is exactly $1 - (N - c)/E$, for N ligands, E edges and c connected components, and so is small wherever a network is sparse: it runs from 0.12 to 0.46 with a median of 0.31, and no benchmark system exposes more than half of its own error space to the test. The eight flagged systems span essentially that whole range, from 0.12 to 0.46, rather than concentrating at the auditable end: a system is flagged for the error it carries, not for how much of that error is in view. `brd4`, flagged on two of the three replicates, sits at the bottom of the table with a single independent cycle: its reduced χ^2 is a one-degree-of-freedom statistic, which is the quantitative form of the caution Section 6.1 places on it. Rows are ordered by auditable fraction, and the flag marks are recomputed from the three replicates by the same code that draws Figure 7, not transcribed from the main text.

What was pre-registered. Several arguments in this article turn on what was frozen before the computation rather than chosen after it. The design file governing the decomposition increment was committed to the article’s repository before the commit that computes any of the quantities it governs, and it is released with the analysis code; the Data and Software Availability statement says where. It fixes three items and no others: H1, the visible fraction of the measured error against experiment with its chance level and its success criterion; H2, the influence-ranked repair race; and H3, the auditability map, which it labels descriptive and gives neither threshold nor verdict. Both success criteria are conjunctive, so a single conjunct firing is not a success. H1 is labelled a quantification backed by an identity rather than a risky test, and says why: experimental affinities are per-ligand, so $\Pi\epsilon$ reduces to the closure residual and the visible fraction is an identity, with only the comparison against the chance level at risk. The confound the design cannot resolve is declared in it: experimental measurement error and the stereo-blind affinity join are both per-ligand and therefore also gradient fields, so the design cannot separate per-ligand force-field bias from per-ligand experimental or annotation error, and the conclusion it licenses is the first of those and not the second. H2 is declared in the design file to be a second look at data that had already produced one null, and its result carries that status.

The other pre-specified constants this article invokes are recorded in the released analysis code and in the pre-registration for the experiment that fixed them, not in that file: the $|z|$ -rule conjunctive criterion, the curl-leverage falsifier rule, the fixed-cutoff head-to-head criterion, the 1.0 kcal/mol cutoff itself, the swept scale grid with its two frozen readouts, the per-edge granularity of the self-calibrated null, and the close-the-loop criterion. Three of them live only in code, and the paragraph below audits those three against the descriptions printed here.

Audit of the frozen criteria that live only in code. Three criteria are recorded in the analysis code and not in the design file, and each is audited here against the description printed in this article. The curl-leverage falsifier is confirmed if and only if $\rho > 0$ with a within-system block-permutation $p < 0.05$ over 10^4 permutations; the code implements that rule and the supplement prints it. The fixed-cutoff head-to-head requires the calibrated test’s detection area to exceed both baselines, implemented as a paired bootstrap of $\text{AUC}(A) - \max(\text{AUC}(B), \text{AUC}(C))$ with a win declared only when the point difference is positive and the lower confidence bound excludes zero; the printed description matches, and the tie follows from the +0.140 [−0.008, 0.316] against the stronger baseline. The $|z|$ -rule conjunctive criterion matches the code as printed here, requiring Stouffer

$p < 0.05$ and a guided ΔMUE above the 95th percentile of its own random null in a majority of grounded systems, and that is the upper tail because ΔMUE is defined as $\text{MUE}(\text{full}) - \text{MUE}(\text{after})$, so an improvement is positive. Its frozen file names the lower tail, "below the 5th percentile", which is the same event stated in the opposite sign convention; the code’s variable name follows the file rather than the statistic. No result depends on the discrepancy. The frozen file is unaltered, and the description printed in this article is the one the code implements.

The visible fraction is metric-dependent (Section 4). The visible fraction f , the share of the error against experiment lying in the auditable subspace (Section 4), is a ratio of squared norms, so it depends on the metric those norms are taken in, and the main text quotes it in the variance-weighted metric the closure statistic itself uses. Three readings of the same four sub-networks are collected in Table S2, each with its own chance level and with the range the fraction takes when one edge is deleted at a time. The whitened reading takes both quantities in the variance-weighted metric the closure statistic lives in, against dof/E . The isotropic reading keeps that fraction and measures it against $\text{tr}(\Pi W)/\text{tr}(W)$, the chance level an error isotropic in kcal/mol would produce under the same projector; it gives the weakest margins, and it is the reading the abstract and the main text quote. The unweighted reading recomputes the projector itself in the unweighted metric, $\Pi = I - A(A^\top A)^+ A^\top$ on the raw incidence matrix A , so neither the fraction nor its chance level carries any variance weighting; there the pooled fraction rises from 0.6% to 5.45%. The worst single cell in the whole table is **bace** under the isotropic reading with one edge deleted, at a factor of 1.36 below its chance level. Both the fraction and the chance level are recomputed on every deletion, since deleting an edge moves the projector as well; **make figHodge** produces the table. **cdk8**’s flag arises from a naming collision between two releases rather than from either release’s simulations (Section 6), so the pooled figure is given both ways: 0.0060 whitened and 0.0545 unweighted over the four systems, 0.0060 and 0.0610 with **cdk8** dropped, so the whitened headline does not depend on it. Recomputing the ratio on medians rather than sums, which removes any dependence on a few large residuals, gives 0.0034, 0.0068, 0.0052 and 0.0015, the same order throughout; and **bace**, the one grounded system whose ligand set contains no enantiomer pair collapsed by the stereo-blind affinity join, shows the effect as plainly as the three that do. Every reading puts f below its own chance level; what changes is by how much, and the conclusion the main text draws is the weakest of them.

The label source can also be changed. The measurement uses a stereo-blind, target-wide ChEMBL search under the coverage rule, and the OpenFF protein–ligand benchmark of the downstream audit below (Table S11) carries curated per-edge experimental $\Delta\Delta G$ for three of the four systems (**bace** has none). Matching those to the network by canonical SMILES recovers 57%, 98% and 91% of the curated edges for **cdk8**, **hif2a** and **p38**. The matched sub-networks are much smaller than the grounded ones and **p38**’s retains no cycle, so the check is weak, but where it can be run it agrees: the visible fraction on curated labels is 0.0007 for **cdk8** against a chance level of 0.071, and 0.0006 for **hif2a** against 0.147. Comparing those two figures against what the ChEMBL labels give is only meaningful on identical edges. Against the full systems’ ChEMBL fractions, measured over 31 and 59 edges rather than over these 14 and 34, the curated reading is the lower one; restricted to the same edges it is the higher one, 0.00071 and 0.00063 against 0.00016 and 0.00021, by factors of 4.4 and 2.9. The direction on identical edges is the one the decomposition predicts. A per-ligand label error is a gradient field, the whitened projector annihilates it exactly, and it therefore lands entirely in the denominator of f and pushes the fraction down. The stereo-blind, target-wide ChEMBL join is the noisier of the two label sources, and on these same edges its median unsigned error against experiment is the larger of the two, 1.61 against 0.56 kcal/mol on **cdk8** and

Table S2: **The visible fraction under three metric conventions.** For each grounded system: edges E , independent cycles ν , the auditable share of that system’s error against experiment, the chance level it is measured against, their ratio, the range the fraction takes when one edge is deleted at a time, and the worst margin over those deletions with both the fraction and the chance level recomputed on each. *Whitened* takes both in the variance-weighted metric the closure statistic lives in, against ν/E . *Isotropic* keeps that fraction and measures it against the chance level an error isotropic in kcal/mol would produce under the same projector, $\text{tr}(\Pi W)/\text{tr}(W)$. *Unweighted* recomputes the projector in the unweighted metric, so neither the fraction nor its ν/E chance level carries any variance weighting. Every cell of every convention, and every single-edge deletion, sits below its own chance level; the weakest is **bace** isotropic under deletion. Pooled over the four systems the fraction is 0.0060 whitened and 0.0545 unweighted, and 0.0060 and 0.0610 with **cdk8** dropped. Produced by `make figHodge`.

system	E	ν	metric	visible	chance	margin	leave-one-out	worst
cdk8	31	9	whitened	0.0065	0.2903	44.8×	[0.0034,0.0113]	23.60×
			isotropic, kcal/mol	0.0065	0.1273	19.6×	[0.0034,0.0113]	7.36×
			unweighted	0.0286	0.2903	10.2×	[0.0178,0.0324]	8.22×
hif2a	59	19	whitened	0.0017	0.3220	187.2×	[0.0010,0.0029]	106.60×
			isotropic, kcal/mol	0.0017	0.0510	29.6×	[0.0010,0.0029]	13.21×
			unweighted	0.0172	0.3220	18.7×	[0.0127,0.0188]	16.53×
p38	51	18	whitened	0.0086	0.3529	41.0×	[0.0033,0.0132]	25.81×
			isotropic, kcal/mol	0.0086	0.1240	14.4×	[0.0033,0.0132]	1.46×
			unweighted	0.1371	0.3529	2.6×	[0.0285,0.1499]	2.31×
bace	26	5	whitened	0.0460	0.1923	4.2×	[0.0027,0.0628]	2.55×
			isotropic, kcal/mol	0.0460	0.0919	2.0×	[0.0027,0.0628]	1.36×
			unweighted	0.0564	0.1923	3.4×	[0.0033,0.0707]	2.26×

1.24 against 1.05 on **hif2a**; the noisier source should therefore give, and does give, the smaller visible fraction. What the two label sets supply is a consistency check on shared calculated values, with the cleaner set giving the larger visible fraction, and not two independent confirmations of either. Lifting the pooled fraction to its chance level by annotation noise alone would require that noise to carry almost all of the total squared standardized error; how much it would have to carry, per system as well as pooled, and how much a literature-scale label noise can actually supply, are measured below (Figure S7). **bace**’s margin is the exception and is why that number does not appear in the abstract.

The visible fraction on curated labels, over the benchmark (Figure S6, Table S3). The three-system comparison above runs on three sub-networks the closure test itself selected. The same measurement runs over the whole curated set. Fifteen of the benchmark’s targets carry curated per-edge experimental $\Delta\Delta G$, and fourteen of those names also appear in the replicate benchmark (**pde2** does not); after the join, eight of the fourteen have a matched sub-network that still carries a cycle and so a measurement at all. Pooled over those eight, across 205 edges and 31 residual degrees of freedom, the visible fraction is 0.00052, against a dof/E chance level of 0.151 and an isotropic-in-kcal/mol chance level of 0.0137: a factor of 289 below the first and 26 below the second. Split by the flag, the two systems the closure test flags give 0.00071 over 63 edges and the six it does not give 0.00042 over 142. The systems the test does not select show the effect at least as strongly as the ones it does, $35\times$ below the isotropic chance level against $17\times$, so the measurement is a statement about the benchmark rather than a property of test-selected networks. Every system with a cycle, taken singly, sits one to three orders of magnitude below its own chance level (Table S3). The join rule, the pooling rule and a recovery threshold of 0.50 – recovery being the share of a system’s curated edges whose two ligands both resolve into the network – were fixed before any of the eleven systems beyond the three above was measured.

This does not replace the four-system measurement of Section 4 and is not offered as doing so. **bace** is absent from the curated benchmark entirely, **p38**’s matched sub-network is acyclic and contributes nothing, and the expanded set carries 31 residual degrees of freedom against the original’s 51. The labels are different labels and the sub-networks are different sub-networks, so the pooled figures are two grounded measurements of different things and neither corrects the other. The expanded pooled figure is $11.5\times$ smaller than the four-system one, and three things change at once to produce that: eleven new systems enter, the matched sub-networks are smaller and sparser in cycles than the full ones, and the label source changes. Only the third is isolated above, on identical edges, and it pushes the other way. What carries across both is the ordering: on every network that carries a cycle, under either label source, the visible fraction sits far below its own chance level.

What the curated join cannot measure, and two ways of keying it (Table S3). Four systems with a real matched sub-network carry no independent cycle in it: **cdk2** ($E = 9$), **cmet** ($E = 2$), **p38** ($E = 16$) and **tyk2** ($E = 10$). A sub-network with no cycle has nothing for a closure diagnostic to see and nothing for the projector to keep, so its visible fraction is identically zero by construction; that is not evidence, and those four contribute to neither side of the pool. Two more join no curated edge at all. **pfkfb3** recovers 2% because the replicate benchmark carries only two **pfkfb3** edges against the curated table’s 45, and **thrombin** recovers none because the benchmark’s prepared thrombin ligands are protonated where its curated table is neutral and the exact-structure rule refuses the join. Repeating the entire measurement with formal charges stripped from both sides – a rule chosen after seeing that result, and reported as a sensitivity analysis for that reason –

gives **thrombin** 14 matched edges. Under the name key those change the pooled figures by nothing at all, because a larger acyclic sub-network is still an acyclic sub-network. Under the structure key used here they do not: merged, they carry two cycles and a visible fraction of 0.115, and the pooled figure moves to 0.00445. That is an artefact of the key. The releases name one molecule in several poses, **3b** in the JACS set against **3b pose 1** in the water set, so a 2D structure merges nodes a release deliberately keeps apart and the cycles it creates assert that two poses of two preparations have one free energy. It is the mirror of the **cdk8** collision, and the reason a name not being an identifier does not make a 2D structure one. The measurement never meets it: the frozen exact rule refuses every **thrombin** join on protonation state, so the system contributes nothing to the figures reported above. The declared recovery threshold likewise changes nothing here: the two systems it excludes already contribute no matched edge. Both readings are reported; neither replaces the other. Restricting a network to the edges a curated table happens to cover also removes cycles: the pooled dof/E here is 0.151 against the whole benchmark’s 0.325, which is why both chance levels are carried in every row rather than one.

Two keying decisions move the numbers. The first is how a network ligand name is resolved to a structure. **cdk8**’s two releases reuse seven ligand names for chemically different molecules; resolving each edge through the prepared input structures of its own release, rather than through one name-to-structure table per system in which the second release silently overwrites the first, recovers 44 of 46 curated edges rather than 57% of them and gives a 29-edge matched sub-network rather than a 15-edge one. **cdk8**’s visible fraction rises from 0.0007 to 0.0023 as a result, against chance levels of 0.172 and 0.077. The release-local rule is the less favourable of the two for the argument here and is the one used throughout. The second decision was undeclared in the protocol above, which fixes how edges are matched but not how the matched sub-network’s nodes are identified, and the same finding settles it. Keying nodes by ligand name gives **ptp1b** $f = 0.0023$; keying them by resolved structure, which merges two names in different releases that denote the same molecule and knits two components into one, gives 0.0159, and the worst margin any single-edge deletion leaves under the isotropic chance level falls from $10.3\times$ to $2.0\times$. **ptp1b** is the only system of the eight whose matched sub-network changes at all under the alternative key. The figures quoted here and in Table S3 are the structure-keyed ones, which are also the less favourable of the two: merging nodes adds cycles, so the pooled fraction rises from 0.00046 to 0.00052 and the not-flagged pool from 0.00032 to 0.00042. Both readings are generated by **make figGround** and recorded in full.

The label-noise floor under the visible fraction (Figure S7). Experimental and annotation error is a property of a ligand, so it enters a per-edge error as a difference of per-ligand terms, $\epsilon_e \rightarrow \epsilon_e + b_j - b_i$: a gradient field, which the whitened residual projector annihilates. Injecting such a field at a per-ligand σ of 0.93 kcal/mol into the four grounded sub-networks, 200 seeded draws each, moves the numerator $\|\Pi\tilde{\epsilon}\|^2$ by at most 1.96×10^{-11} of its own size over all 800 draws, against a tolerance of 10^{-10} fixed before the measurement was run, and by at most 1.3×10^{-13} measured against the injected field’s own scale; the denominator meanwhile grows by 1.6 to $2.2\times$ at the median. Label noise therefore contributes nothing to the numerator of f and all of it to the denominator, and any correction for it can only move f upward. The 0.6% the main text reports is a floor on the share of the error attributable to the calculation’s own bias, not an estimate of it.

How much room that leaves is exact arithmetic rather than a fit. If a share r of the measured per-edge error variance is label noise, the denominator shrinks by $s = 1/(1 - r)$ and f rises to sf , the numerator being fixed. Pooled, f would first reach its chance level at a shrinkage of $50.6\times$ in the whitened metric and $14.1\times$ in the isotropic one. That pooled crossing is not the figure to quote on its own, because per system the room is far from uniform: **bace** needs $4.2\times$ whitened

Table S3: **The visible fraction on curated labels, over the benchmark.** For each target whose curated per-edge experimental $\Delta\Delta G$ the replicate benchmark can be joined to: curated edges E_{cur} , the share of them whose two ligands both recover into the network (ρ), and the matched sub-network’s ligands N , edges E , components c and independent cycles dof. Then the visible fraction f of the error against those labels, its dof/ E chance level, the chance level $\text{tr}(\Pi W)/\text{tr}(W)$ an error isotropic in kcal/mol would produce under the same projector, the range of f over single-edge deletions, the worst margin chance/ f any deletion leaves under each chance level, and the median $|\epsilon|$ against experiment. Both the fraction and its chance level are recomputed on every deletion, since deleting an edge moves the projector as well. f and its deletion range are given in units of 10^{-3} . A dagger marks the systems the closure test flags. The systems whose matched sub-network carries no cycle, and so no measurement, are named below the rule.

system	E_{cur}	ρ	N	E	c	dof	$f/10^{-3}$	dof/ E	iso	f range/ 10^{-3}	worst dof/ E	margin iso	$ \epsilon $
cdk8 [†]	46	0.96	29	29	5	5	2.28	0.172	0.077	[1.59, 2.64]	54×	6.2×	1.08
eg5	43	1.00	26	32	3	9	2.07	0.281	0.095	[1.04, 4.75]	54×	12.0×	0.92
hif2a [†]	51	0.98	33	34	4	5	0.63	0.147	0.005	[0.28, 1.00]	151×	5.4×	1.05
mc11	36	1.00	23	21	3	1	0.26	0.048	0.007	[0.26, 0.40]	124×	22.9×	0.60
ptp1b	36	0.94	17	19	2	4	15.86	0.211	0.061	[2.32, 19.25]	9×	2.0×	1.15 No
shp2	37	1.00	16	12	5	1	0.08	0.083	0.009	[0.08, 0.24]	372×	42.5×	1.02
syk	66	0.97	39	35	8	4	0.38	0.114	0.007	[0.10, 0.70]	168×	12.0×	0.59
tnks2	36	0.86	25	23	4	2	0.03	0.087	0.003	[0.00, 0.05]	1558×	22.8×	0.64
pooled, all (8 systems)			205			31	0.52	0.151	0.014				
pooled, flagged (2 systems)			63			10	0.71	0.159	0.012				
pooled, not flagged (6 systems)			142			21	0.42	0.148	0.015				

measurement, matched sub-network acyclic: **cdk2** ($E = 9$, dof = 0), **cmet** ($E = 2$, dof = 0), **p38** ($E = 16$, dof = 0), **tyk2** ($E = 10$, dof = 0). No measurement, no curated edge joined: **pfkfb3** ($\rho = 0.02$), **thrombin** ($\rho = 0.00$).

and $2.0\times$ isotropic, and under the least favourable single-edge deletion **p38** needs $1.46\times$ isotropic and **bace** $1.36\times$. Against those, three declared levels of label noise say what a correction could supply. The first, L1, is the measured literature quantity: mixed public affinity data reproduce across laboratories at $0.68 \log_{10}$ units per ligand [20], which at 1.3642 kcal/mol per $\log_{10}$ unit is 0.93 kcal/mol per ligand and 1.31 kcal/mol per edge. L2 and L3 carry half and a quarter of L1’s per-edge variance and are stated reductions of it rather than independent measurements.

The verdict differs by metric, and the metric it fails in is the one the abstract and the main text quote. In the isotropic metric the bound does not survive a label-noise correction uniformly. **bace** isotropic needs $2.00\times$ and L2 supplies $2.55\times$; **bace** isotropic under the least favourable single-edge deletion needs $1.36\times$, which both L2 and L3 supply; **p38** isotropic under one deletion needs $1.46\times$ and L3, a quarter of the mixed-laboratory variance, supplies $1.80\times$. A plausible label-noise correction therefore erases the isotropic reading on those two systems. In the whitened metric the pooled bound needs a $50.6\times$ shrinkage to be lost and the most any level supplies pooled is $4.13\times$ once each system is held to its own arithmetic, so there it survives by an order of magnitude; the single exception is **bace** under the least favourable single-edge deletion, where the $2.55\times$ needed and the $2.55\times$ L2 supplies coincide to within one per cent. The bound is therefore robust to a label-noise correction in the whitened metric the fraction is defined in, and is not robust in the isotropic metric on **p38** and **bace** – the two systems whose margins were the weakest before any correction was considered.

The isotropic readings just given stand: L2 and L3 erase them, and nothing here refutes those two levels. These data do refute L1, the only level with a literature measurement behind it. A mixed-laboratory per-ligand σ of $0.68 \log_{10}$ implies a label-noise share $r \geq 1$ on **cdk8**, **p38** and **bace**: more label variance than each of those networks’ entire measured error against experiment contains, which is arithmetically impossible, and is reported as impossible and not clipped. That is direct evidence, from the data and not from an assumption imported to protect the bound, that $0.68 \log_{10}$ is an upper bound and not an estimate for this article’s grounding, which takes one ChEMBL target and one assay type per system and never mixes IC_{50} with K_i .

Two numbers in the same arithmetic land close together without being the same quantity. The $14.13\times$ that L1 supplies pooled without the per-system cap sits within one per cent of the $14.08\times$ needed to reach the pooled isotropic chance level. They are different quantities, and L1 is in any case impossible on three of the four systems taken singly.

None of this measures the label noise in these particular ChEMBL series; it bounds what a literature-scale label noise could do. A direct measurement would need replicate assay values per ligand, which the public sources do not carry.

The system column carries the benchmark’s own name for a target’s edge network, which is not a protein: the 49 names span 41 protein labels, **bace** contributing four and **hsp90** four, so a count of flagged systems is not a count of flagged targets. Nor is a name always one released data set: eight of them combine two or more, which the cross-release paragraph below takes up. **ciordia_retro** is released under the **asap_mix** group, and the release’s own statement is the only target assignment used for it.

The component column records the number of connected pieces of a system’s edge network. Six of the 48 systems are disconnected (**cdk2**, **jnk1**, **mc11**, **p38**, **ptp1b**, **thrombin**), each into two pieces, so the components sum to 54: the benchmark’s edge lists join congeneric series that share a target without joining them to each other. Nothing in the closure statistic is affected, since dof is computed as E minus the incidence rank throughout and that rank already accounts for the extra kernel direction. Elsewhere disconnection does matter. The offset that Theorem 2(iv) says the first anchor removes is one offset per component, so a disconnected system needs one anchor in each piece before any further anchor resolves a dimension. The grounded networks of Section 4

Table S4: **What each benchmark system can be audited for.** For the 48 systems carrying at least one independent cycle: N ligands, E edges, c connected components, ν independent cycles, the auditable fraction ν/E of that system’s error space, the number of bridge edges (br), which lie in no cycle and carry no evidence at any magnitude, and the median shift $\tilde{\delta}^*$ in kcal/mol at which the closure statistic carries unit noncentrality, over that system’s auditable edges. Rows are ordered by auditable fraction ascending. A dagger marks the eight systems the Benjamini–Hochberg test flags on at least one of the three replicates. Values are replicate-0 and are produced by the same run of the released analysis code that draws Figure 7. Note that $\tilde{\delta}^*$ is not a detection threshold: reaching 80% power at $\alpha = 0.05$ needs a shift of 2.8 to 4.7 times it, and it is carried on the raw reported standard errors.

system	N	E	c	ν	ν/E	br	$\tilde{\delta}^*$	system	N	E	c	ν	ν/E	br	$\tilde{\delta}^*$
brd4 [†]	8	8	1	1	0.12	0	0.49	mc11	54	76	2	24	0.32	1	0.91
taf12	8	8	1	1	0.12	0	0.48	hif2a [†]	41	59	1	19	0.32	1	0.78
scyt_dehyd	7	7	1	1	0.14	0	0.43	shp2	26	37	1	12	0.32	2	1.14
chk1	13	15	1	3	0.20	1	1.29	syk	46	67	1	22	0.33	5	1.25
t4_lysozyme	12	14	1	3	0.21	0	0.39	bace1	3	3	1	1	0.33	0	0.64
hsp90_kung	11	13	1	3	0.23	0	0.56	dlk	5	6	1	2	0.33	0	2.09
liga	11	13	1	3	0.23	1	0.95	factor_xa	3	3	1	1	0.33	0	2.46
jnk1	24	29	2	7	0.24	3	0.93	ptp1b	26	36	2	12	0.33	5	1.39
ephx2	4	4	1	1	0.25	0	3.85	renin [†]	29	42	1	14	0.33	0	0.64
hsp90_single_ring	7	8	1	2	0.25	0	0.53	thrombin	37	54	2	19	0.35	0	0.55
hsp90_woodhead	4	4	1	1	0.25	1	0.37	keranen_p2	12	17	1	6	0.35	0	1.27
irak4_s3	4	4	1	1	0.25	1	1.60	jak2_set1	10	14	1	5	0.36	1	0.74
urokinase	4	4	1	1	0.25	0	0.41	p38 [†]	40	60	2	22	0.37	5	1.02
tnks2	27	35	1	9	0.26	0	0.47	hiv1_protease	13	19	1	7	0.37	0	0.46
faah [†]	24	31	1	8	0.26	3	0.88	cdk2	19	27	2	10	0.37	0	0.92
bace_ciordia_prospective	9	11	1	3	0.27	0	0.68	eg5	28	43	1	16	0.37	3	1.51
bace [†]	36	49	1	14	0.29	2	0.57	ltk	6	8	1	3	0.38	0	1.63
bace_p3_arg368_in	21	28	1	8	0.29	8	0.71	tyk2	19	29	1	11	0.38	1	0.68
hsp90_2rings	6	7	1	2	0.29	0	0.13	itk	4	5	1	2	0.40	0	1.24
jak1	6	7	1	2	0.29	1	1.60	cmet	24	39	1	16	0.41	0	1.12
mup1	6	7	1	2	0.29	0	0.28	jak2_set2	8	12	1	5	0.42	0	1.71
hne	17	23	1	7	0.30	0	1.25	egfr	5	7	1	3	0.43	0	2.90
galectin	26	36	1	11	0.31	0	0.61	irak4_s2	5	7	1	3	0.43	0	0.98
ciordia_retro [†]	32	45	1	14	0.31	0	0.68	cdk8 [†]	35	63	1	29	0.46	3	0.98

inherit the same structure, `cdk8` and `bace` being disconnected once restricted to ligands carrying affinities. Their mean unsigned error against experiment is reported after a single global mean alignment, which is the conservative convention: aligning each component to its own experimental mean instead, which uses more experimental information than a prediction has access to, lowers the error by 0.067 and 0.014 kcal/mol respectively and leaves the two connected systems unchanged.

The cost of a network that can be audited, per system (Table S5). The design rule of Section 3 is the prospective counterpart of the table above, and Table S5 gives it per system. Two kinds of number sit in it and the table keeps them apart. The augmentation counts a_b and a_2 , 29 and 36 added ligand pairs over the benchmark, are pure topology: they assume nothing whatever about the edges they add, are certified minimum against the Eswaran–Tarjan augmentation bound, and were checked independently by brute force on 39 of the 40 (system, target) pairs that need any edge at all, every subset of one fewer edge failing on each. The further columns, the edges a greedy design adds to bring a system’s median δ^* below 1.0, 0.75 and 0.5 kcal/mol, assume each

new edge carries that system’s median reported variance, and are an achievable cost rather than a proven minimum. That assumption is what the third column of the sensitivity check tests: sweeping the assumed variance by a factor of two in each direction leaves the verdict and the count stable to within one edge on 93 of 96 system-by-scale comparisons. Two consequences belong with the rule. Removing bridges buys coverage and not resolution: on three of the 21 systems that need any augmentation the median δ^* taken over all edges is *worse* after augmentation than the median over auditable edges alone that Table S4 reports, because the bridges that median silently excludes are now inside it. The second is that $h_e \leq 1$ pointwise gives $\delta_e^* \geq se_e$, so no topology can push an edge below its own reported standard error; on a complete graph over N ligands $h_e = 1 - 2/N$, a second floor for small systems. Measuring every remaining ligand pair still leaves nine systems above 1.0 kcal/mol and nineteen above 0.5, which is the bound of Theorem 2 appearing in a design question rather than in a diagnostic one.

Between-preparation variance: what the multi-release systems can and cannot supply (Figure S8). Three passages of this article need the same missing quantity: the variance between independent *preparations* of one target, relative to the sampling variance the reported bar carries. The three are the disagreement with Wade et al. [35] over whether that bar is conservative, the observation that nothing flags once every bar is widened by half, and the caveat that the three repeats share one starting structure. The benchmark appears to offer it in one place. Eight system names union edges from more than one separately released data set, and where two releases share ligands the union carries cycles that exist in neither release alone, which close only if the two preparations agree; splitting the closure statistic into its within-release and across-release blocks would then measure the excess directly. That split, run over every multi-release system and all three repeats, shows that the public data do not supply the quantity. Two shared ligands are needed before the union carries an across-release cycle, and only two of the eight reach that: in the other six (*cdk2*, *jnk1*, *mc11*, *p38*, *ptp1b*, *thrombin*) the releases share at most one ligand, the union is disconnected or joined by a tree, and no measurement exists at any magnitude.

Of the two that remain, one is not a join between preparations at all. *cdk8*’s across-release block is large, at a reduced χ^2 of 7.77, 4.79 and 8.41 over the three repeats against a null of exactly 1, and 5.8 to 8.2 times its own within-release block. That excess is the naming collision of Section 6 and not a disagreement between two preparations: its two releases reuse seven ligand names for chemically different molecules, so every spanning cycle asserts an equality between distinct ligands. Disambiguating by release removes all six across-release cycles on every repeat and takes the system’s degrees of freedom from 29 to the within-release total of 23, which is the correction this article already applies. The across-release excess must not be read as a between-preparation estimate, and it is not used as one anywhere here.

That leaves *tyk2* as the only genuine cross-release join in the benchmark: two shared ligands, one across-release cycle. Its across-release block sits at a reduced χ^2 of 0.18, 0.70 and 0.04 over the three repeats, $p = 0.67$ at the first, with a ratio to the within-release block of 0.2, 0.6 and 0.0. The null it is read against is the right one: synthetic draws on each system’s real topology and real reported bars, with one node potential per shared ligand and independent noise of the reported size, put *tyk2*’s block at a mean reduced χ^2 of 1.069 over 2000 draws per repeat, rejecting at 5.7 to 5.8 per cent, and *cdk8*’s at 1.001. A second preparation of *tyk2* therefore adds nothing detectable beyond sampling, on one cycle, which is far too little to estimate a variance from. The public data contain no usable estimate of between-preparation variance, and the disagreement with Wade et al. [35] and the widened-bar observation are scoped by that absence rather than settled by it. Four further limits apply to any future reading of this split and none is removable from these

Table S5: **The cost, in added edges, of a network that can be audited.** For the 48 benchmark systems carrying at least one independent cycle: E edges and the number of bridge edges (br), then two certified-minimum augmentation counts. a_b is the smallest number of new ligand pairs that leaves NO bridge, so that every edge lies on a cycle and carries some evidence; it treats a system’s connected components separately, which is all auditability requires, since the closure fit already absorbs one offset per component. a_2 additionally joins those components, leaving the whole network 2-edge-connected. Both are pure topology and assume nothing about the new edges. The last three columns are the FURTHER edges a greedy design adds, on top of a_2 , to bring the median δ^* over that system’s own E edges below 1.0, 0.75 and 0.5 kcal/mol, each new edge assumed to carry that system’s median reported variance; unlike a_b and a_2 these are an achievable cost and not a proven minimum, since the design is greedy. - marks a target no topology can reach, because even measuring every remaining ligand pair leaves the median above it, and $> 2E$ a target not reached within the explored budget. Across the benchmark 29 added edges remove all 48 bridges and 36 leave every system 2-edge-connected, 2.5% and 3.1% of the 1143 edges already run. δ^* is the shift at unit noncentrality and not a detection threshold; 80% power at $\alpha = 0.05$ needs 2.8 to 4.7 times it, so these are targets on a resolution scale. Values are replicate-0 and come from the same run of the released analysis code, `make figDesign`. A dagger marks the six systems the cycle-closure test flags.

system	E	br	a_b	a_2	+1.0	+0.75	+0.5	system	E	br	a_b	a_2	+1.0	+0.75	+0.5
p38 [†]	60	5	3	4	1	13	-	ciordia_retro	45	0	0	0	0	0	4
bace_p3_arg368_in	28	8	3	3	1	1	12	galectin	36	0	0	0	0	0	2
ptp1b	36	5	2	3	62	-	-	hiv1_protease	19	0	0	0	0	0	0
cdk2	27	0	0	2	0	2	27	hsp90_2rings	7	0	0	0	0	0	0
cdk8 [†]	63	3	2	2	0	45	-	hsp90_kung	13	0	0	0	0	0	1
faah [†]	31	3	2	2	0	1	8	hsp90_single_ring	8	0	0	0	0	0	1
jnk1	29	3	2	2	0	2	9	irak4_s2	7	0	0	0	0	-	-
mcl1	76	1	1	2	0	1	3	mup1	7	0	0	0	0	0	0
thrombin	54	0	0	2	0	0	1	renin	42	0	0	0	0	0	2
syk	67	5	2	2	2	5	58	scyt_dehyd	7	0	0	0	0	0	0
eg5	43	3	2	2	-	-	-	t4_lysozyme	14	0	0	0	0	0	0
bace [†]	49	2	1	1	0	0	1	taf12	8	0	0	0	0	0	0
hif2a [†]	59	1	1	1	0	1	12	tnks2	35	0	0	0	0	0	0
hsp90_woodhead	4	1	1	1	0	0	0	urokinase	4	0	0	0	0	0	0
jak2_set1	14	1	1	1	0	0	2	cmet	39	0	0	0	1	16	-
liga	13	1	1	1	0	0	4	keranen_p2	17	0	0	0	1	3	-
tyk2	29	1	1	1	0	0	2	btb	8	0	0	0	2	-	-
chk1	15	1	1	1	2	16	24	hne	23	0	0	0	3	>2E	-
shp2	37	2	1	1	2	7	-	dlk	6	0	0	0	-	-	-
irak4_s3	4	1	1	1	-	-	-	egfr	7	0	0	0	-	-	-
jak1	7	1	1	1	-	-	-	ephx2	4	0	0	0	-	-	-
bace1	3	0	0	0	0	0	-	factor_xa	3	0	0	0	-	-	-
bace_ciordia_prospective	11	0	0	0	0	0	3	itk	5	0	0	0	-	-	-
brd4 [†]	8	0	0	0	0	0	0	jak2_set2	12	0	0	0	-	-	-

data: an across-release block confounds preparation with force field, protonation, starting structure and mapping, so it bounds the preparation term from above rather than isolating it; two separately released data sets are not two replicates of one protocol; the counts are small enough that a single edge moves them, which the leave-one-edge-out ranges emitted with the figure quantify; and the split is not independent of what this article already reports, since the one system carrying most of the arithmetic is one the closure test already flags for the reason given above.

Close-the-loop accuracy comparison (Figure S10). On the four flagged systems groundable against public ChEMBL affinity, guided removal of the QC-flagged high- $|z|$ edges ties a size-matched random removal on MUE-vs-experiment (combined Stouffer $p = 0.48$). Removing the highest- $|z|$ edges lowers the closure statistic (Figure 6A), but that is arithmetic and not evidence, because the statistic is the sum of exactly the per-edge quantities being removed in descending order (Section 6). The comparison adds that the same removal yields no improvement against experiment. The reach demonstrated for the closure test therefore stops at internal edge-level consistency and does not extend to force-field-limited absolute accuracy, the same scope boundary as the clean **eg5** system. Throughout, ΔMUE is oriented as $\text{MUE}(\text{full}) - \text{MUE}(\text{after})$, the same orientation as Figure 7C.

Fixed-cutoff head-to-head and χ^2 reconciliation (Figure S11). Two pre-registered checks were run on the same 1143 edges of the 48 cyclic systems. The first is the head-to-head: the calibrated test, scored by area under the receiver-operating-characteristic curve (AUC) against the reproducible-systematic magnitude defined in Computational details, scores 0.865. Against a fixed 1.0 kcal/mol hysteresis cutoff’s 0.575 the paired bootstrap difference is +0.290 [+0.094, +0.490]; against a fixed-se χ^2 test’s 0.725 it is +0.140 [−0.008, +0.316], straddling zero. The pre-registered criterion required the calibrated test to exceed both, so the verdict is a tie. The ordering within that tie is not a discrimination gain over the fixed cutoff in standard use: the scored label standardizes each residual by the same per-edge reported variance the calibrated rule weights by, so the AUC ordering is monotone in a rule’s proximity to that variance and the comparison ranks nothing. The second check tested whether correlation among residuals of edges sharing a ligand endpoint explains the apparent gap between the inverted-median factor, $1.71\times$ obtained as $1/\sqrt{0.34}$ from the median reduced χ^2 , and the replicate ratio of $1.41\times$ measured against seed and velocity repeats. It does not: the excess correlation is −0.0079 [−0.0314, +0.0164], and the Benjamini–Hochberg flag set is unchanged.

The two inverted-median factors quoted in this article are the same functional on two populations and do not conflict. The pre-registered constant $1.71\times$ is $1/\sqrt{0.34}$ from the median reduced χ^2 over all 48 cyclic systems, the population of Figure 5; the $1.67\times$ annotated inside panel B is $1/\sqrt{0.361}$ over the 46 systems this second check admits, since estimating the shared-node null needs at least one disjoint edge pair per system and **bace1** and **factor_xa** are three-edge triangles that have none. Dropping those two raises the median.

Neither the pre-registered constant nor the factor annotated in panel B is a conservatism estimate, because the premise of the second check is unsound: by Theorem 1 the closure statistic identifies the closure-weighted mean of c_e^{-2} and not a median, so $1/\sqrt{\text{median}}$ does not estimate the same functional as the replicate ratio. There is accordingly no physical gap for shared-node correlation to explain. The pre-registered constants are retained as the pre-registered record, and the verdict above is reported as run and not revised. On replicate 0 the calibrated test flags 6 of 48 systems, the fixed-se χ^2 test 11, and the fixed cutoff 26, and the three flag sets are strictly nested (calibrated $\subset$ fixed-se $\subset$ fixed cutoff), so on that replicate the calibrated test catches no system the fixed cutoff misses and instead removes 20 of its flags. The strict chain is a property

of that replicate and not of the benchmark. On replicate 1 the counts are 6, 13 and 24 and two links fail: `ciordia_retro` is flagged by the calibrated test and not by the fixed-se test, and `mup1` by the fixed-se test and not by the cutoff. On replicate 2 the counts are 3, 11 and 24 and the chain holds. The calibrated set lies inside the cutoff’s on all three, which is the only containment any claim here rests on. Under the release’s own blocks rather than the by-target grouping, the same replicate-0 test over 56 testable blocks gives 6, 10 and 29, the chain nests, and the calibrated test removes 23 of the cutoff’s flags. Whether that removal is an improvement is what the head-to-head above cannot determine, for the reason given there. The 1.0 kcal/mol cutoff was pre-registered at the field-standard value before this comparison was run, and no sweep over alternative thresholds was performed, so the comparison above is specific to that one fixed rule.

The flag sets under arbitrary dependence (Figure S12). The correction reported throughout is Benjamini–Hochberg, whose guarantee needs a dependence assumption between systems. Benjamini–Yekutieli [6] needs none, at the cost of running the same step-up rule at α/H_m with $H_{48} = 4.4588$, that is at 0.0112 rather than 0.05. It returns `{bace, brd4, cdk8, faah, hif2a, p38}` on replicate 0, identical to the published set; `{bace, cdk8, renin}` on replicate 2, again identical; and on replicate 1 `{bace, ciordia_retro, hif2a, p38, renin}`, which drops `brd4` from the Benjamini–Hochberg set of six. The published set therefore does not depend on the dependence assumption, and the variation of the flag set across replicates is unchanged by removing it. `make figStab` emits the three sets.

Per-system test statistics behind the flag sets (Tables S6 and S7). The flag sets summarized in Figure S12 and in Section 6.1 are thresholded Benjamini–Hochberg adjusted p -values, so the underlying continuous quantities are reported in full rather than only their dichotomization. For each system, a generalized-least-squares cycle-closure fit under the calibrated per-edge null gives a χ^2 on ν degrees of freedom, ν being the number of independent cycles in that system’s edge network; the reduced value $\chi^2_\nu = \chi^2/\nu$ and the raw upper-tail probability follow, and those raw p -values are corrected for testing 48 systems by the Benjamini–Hochberg procedure at a false-discovery rate of 0.05, giving the adjusted q -values tabulated below. The $q \leq 0.05$ rule reproduces the released implementation exactly, and that correspondence is asserted on every set the analysis emits. Each of the benchmark’s three protocol replicates is analysed independently: the fit is rerun from scratch on that replicate’s edges and the correction is applied within that replicate alone, so the three q columns are not a joint error rate across replicates. All values in both parts of the table are produced by the same run of the released analysis code that draws Figure S12.

Out-of-sample behaviour of the calibrated null (Figure S13). Each candidate held-out statistic is calibrated against a no-error null before use, and the realized level of every candidate is reported. Two of the candidates fire far above nominal and are not used; the remaining two hold their nominal level and are the ones carried into the selection comparisons.

Two-sided self-calibration of the per-edge error bar (Figure S14). Seven arms are compared: the nominal test, which uses the reported standard errors unchanged; the two-sided correction at the pre-specified per-edge granularity, which corrects each edge by its own ratio; a coarser variant correcting each system by its aggregate ratio; a variant clipping the ratio to $[1/3, 3]$; a variant dividing the replicate standard deviation by the constant $c_4(3) = 0.886$, the expectation of the sample standard deviation at three draws, which widens the bars; a variant measuring the ratio on the two replicates the test does not read; and, at the same per-edge granularity, the one-sided correction

Table S6: **Per-system cycle-closure statistics and Benjamini–Hochberg adjusted q -values, per replicate (part 1 of 2).** Reduced χ^2 and adjusted q -value from the calibrated cycle-closure test at a false-discovery rate of 0.05, computed independently on each of the three protocol replicates, for the 48 systems with at least one independent cycle; ν is the number of independent cycles. Bold marks $q \leq 0.05$; the final column lists the replicates on which the system is flagged, and a dash where it never is. Edge and cycle counts are the replicate-0 values. Rows are ordered by replicate-0 q ascending. The remaining 24 systems continue in Table S7.

system	edges	ν	replicate 0		replicate 1		replicate 2		flagged
			χ^2_ν	q	χ^2_ν	q	χ^2_ν	q	
bace	49	14	5.57	3.2×10^{-9}	7.33	7.0×10^{-14}	7.10	2.9×10^{-13}	0, 1, 2
cdk8	63	29	2.68	6.1×10^{-5}	1.45	2.8×10^{-1}	2.62	1.1×10^{-4}	0, 2
brd4	8	1	15.40	1.4×10^{-3}	8.05	3.6×10^{-2}	4.94	1.7×10^{-1}	0, 1
faah	31	8	3.71	2.0×10^{-3}	0.45	1.0×10^0	1.47	7.1×10^{-1}	0
hif2a	59	19	2.53	2.0×10^{-3}	2.96	2.4×10^{-4}	2.05	5.4×10^{-2}	0, 1
p38	60	22	2.40	2.0×10^{-3}	2.48	1.7×10^{-3}	1.81	1.1×10^{-1}	0, 1
mcl1	76	24	1.82	5.5×10^{-2}	1.79	6.8×10^{-2}	1.04	1.0×10^0	–
bace_p3_arg368_in	28	8	1.88	3.5×10^{-1}	1.88	2.8×10^{-1}	2.32	1.4×10^{-1}	–
taf12	8	1	3.21	3.9×10^{-1}	0.15	1.0×10^0	0.33	1.0×10^0	–
liga	13	3	2.10	4.7×10^{-1}	0.76	1.0×10^0	1.80	6.9×10^{-1}	–
thrombin	54	19	1.30	7.4×10^{-1}	0.86	1.0×10^0	0.61	1.0×10^0	–
renin	42	14	1.29	8.1×10^{-1}	2.81	3.1×10^{-3}	3.32	3.8×10^{-4}	1, 2
ciordia_retro	45	14	0.89	1.0×10^0	3.86	3.1×10^{-5}	0.40	1.0×10^0	1
tyk2	29	11	0.79	1.0×10^0	1.06	1.0×10^0	0.78	1.0×10^0	–
ephx2	4	1	0.77	1.0×10^0	1.67	7.7×10^{-1}	1.33	9.3×10^{-1}	–
shp2	37	12	0.74	1.0×10^0	1.21	8.7×10^{-1}	0.64	1.0×10^0	–
itk	5	2	0.68	1.0×10^0	0.08	1.0×10^0	0.10	1.0×10^0	–
tnks2	35	9	0.63	1.0×10^0	0.43	1.0×10^0	2.07	1.7×10^{-1}	–
mup1	7	2	0.55	1.0×10^0	3.05	2.8×10^{-1}	0.29	1.0×10^0	–
scyt_dehyd	7	1	0.54	1.0×10^0	1.34	8.5×10^{-1}	1.23	9.3×10^{-1}	–
cdk2	27	10	0.48	1.0×10^0	1.33	7.7×10^{-1}	1.22	9.3×10^{-1}	–
eg5	43	16	0.48	1.0×10^0	0.40	1.0×10^0	0.30	1.0×10^0	–
galectin	36	11	0.37	1.0×10^0	0.22	1.0×10^0	0.88	1.0×10^0	–
jak2_set1	14	5	0.35	1.0×10^0	0.83	1.0×10^0	0.15	1.0×10^0	–

Table S7: **Per-system cycle-closure statistics and Benjamini–Hochberg adjusted q -values, per replicate (part 2 of 2).** Table S6 continued: the remaining 24 of the 48 systems, in the same order and with the same conventions. No system in this part is flagged on any replicate.

system	edges	ν	replicate 0		replicate 1		replicate 2		flagged
			χ^2_ν	q	χ^2_ν	q	χ^2_ν	q	
hsp90_2rings	7	2	0.34	1.0×10^0	0.29	1.0×10^0	0.12	1.0×10^0	–
factor_xa	3	1	0.31	1.0×10^0	0.01	1.0×10^0	0.02	1.0×10^0	–
keranen_p2	17	6	0.29	1.0×10^0	0.19	1.0×10^0	0.04	1.0×10^0	–
chk1	15	3	0.28	1.0×10^0	0.27	1.0×10^0	0.69	1.0×10^0	–
hiv1_protease	19	7	0.28	1.0×10^0	0.43	1.0×10^0	0.94	1.0×10^0	–
irak4_s2	7	3	0.25	1.0×10^0	0.15	1.0×10^0	0.04	1.0×10^0	–
t4_lysozyme	14	3	0.25	1.0×10^0	0.57	1.0×10^0	2.14	5.0×10^{-1}	–
ptp1b	36	12	0.21	1.0×10^0	0.20	1.0×10^0	0.58	1.0×10^0	–
btk	8	3	0.19	1.0×10^0	2.15	4.0×10^{-1}	1.19	9.3×10^{-1}	–
hsp90_kung	13	3	0.19	1.0×10^0	0.18	1.0×10^0	0.83	1.0×10^0	–
cmet	39	16	0.17	1.0×10^0	0.36	1.0×10^0	0.42	1.0×10^0	–
hsp90_single_ring	8	2	0.16	1.0×10^0	0.02	1.0×10^0	0.24	1.0×10^0	–
hne	23	7	0.13	1.0×10^0	0.28	1.0×10^0	0.07	1.0×10^0	–
urokinase	4	1	0.13	1.0×10^0	0.01	1.0×10^0	0.25	1.0×10^0	–
jnk1	29	7	0.12	1.0×10^0	0.12	1.0×10^0	1.19	9.3×10^{-1}	–
syk	67	22	0.11	1.0×10^0	0.20	1.0×10^0	0.25	1.0×10^0	–
egfr	7	3	0.07	1.0×10^0	0.01	1.0×10^0	0.02	1.0×10^0	–
irak4_s3	4	1	0.07	1.0×10^0	0.07	1.0×10^0	0.05	1.0×10^0	–
jak2_set2	12	5	0.06	1.0×10^0	1.02	1.0×10^0	0.49	1.0×10^0	–
bace1	3	1	0.05	1.0×10^0	1.09	8.9×10^{-1}	0.43	1.0×10^0	–
dlk	6	2	0.02	1.0×10^0	0.06	1.0×10^0	0.05	1.0×10^0	–
jak1	7	2	0.02	1.0×10^0	0.32	1.0×10^0	0.10	1.0×10^0	–
bace_ciordia_prospective	11	3	0.01	1.0×10^0	0.07	1.0×10^0	0.04	1.0×10^0	–
hsp90_woodhead	4	1	0.01	1.0×10^0	0.07	1.0×10^0	0.15	1.0×10^0	–

that inflates only where the measurement says the bar is too tight, which is the direction of Table S9 applied per edge rather than per system. The constant $c_4(3)$ is unrelated to the per-edge calibration ratio c_e . Table S8 tabulates the four granularity-by-direction combinations.

Simulation protocols. The two protocols the main text names are specified here.

The pure-null simulation behind the realized levels. Every realized level quoted for a self-calibrated arm, in Section 6, in Section 8 and in the last column of Supplementary Table S8, is measured on a synthetic benchmark that carries no systematic error, so that every flag it produces is a false positive by construction. The topologies are the real ones: the same 48 systems the closure test admits, each with its own ligand-pair edge list and its own admission by the same rule (≥ 3 edges, ≥ 1 independent cycle), which depends on topology alone. The variances are the real ones: each edge keeps the per-edge reported standard error it carries, and that value is then declared to be the edge’s true sampling standard deviation, which makes the reported bars perfectly calibrated by construction. The draw law is the null: every node potential is exact, so each edge’s true $\Delta\Delta G$ is exactly the difference of its endpoints’ potentials, and an observation is that value plus an independent Gaussian of the edge’s own standard deviation. There is no systematic component anywhere and no correlation between edges. The replicate structure mirrors the benchmark’s: three independent draws per edge, of which the first plays the role of replicate 0 and is the one the closure statistic is computed on, while all three are read by the arm’s own calibration formula exactly as that arm reads the real replicates (the held-out arm reads only the second and third). Each arm’s correction, tail probability and Benjamini–Hochberg step-up at $\alpha = 0.05$ are then the identical code paths used on the real data. The simulation is run for 400 draws, so a realized probability is measured to a Monte-Carlo standard error of at most $\sqrt{0.25/400} = 0.025$, and the generator is seeded once for the whole sweep, which makes the reported levels reproducible.

The reproducible-systematic magnitude used as ground truth. Detection comparisons need a label that does not come from the detector being scored, and the only one these data supply is reproduction across the three protocol repeats. It is defined as follows. Each replicate’s network is fitted separately by the same generalized least squares, giving for replicate k and edge e a standardized residual $z_e^{(k)} = r_e^{(k)} / \sqrt{V_e^{\text{rep}}}$. The three signed residuals of an edge are averaged, $\bar{z}_e = \frac{1}{3} \sum_k z_e^{(k)}$, so that a residual reproducing in sign and magnitude survives the average while sampling noise cancels toward zero, and the edge’s score is $|\bar{z}_e|$. A system’s score is the median of $|\bar{z}_e|$ over its edges. This is the quantity the detectors of the head-to-head comparison are ranked against, by Mann–Whitney AUC of the binary flag against the per-system score. It uses only replicate residuals and no detector output, but it is not independent of the rules being scored, and the dependence is the reason no ordering is read out of that comparison (Section 6): $z_e^{(k)}$ divides by V_e^{rep} , the same per-edge reported variance the calibrated rule uses as its weight, so the label rewards a rule for standing close to that variance. “Edges whose sign and magnitude reproduce across independent replicates”, wherever that phrase appears above, means edges with a large $|\bar{z}_e|$ in this sense.

Model architectures. The two learned models the main text compares against or uses as an amortized predictor are specified here in full.

Amortized predictor. The predictor used in the acquisition, stopping and reward experiments, referred to elsewhere in this article as the trunk, is small, and it is not an equivariant network.

A ligand is featurized as a 1024-bit ECFP4 Morgan fingerprint (radius 2, chirality flag off) concatenated with six achiral RDKit descriptors: molecular weight, Crippen log P , topological polar surface area, and counts of hydrogen-bond donors, hydrogen-bond acceptors and rotatable bonds. An edge is represented by the difference of its two endpoint feature vectors. A single parity-odd coordinate is appended, the difference of a summed signed volume over the assigned stereocentres of an ETKDGV3 embedding, giving 1031 features per edge; every other coordinate is invariant under reflection, so a readout without it would be constant on an enantiomer pair. The regressor is a bootstrap deep ensemble of eight multilayer perceptrons, each with two hidden layers of 64 units and SiLU activations and a scalar output, trained for 300 epochs on a squared-error objective with Adam at learning rate 5×10^{-3} and weight decay 10^{-4} , on inputs standardized by the training mean and standard deviation. Each member is seeded independently and fitted to its own bootstrap re-sample; the epistemic term is the spread across members, and the total σ combines it in quadrature with the aleatoric floor before a conformal rescaling.

Learned-variance foil. The learned head the calibrated bar is compared against is a separate and much smaller model from the amortized predictor above, and it is trained on the controlled panel rather than on the benchmark. Each training example is one simulated edge, from which six features are formed: the means and standard deviations of the forward and reverse work samples, the absolute difference of the two means, and the normalized overlap scalar. Including the overlap scalar gives the head the same overlap information the closed form uses, so the comparison is fair. The network is a multilayer perceptron with two hidden layers of 32 units and SiLU activations and two outputs, a mean and a log-variance, trained for 1500 steps of Adam at learning rate 5×10^{-3} on a Gaussian negative log-likelihood, with inputs standardized by the training mean and standard deviation. The realistic-budget arm trains on 200 edges of 20 work samples each; the large-budget arm raises that to 4000 edges, twenty times larger, and the corrected arms replace the objective by the β -NLL of Seitzer et al. [32] at $\beta = 0.5$ or average five independently seeded heads.

Granularity and direction of the self-calibrated null (Table S8). The self-calibrated null has two independent design choices, and the flag count depends on both. The *granularity* is whether an edge is corrected by its own replicate-measured ratio or by the ratio pooled over its whole system; per edge was pre-specified. The *direction* is whether the correction is applied only where the measurement says the bar is too tight, which can only remove flags, or in whichever direction the measurement points, which can also add them; the one-sided direction is the one the circularity concern of Section 6 calls for, and is the direction of Table S9. Table S8 reports all four combinations on the identical population, together with each arm’s realized probability of at least one false flag under a pure-null simulation in which every flag is a false positive by construction. The count of surviving flags ranges from two to sixteen across the four, so no single self-calibrated flag list is a stable object, and none of the four arms is reported as a flag set. Only the pre-specified per-edge one-sided arm holds its level: the two-sided arms fire far above nominal because rescaling an edge by its own replicate spread multiplies the statistic by σ^2/s^2 , whose expectation $\nu_{\text{rep}}/(\nu_{\text{rep}} - 2)$ is undefined at the $\nu_{\text{rep}} = 2$ replicate degrees of freedom three replicates provide. Taken together, the two-sided arms bound how far a flag list can move under calibration error of the size actually present, and the one-sided per-edge arm gives the conservative count: two of the six survive being retested against their own measured width. The realized levels are measured over 400 draws of the pure-null simulation specified above, so a realized 0.003 is a single flag and is not resolved from zero.

Table S8: **Self-calibrated cycle-closure flags by correction granularity and direction.** All four arms use the identical population (replicate 0, the 48 systems with at least one independent cycle) and the identical Benjamini–Hochberg level $\alpha = 0.05$; they differ only in whether the replicate-measured calibration ratio is applied per edge or pooled per system, and in whether it is applied only where it loosens a bar (one-sided) or in whichever direction the measurement points (two-sided). “published six” counts how many of the flags of Figure 5 the arm retains; “entering” lists the systems it adds. The last column is the realized probability of at least one false flag over 400 draws of a perfectly-calibrated synthetic null, against the nominal 0.05, with its exact binomial interval.

granularity	direction	flagged	published six	entering	null $P(\geq 1)$
reported se, uncorrected		6	all six	none	0.040 [0.023, 0.064]
per edge	one-sided	2	bace, cdk8	none	0.003 [0.000, 0.014]
per edge	two-sided	7	4 of 6	irak4_s2, thrombin, tyk2	0.588 [0.538, 0.636]
per system	one-sided	6	all six	none	not measured
per system	two-sided	16	all six	ciordia_retro, cmet, galectin, irak4_s2, keranen_p2, mcl1, renin, thrombin, tnks2, tyk2	0.240 [0.199, 0.285]

The per-edge two-sided arm retains **bace, brd4, cdk8** and **p38** and drops **faah** and **hif2a**. The per-edge one-sided arm loses more, retaining **bace** and **cdk8** alone; the two per-system arms retain all six. The per-system one-sided arm is the pre-registered circularity check of Table S9, whose harness reports the flag set rather than a synthetic-null level, so no realized probability is available for it; a one-sided correction can only widen bars, and the flag sets of both one-sided arms here are contained in the uncorrected one.

QC sweep under the real learned- σ profile (Figure S15). Figure 5B’s uniform $\times 0.15$ shrink is a stress test; here the learned head’s measured, overlap-dependent miscalibration profile (Figure S2: ratio $0.09\text{--}0.20\times$ across overlap $0.26\text{--}0.78$) is propagated through the identical GLS + Benjamini–Hochberg pipeline, per edge, by percentile-rank transfer onto each of 1143 (of the 1145; the sole cycle-free system contributes the other two) real edges’ own OpenFE **smallest_overlap** (an approximation: the two overlap statistics are on different scales, so only the profile’s ordering and spread are transferred, not its raw values). The rank transfer approximates how the head’s measured overconfidence would vary across these edges; it is not the head’s own output on this benchmark. Four arms are run on the same 48-system, 1143-edge network. The calibrated sandwich ($\times 1$) flags 6/48 (12%); the real profile (primary) flags 42/48 (88%), $7.00\times$ the calibrated count, past the pre-registered $2\times$ threshold; a shuffled-profile control (the identical per-edge ratios with their association to overlap destroyed, seed 20260808) also flags 42/48; and the uniform $\times 0.15$ stress test again flags 42/48. The three shrinking arms are not identical sets of systems (real vs. shuffled and real vs. uniform each swap exactly one system, **bace1** for **egfr**), so the per-edge ratio assignment is heterogeneous. The shuffled-profile control, the identical multiset of per-edge ratios with the association to overlap destroyed, also flags 42 of 48: the profile’s entire measured range ($0.09\text{--}0.20\times$) is severely overconfident, every ratio is at most 0.20, so even the mildest value inflates an edge’s χ^2 contribution by at least $\sim 25\times$, and which systems cross the Benjamini–Hochberg threshold is set by the magnitude of the shrink rather than by its structure. The whole assignment is deterministic: each edge’s transferred ratio is a fixed function of its own overlap percentile, the shuffled control draws its permutation from the recorded seed, and the fit at each arm is a closed-form recomputation, so re-running reproduces the four arm counts exactly.

Dose-response of the QC to σ miscalibration (Figure S16). Figure 5B and Supplementary Figure S15 each characterize the closure test at a small number of scale points ($\times 1$ calibrated, $\times 0.15$ stress test / real profile); here a pre-registered, frozen grid of 16 uniform se scale factors ($2.0\text{--}0.10\times$) is swept through the identical GLS + Benjamini–Hochberg pipeline on the same 48-system OpenFE network. Two readouts were frozen in advance: s_{onset} , the largest scale that flags strictly more systems than calibrated, and s_{50} , the largest scale at which at least half the systems are flagged. $s_{\text{onset}} = 0.9\times$ ($7/48$ vs. the calibrated $6/48$) is exactly one system crossing the Benjamini–Hochberg threshold. That readout is grid-determined rather than a measurement of this benchmark: because the flagged count is monotone in the scale, a strictly-exceeds-calibrated onset returns whichever grid point below 1.0 sits just past the first system’s threshold crossing. Independent probing places that system’s crossing as high as $s = 0.995\times$, so a finer grid would return $s_{\text{onset}} = 0.995\times$, and in the continuum limit $s_{\text{onset}} \rightarrow 1.0\times$ for any benchmark with a system near the cutoff. It is reported as the pre-registered readout. The informative readout is magnitude-based, such as the largest scale that doubles the calibrated count, which here is $0.7\times$ at $12/48$. The substantive change is mid-curve: from $0.7\times$ to $0.3\times$ the flagged fraction climbs from $12/48$ (25%) to $30/48$ (62%), reaching half the benchmark at $s_{50} = 0.3\times$ (the true 50% crossing is bracketed by the adjacent $0.35\times$ and $0.30\times$ grid points, which flag $23/48$ and $30/48$ respectively). The learned head’s measured, rank-transferred miscalibration band ($0.09\text{--}0.20\times$; Figure S2, Supplementary Figure S15) lies entirely below s_{50} , so the head is not near the transition but well inside the saturated regime. The over-wide half of the grid is not flat either: relative to the calibrated $6/48$, a bar only 30% too wide ($1.3\times$) already loses five of the six flagged systems ($1/48$), and any wider bar ($1.5\times$, $2.0\times$) flags none, so the requirement is calibration, not conservatism. Two consistency checks confirm that the sweep reproduces the published test path: the $\times 1.0$ row reads $6/48$, matching Figure 5A’s calibrated count, and the $\times 0.15$ row reads $42/48$, matching Figure 5B and Supplementary Figure S15’s uniform/real-profile arms. The response to the shrink magnitude is graded rather than threshold-like. The closure test is therefore selective over roughly a factor of three of σ under-estimation below the calibrated scale, and over-wide σ costs power in the other direction, so the calibrated scale is a two-sided optimum rather than a plateau. The sweep is deterministic: the sixteen scale factors are a frozen grid and the test at each of them is a closed-form recomputation on fixed inputs, so re-running reproduces the entire curve exactly.

Per-system flag robustness to each system’s own calibration error (Table S9). The Figure 5 flag set raises a mild circularity concern: the per-system calibration of the sandwich null, measured independently against across-replicate spread (Figure 3), is heterogeneous, and four of the six flagged systems are themselves locally overconfident (**brd4** $0.76\times$, **faah** $0.90\times$, **bace** $0.96\times$, **hif2a** $0.96\times$; only **cdk8** $1.41\times$ and **p38** $1.43\times$ are conservative), so a flag could partly reflect that system’s own too-tight bars rather than genuine physical inconsistency. That concern is tested directly. For replicate 0, over the 48 systems with ≥ 1 independent cycle (the identical population as Figure 5), each system’s per-edge se is inflated by $1/\text{ratio}$ where its own ratio is below 1 (systems with ratio ≥ 1 are already conservative and are left unchanged), and the Benjamini–Hochberg test is recomputed over the whole set under this self-calibrated null. The criterion was fixed in advance: the result stands if every nominally-flagged system remains flagged, and is partial or negative if only some or none do. The population, the inflation rule and $\alpha = 0.05$ were fixed before this run. All six flags remain under their own self-calibrated null (Table S9), each self-calibrated reduced χ^2 ($2.36\text{--}8.97$) remaining far above the level a sampling-consistent network of this size and degree-of-freedom profile shows (Section 6, main text). This check pools the ratio over a whole system; the same one-sided inflation applied at the per-edge granularity this work pre-specified retains two of the

Table S9: **Per-system reduced χ^2 before and after inflating each system’s own per-edge standard error by its own measured calibration error.** “ratio” is the per-system root-mean-square ratio of reported standard error to across-replicate standard deviation of binding $\Delta\Delta G$, at $n = 3$. χ^2_{nom} is the nominal reduced χ^2 ; χ^2_{self} inflates that system’s own standard error by $1/\text{ratio}$ where $\text{ratio} < 1$ and leaves it unchanged otherwise. Rows are the six flagged systems.

System	ratio	χ^2_{nom}	χ^2_{self}	Still flagged
brd4	0.76	15.40	8.97	yes
bace	0.96	5.57	5.13	yes
faah	0.90	3.71	2.99	yes
cdk8	1.41	2.68	2.68	yes (unchanged; already conservative)
hif2a	0.96	2.53	2.36	yes
p38	1.43	2.40	2.40	yes (unchanged; already conservative)

six, and Table S8 reports both granularities in both directions. The distribution-free cross-replicate reproduction check of Figure 6B is a function of raw residual correlation only, never of se, so it does not depend on this null and is not affected by this concern in either direction. The per-system ratio in Table S9 is the root-mean-square reported standard error over the root-mean-square across-independent-replicate SD, pooled per system, not a naive per-system average of per-edge ratios or a naive ratio of arithmetic means, which for three of the six systems below would sit on the opposite side of $\text{ratio} = 1$. The ratios are raw $n=3$ SD ratios. Their residual small-sample bias is upward and target-specific, of order the reciprocal of the edge count, so debiasing would move all four of the locally overconfident flagged systems (**brd4** $0.76\times$, **faah** $0.90\times$, **bace** and **hif2a** $0.96\times$) further below one rather than toward it; the raw ratios used here are generous to those systems rather than the reverse. The computation is deterministic, and the full 48-system table is released with the analysis code.

The downstream audit: what is under test (Table S11). The test of Section 6 runs on a network’s own residuals. Whether the same free calibration also improves downstream machine learning is a separate question, and the six tasks collected in Table S11 are the audit of it. Three performance tasks, interval sharpness, commit-to-synthesis selection and reward re-ranking over a measured congeneric set, run with the commit-trust decision task on the public OpenFF protein–ligand benchmark [15] (FEP+ and Merck data [31, 36], experimental $\Delta\Delta G$ labels, both-endpoint scaffold-disjoint out-of-distribution splits); edge ranking in active learning and the stopping decision run in controlled simulations. The amortized σ under test is the predictive uncertainty of the amortized $\Delta\Delta G$ predictor, the trunk specified above, on unseen $\Delta\Delta G$, $\sigma_{\text{total}} = \gamma\sqrt{\sigma_{\text{ens}}^2 + \sigma_{\text{ale}}^2}$: deep-ensemble epistemic spread and benchmark aleatoric floor, conformally scaled at miscoverage $\alpha = 0.1$. It is *not* the per-edge BAR sampling standard error of Figure 2 or its learned-variance baseline of Figure S2; a bare “ σ ” anywhere in this audit means σ_{total} .

Sharpness, ranking and selection (Figures S18 and S19A). On real held-out residuals of that trunk at matched ~ 0.90 marginal coverage the decomposed interval is slightly wider than flat split-conformal (width ratio $0.92\times$), because epistemic spread is small against the large irreducible out-of-distribution error; its conditional calibration is comparable to a fair Mondrian-by-target conformal and the advantage is not significant (ours–split -0.044 $[-0.121, 0.005]$). The synthetic illustration above, whose generative structure the decomposition knows, is up to $2.2\times$ sharper than split-conformal; that does not transfer. For edge ranking, sandwich against naive weights differ

Table S10: **Simulated stopping under a swept assumed error.** Assumed posterior standard error as a multiple of the true one, against oracle calls at the stopping rule, top- k correctness at that stop, and the mean absolute gap between claimed confidence and actual correctness over the budget; 60 seeds. Starred rows are the two arms of Figure S20.

assumed se / true se	calls at stop	correct at stop	mean claimed – actual
1.06 $\times$ (conservative)	30.8	0.83	0.027
1.00 $\times$ (calibrated)*	29.6	0.82	0.022
0.94 $\times$	28.4	0.82	0.020
0.80 $\times$	25.4	0.82	0.029
0.60 $\times$	21.1	0.77	0.059
0.40 $\times$	17.9	0.68	0.093
0.20 $\times$	14.9	0.60	0.127
0.15 $\times$ (stand-in)*	14.4	0.60	0.137

by 0.98 $\times$ [0.93, 1.03] on calls-to-top- k (above): in a connected graph the GLS contrast mean is unbiased for any positive weights, so the weight value is second-order for ranking. So too for commit selection at matched volume, where the σ -aware ranking does not beat a raw- μ ranking (-0.006 [$-0.016, 0.007$]). A calibrated bar informs whether to act and not which action ranks first.

Commit trust against recalibrated baselines (Figure S19B). Fixing the ensemble mean and varying only the uncertainty in a commit rule (commit iff the lower-confidence bound $\mu - \kappa\sigma \geq \tau$), an overconfident epistemic-only σ over-claims badly: its realized-versus-claimed correctness shortfall is $+0.182$ [0.132, 0.225] larger than the decomposed σ ’s. That number quantifies miscalibration; it does not rank the ways of curing it. On the same scale, temperature scaling [14] of that overconfident σ leaves $+0.034$ [$-0.018, 0.081$]. Split-conformal recalibration of it leaves -0.029 [$-0.077, 0.007$]. Both straddle zero, in opposite directions. Trustworthy decisions need calibration, and all three routes supply it. Neither does this task separate them on cost, since all three fit a scalar on the same held-out split; the closed form’s cost advantage is the one Section 5 measures, a per-edge variance computed rather than learned, and none of these six tasks tests it.

Reward re-ranking over a measured congeneric set (Figure S17). Driving a trajectory-balance GFlowNet [5, 27] over **eg5** (the most σ -varying evaluable target), the calibrated lower-confidence-bound (LCB) reward lowers the true hit-rate, 0.140 against a raw predicted-mean reward’s 0.249 (cal–raw -0.109), because on this set σ tracks chemical novelty ($\text{corr}(\sigma_{\text{total}}, \text{affinity}) = +0.42$) rather than low quality. An LCB reward improves the hit-rate only where σ flags low-quality candidates (Table S11 below).

The stopping decision as a function of the assumed error (Table S10). Figure S20 reports one point of a curve. The experiment measures how the stopping decision depends on the error the learner assumes, everything else held fixed, and the whole dependence is tabulated here. The 0.15 $\times$ arm of Figure S20 is a counterfactual stand-in: it imposes on the assumed error the miscalibration measured for a trained per-edge head in Figure S2, and is not an estimator run here. The two arms of the figure keep their own Monte-Carlo stream, so adding the sweep points leaves them unchanged.

The 0.94 $\times$ row is the 6% under-estimate a same-budget estimator pooling single-label edges by overlap reaches on the identical labels the heads were given (Figure S2). At that scale the stopping decision is indistinguishable from the calibrated one, 28.4 calls against 29.6 at the same

0.82 correctness. Correctness at the stop first moves at $0.60\times$, and the halved budget of Figure S20B needs $0.20\times$. The contrast is therefore scoped, exactly as the quality-control collapse of the main text is scoped, to per-edge heads missing by $5\text{--}11\times$, and is not evidence that a same-budget learned estimator would stop early. The one effect that survives a fair comparison is in the opposite direction: a 6% over-estimate spends 30.8 calls for the same correctness, the decision cost of a conservative bar of Figure S19B.

The decision cost of a conservative bar (Figure S19B). Even a conservative bar carries a decision cost: a systematic $\sim 1.41\times$ se inflation is itself a (safe-direction) miscalibration that, via the LCB rule, biases toward late stopping and under-commit. It errs in the safe direction against sampling noise, and because the independent-replicate SD is itself a lower bound on full reproducibility (Section 5), the inflation may be closer to neutral against the true run-to-run spread once pose and system-preparation variance are included.

Downstream audit summary (Table S11). Table S11 collects, for four downstream performance tasks (interval sharpness, active-learning edge weighting, commit-to-synthesis ranking, and reward re-ranking on the **eg5** congeneric set) and two decision tasks (commit trust and active-learning stopping), the calibrated method’s effect relative to the relevant baseline. The top block shows no measurable performance gain (Figures S18, S3, S19A, S17 above). The bottom block shows what the two decision tasks are measured against: an overconfident σ and a counterfactual stand-in are both far worse, which measures the cost of miscalibration, while against a fair baseline, the same overconfident σ after temperature or split-conformal recalibration and a same-budget pooled estimator at its measured $0.94\times$, the effect is absent (Figures S19B, S20 and Table S10 above).

What the audit establishes. None of the six tasks puts the calibrated σ ahead of a fair baseline. On commit trust the physics σ matches post-hoc recalibration rather than beating it, and the stopping separation is absent at the six per cent a same-budget pooled estimator produces. The audit establishes a precondition rather than an advantage: decisions of this kind need calibration, an overconfident σ corrupts them, and the calibration demonstrably helps on real data inside the network, as the null of the test of Section 6, rather than downstream of it. The graph weight contributes the same way, through structure rather than magnitude, making the gauge directions exactly unidentifiable (Theorem S1) while by Gauss–Markov its value does not improve ranking.

Provenance of the experimental affinities. The grounding against measured affinity uses one ChEMBL target per system, verified to be a human single-protein target before use, and one assay type per system, taken in the order IC_{50} then K_i and never mixed within a system. A flagged system enters the guided-versus-random comparison only if at least 60% of its network ligands map to a measured value. The counts here are grounded ligands, which for two systems exceed the ligand count of the sub-network the measurement runs on by one: an edge enters only when both its endpoints carry an affinity, so a ligand all of whose partners are ungrounded drops out. That is **1_1W7H** on **p38** (35 grounded, 34 in the sub-network) and **CAT-4j** on **bace** (24 and 23); **cdk8** and **hif2a** lose none. The **eg5** grounding of Section 6 predates that coverage-gated join and was assembled differently, per ligand from one named ChEMBL assay rather than by a target-wide structure search, so its row records the assay as well as the target. Table S12 lists what was used.

Table S11: **Effect of the decomposed σ relative to a baseline, across six downstream tasks.** Effects are the calibrated method minus the relevant baseline with a 95% bootstrap confidence interval; “real” denotes OpenFF benchmark out-of-distribution edges, “sim” denotes a controlled simulation. The two decision tasks are each given against an uncalibrated foil and against a fair baseline: the same σ recalibrated, and the assumed error a same-budget pooled estimator reaches.

Downstream task		Data	Effect (calibrated – baseline) [95% CI]			Verdict	
Interval sharpness (out-of-distribution)		real	width	split/decomposed	0.92 $\times$;	no gain	
Edge weighting (active learning)		sim	cond.	calls-to-top- k	−0.044 [−0.121, 0.005]	no gain	
Commit ranking (matched volume)		real	precision		−0.006 [−0.016, +0.007]	tie	
Reward re-ranking (evaluatable set, eg5)		real	true	hit-rate	−0.109 [−0.150, −0.072]	hurts	(directional, single target) ^a
Commit overconfident σ	trust vs	real	shortfall	+0.182 [0.132, 0.225]	better	cost of	miscalibration
Commit temperature-scaled σ	trust vs	real	shortfall	+0.034 [−0.018, +0.081]		tie	
Commit split-conformal σ	trust vs	real	shortfall	−0.029 [−0.077, +0.007]		tie	
Stopping stand-in	vs 0.15 $\times$	sim	82% vs 60% correct at stop			cost of	miscalibration
Stopping estimator (0.94 $\times$)	vs pooled	sim	0.82 vs 0.82 correct, 29.6 vs 28.4 calls			tie	

^aThe reward re-ranking effect is measured on a single evaluatable target (**eg5**) and is directional, and it is not yet established as a general property of calibrated reward re-ranking across chemotypes.

Table S12: **ChEMBL provenance of the experimental affinities.** Target identifier, protein, assay type, and the number of network ligands with a measured value, for every system grounded against measured affinity: the four flagged systems that clear the 60% coverage requirement, and **eg5**, the clean system used for the comparison of accuracy against experiment. Affinities are converted at $RT \ln 10 = 1.364$ kcal/mol and aggregated per ligand by the median pChEMBL value.

system	ChEMBL target	protein	assay	ligands
cdk8	CHEMBL5719	cyclin-dependent kinase 8	IC ₅₀	24
hif2a	CHEMBL1744522	endothelial PAS domain protein 1	IC ₅₀	41
p38	CHEMBL260	mitogen-activated protein kinase 14	IC ₅₀	35
bace	CHEMBL4822	beta-secretase 1	IC ₅₀	24
eg5	CHEMBL4581	kinesin-like protein KIF11	IC ₅₀	28

The **eg5** affinities were taken from a single named ChEMBL ATPase IC₅₀ assay, identifier CHEMBL1101849, queried per ligand; the other four rows come from a target-wide connectivity-based structure search under the 60% coverage rule, for which no single assay identifier applies.

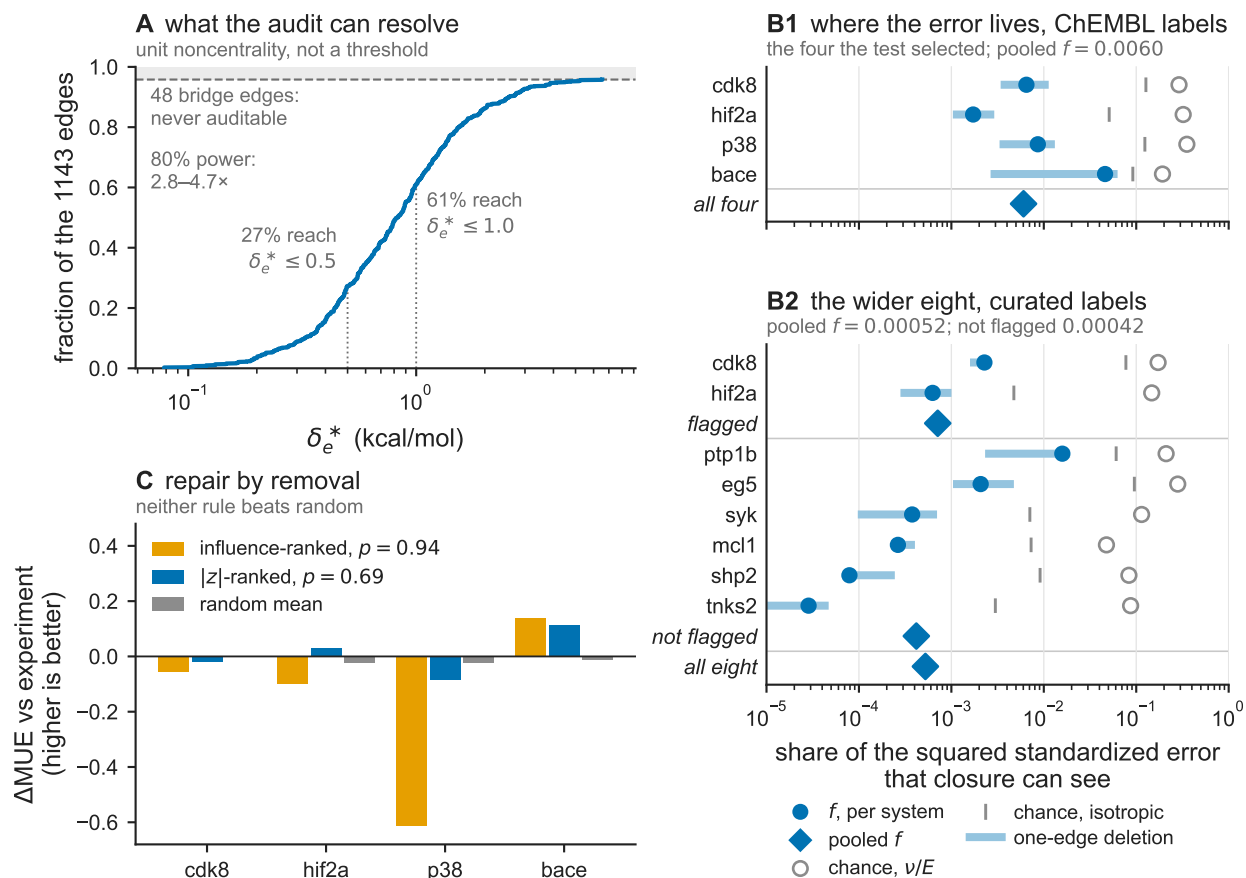


Figure 7: **Auditable capacity and the location of the error.** (A) Cumulative fraction of the 1143 benchmark edges against δ_e^* , the shift at unit noncentrality, with the bridge edges marked as the ceiling they impose. (B) Two grounded measurements of the same quantity, drawn on one shared axis so that they can be read against each other: the fraction f of a system’s error against experiment lying in the auditable subspace, the fraction an unaligned error field would show under two conventions – whitened-isotropic (dof/E) and isotropic in kcal/mol – and the range f takes when one edge is deleted at a time. (B1) The four sub-networks the closure test itself selected, grounded on a stereo-blind ChEMBL join: 167 edges, 51 residual degrees of freedom. (B2) The eight systems whose curated per-edge labels the benchmark carries and whose matched sub-network keeps a cycle: 205 edges, 30 residual degrees of freedom, grouped by whether the test flags them, and the six it does not flag show the effect at least as strongly as the two it does. Neither block corrects the other and the two are not nested: **bace** is absent from the curated benchmark entirely, **p38**’s matched sub-network is acyclic, the edge sets and the labels differ, and B2 carries the fewer residual degrees of freedom. On edges held genuinely identical the curated labels give the *higher* f , which is the direction a per-ligand label error predicts. Italic rows are pooled and carry f alone; the chance levels each pooled figure is measured against are given in the text. (C) Change in mean unsigned error against experiment after removing the same number of edges chosen by influence, by standardized residual, and at random.

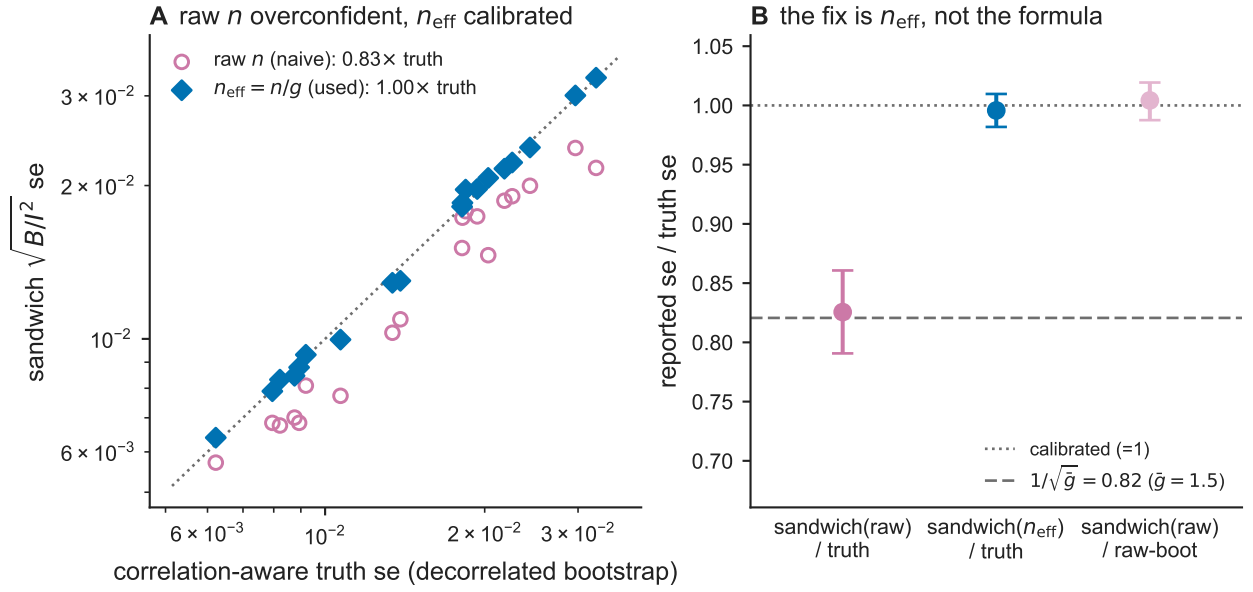


Figure S1: **Sandwich standard error at the raw sample count versus at the effective sample count $n_{\text{eff}} = n/g$, on the real BACE1 λ windows.** (A) Sandwich standard error at n_{eff} and at raw n , plotted against the correlation-aware (decorrelated bootstrap) reference standard error, with the $y=x$ line. (B) Aggregate ratio of sandwich standard error to the reference, at n_{eff} , at raw n , and of the sandwich to its own raw- n -matched bootstrap, with bootstrap confidence intervals; the $1/\sqrt{\bar{g}}$ autocorrelation prediction is marked.

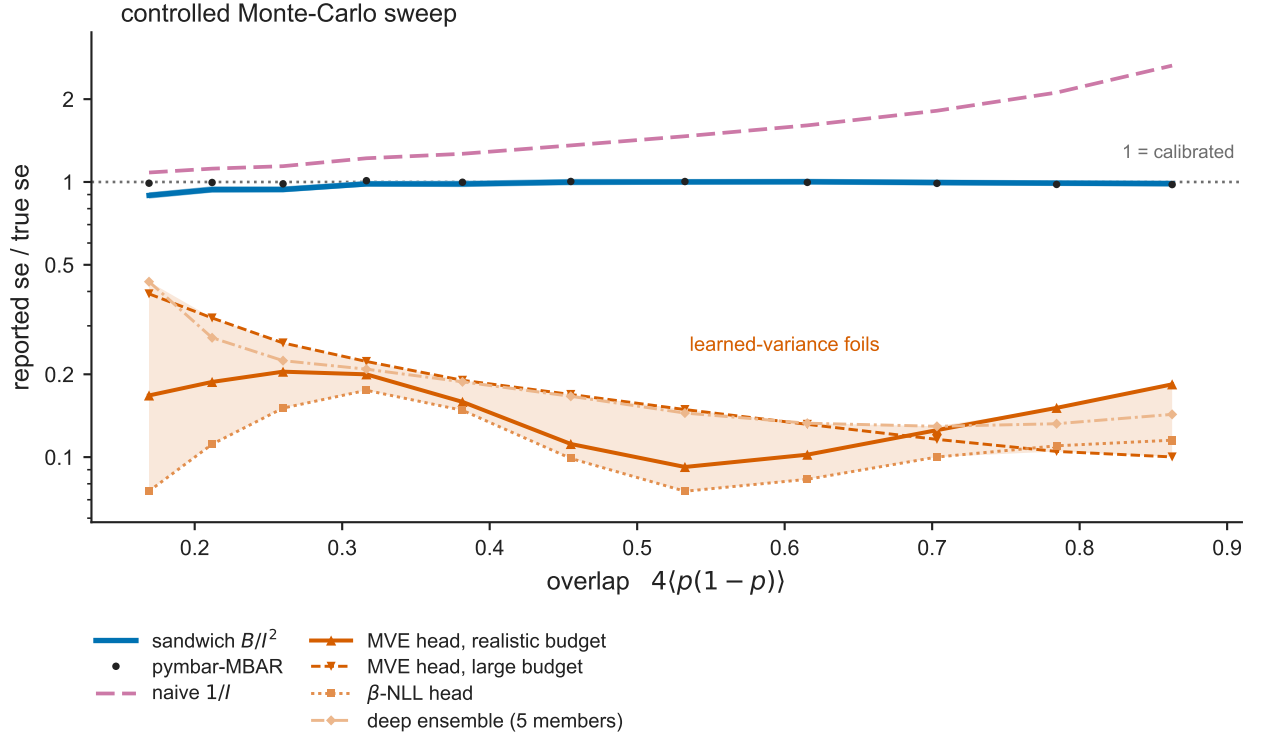


Figure S2: **Reported standard error against the true standard error over a controlled overlap sweep.** Reported standard error divided by the true standard error, versus work-distribution overlap, for the sandwich B/I^2 , pymbar-MBAR, the naive information-equality plug-in $1/I$, a learned mean-variance (MVE) head at a realistic and at a large training budget, a β -NLL head, and a 5-member deep ensemble, in a controlled Monte-Carlo sweep.

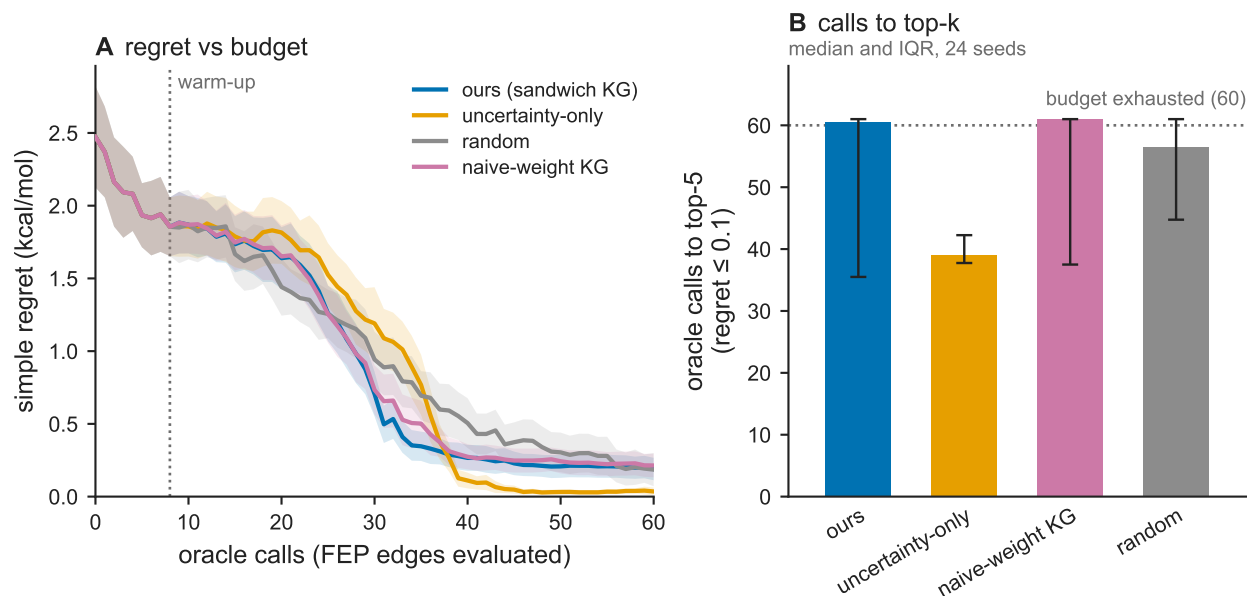


Figure S3: **Simulated active-learning acquisition policies.** (A) Simple regret versus oracle calls (simulated FEP edges evaluated), for a sandwich-weighted knowledge-gradient policy [11], implemented as in Balandat et al. [3] and Gardner et al. [12], a naive-weighted knowledge-gradient policy, an uncertainty-only policy, and random selection, averaged over 24 seeds with 95% confidence bands. (B) Median oracle calls to reach the top- k candidates (regret at or below tolerance), with interquartile-range error bars, for the same four policies.

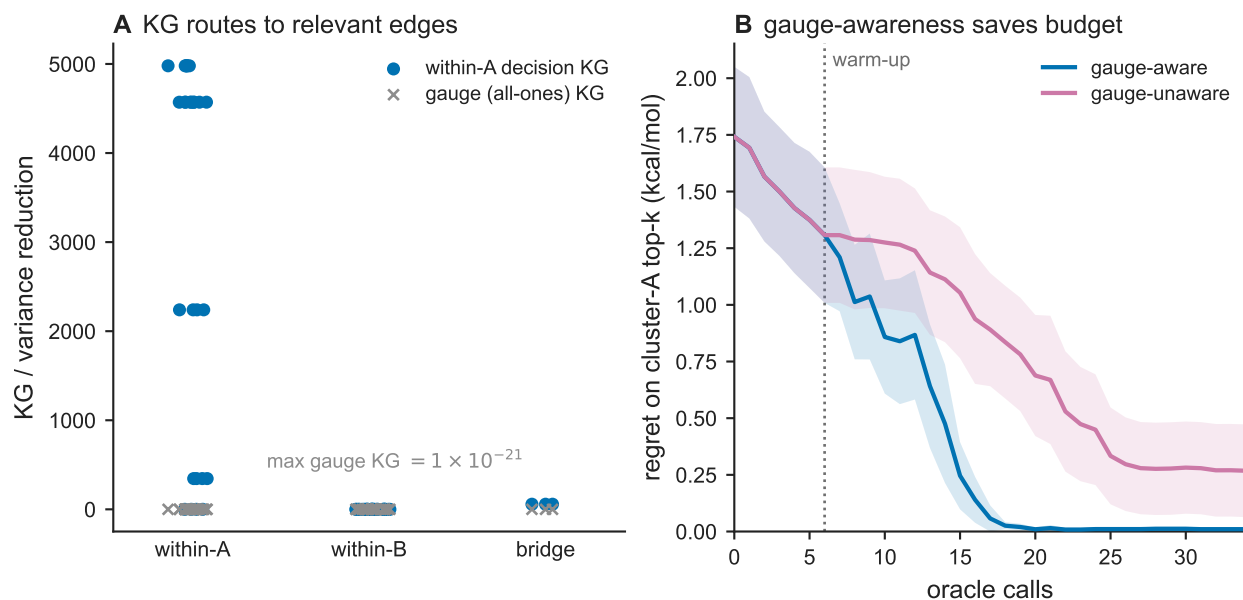
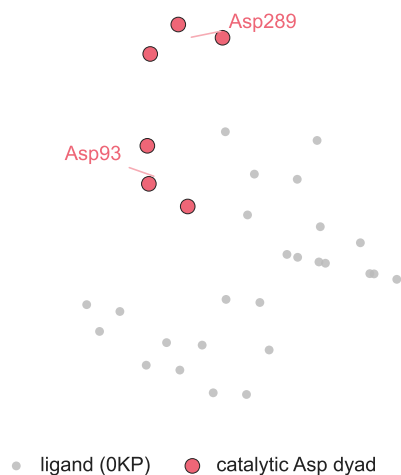


Figure S4: **Knowledge-gradient score on gauge and decision-relevant contrasts (simulation).** (A) Knowledge gradient for the within-cluster decision contrast and for the gauge (all-ones) contrast, by edge type (within-cluster-A, within-cluster-B, bridge). (B) Regret on the cluster-A top- k candidates versus oracle calls, for a gauge-aware and a gauge-unaware acquisition policy, averaged over seeds with 95% confidence bands.

A BACE1 (PDB 4DJW)
catalytic Asp dyad, 2.7/2.7 Å to ligand



B BRD4 BD1 (PDB 3MXF, JQ1)
16 waters ≤ 4.5 Å; Asn140 3.1 Å (KAc)

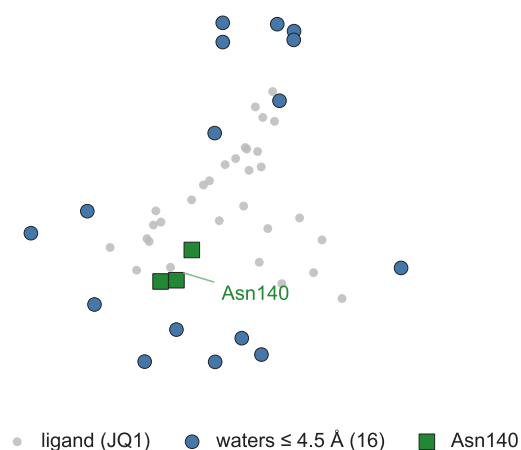


Figure S5: **Structural context of two flagged systems, from public co-crystal structures.** Heavy-atom coordinates. (A) BACE1 (PDB 4DJW): the catalytic aspartic-acid dyad (Asp93/Asp289, red circles) and the bound inhibitor (grey). (B) BRD4 BD1 (PDB 3MXF, in complex with JQ1): crystallographic waters within 4.5 Å of the ligand (blue circles), the recognition residue Asn140 (green squares), and the ligand (grey).

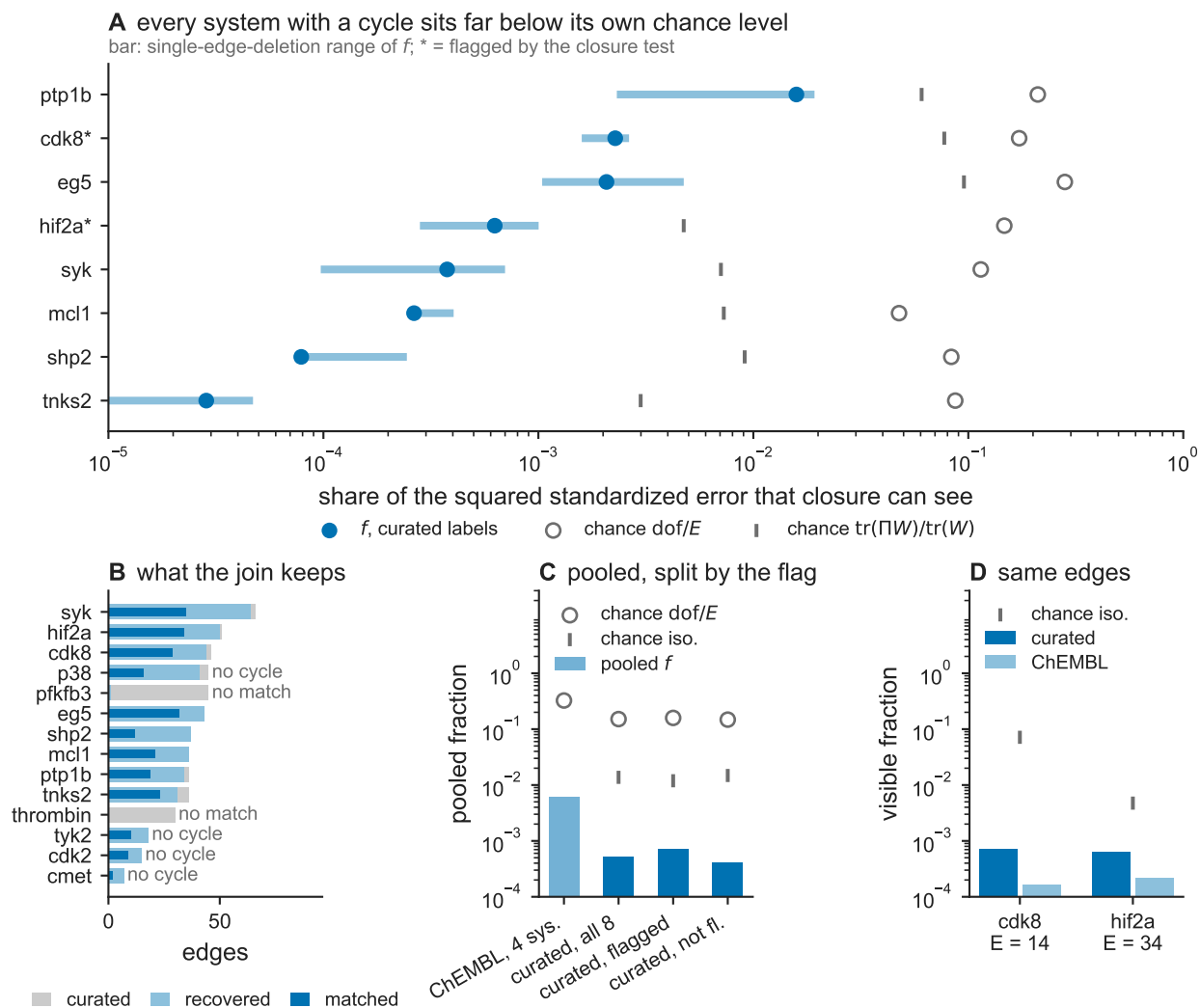


Figure S6: **The visible fraction of the error against curated experimental labels, over the benchmark.** (A) Per system with at least one independent cycle in its matched sub-network: the visible fraction f , the range it takes over single-edge deletions, and its two chance levels, dof/ E and the $\text{tr}(\Pi W)/\text{tr}(W)$ an error isotropic in kcal/mol would produce under the same projector. An asterisk marks a system the closure test flags. (B) What the join keeps, per system: curated edges offered, curated edges whose two ligands both recover into the network, and edges matched into the sub-network the measurement runs on, with the systems that carry no cycle or no matched edge named. (C) The pooled fraction and its chance levels for all eight measurable systems and for the flagged and unflagged systems separately, beside the four-system ChEMBL-grounded figure of Section 4, which is measured on different labels and a different edge set and is shown for scale rather than as a comparison. (D) The two label sources on identical edges, which holds the sub-network fixed: the noisier ChEMBL join gives the smaller fraction, as a gradient-field label error must.

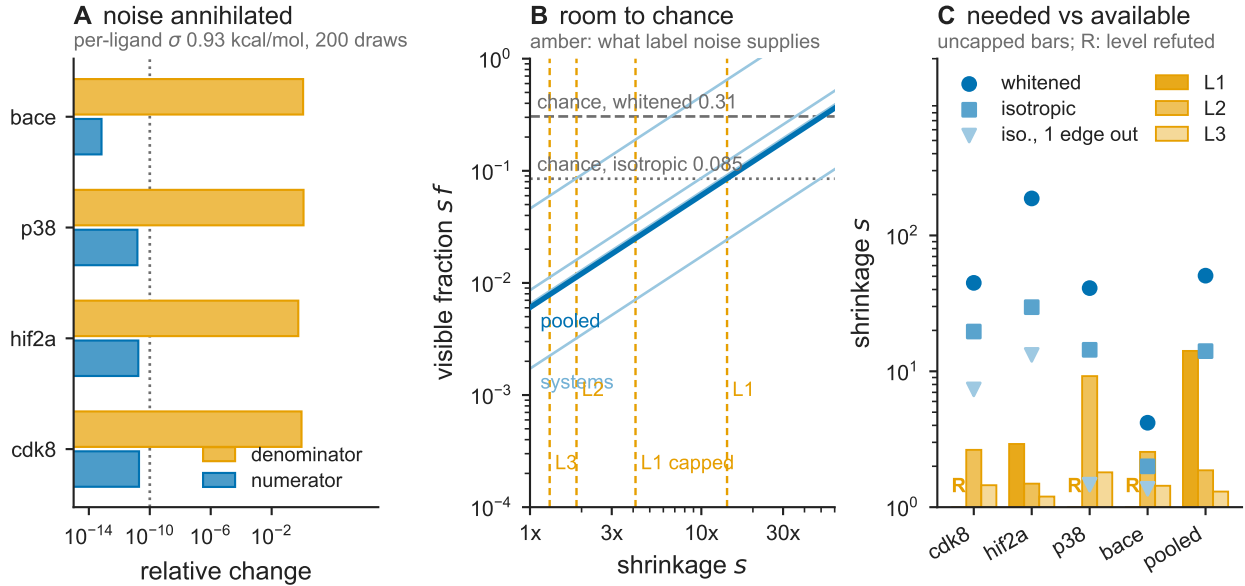


Figure S7: **The label-noise floor under the visible fraction.** (A) A synthetic per-ligand annotation error of standard deviation 0.93 kcal/mol is injected into each grounded network as the gradient field $\epsilon_e \rightarrow \epsilon_e + b_j - b_i$, 200 seeded draws per system: the worst relative change in the numerator $\|\Pi\tilde{\epsilon}\|^2$ over all draws, against the median relative growth of the denominator $\|\tilde{\epsilon}\|^2$, with the exactness tolerance fixed before the run marked. (B) The visible fraction under a denominator shrinkage s , per system and pooled, against the pooled whitened and isotropic chance levels; the vertical rules mark the shrinkage each declared label-noise level supplies. (C) Per system, the shrinkage needed to reach each chance level – whitened, isotropic, and isotropic under the least favourable single-edge deletion – against the shrinkage each level supplies, uncapped. An R marks a level that asserts more label variance than that system’s whole measured error against experiment contains, and is therefore refuted by the data rather than truncated.

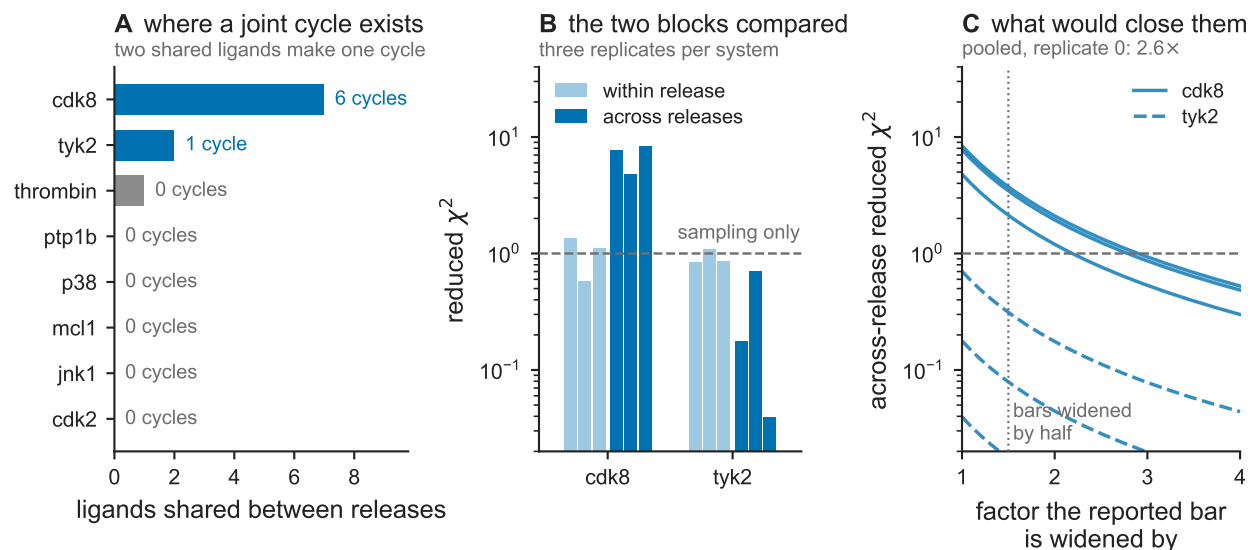


Figure S8: **What the multi-release systems can measure about between-preparation variance.** (A) For each system name that unions more than one separately released data set, the largest number of ligands any two of its releases share, annotated with the number of across-release cycles that creates. Two shared ligands are needed for one cycle. (B) Within-release and across-release reduced χ^2 for the two systems that carry an across-release cycle, three protocol repeats each, against the value sampling alone would give. (C) The across-release reduced χ^2 of the same two systems as every reported standard error is widened by a common factor, with the half-again widening of the main text's stress test marked; scaling every bar by λ divides the statistic by λ^2 exactly and leaves the fit unchanged. *cdk8*'s block measures a reuse of seven ligand names for different molecules between its two releases, not a disagreement between two preparations, and is not a between-preparation reading.

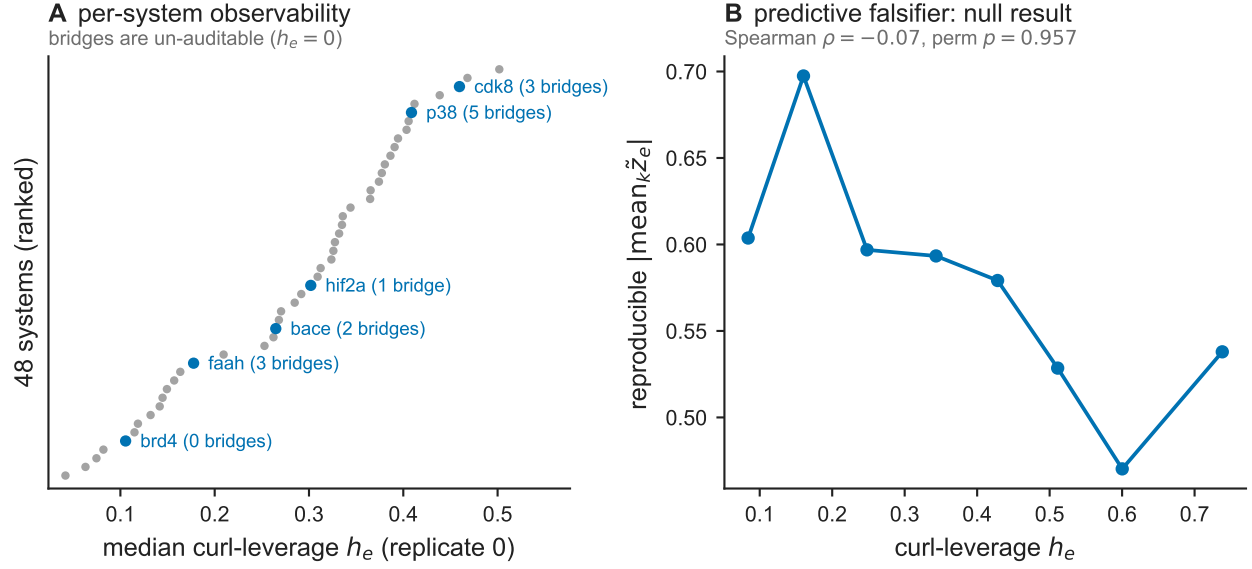


Figure S9: **Per-system curl-leverage, and curl-leverage versus reproducible residual magnitude.** (A) Median curl-leverage h_e per system, ranked, across the 48 systems with at least one independent cycle; the six cycle-closure-flagged systems are labeled with their number of bridge ($h_e = 0$) edges. (B) Curl-leverage h_e , binned into 8 quantiles, versus the mean studentized cross-replicate reproducible residual magnitude in each bin, with the Spearman correlation and permutation p -value.

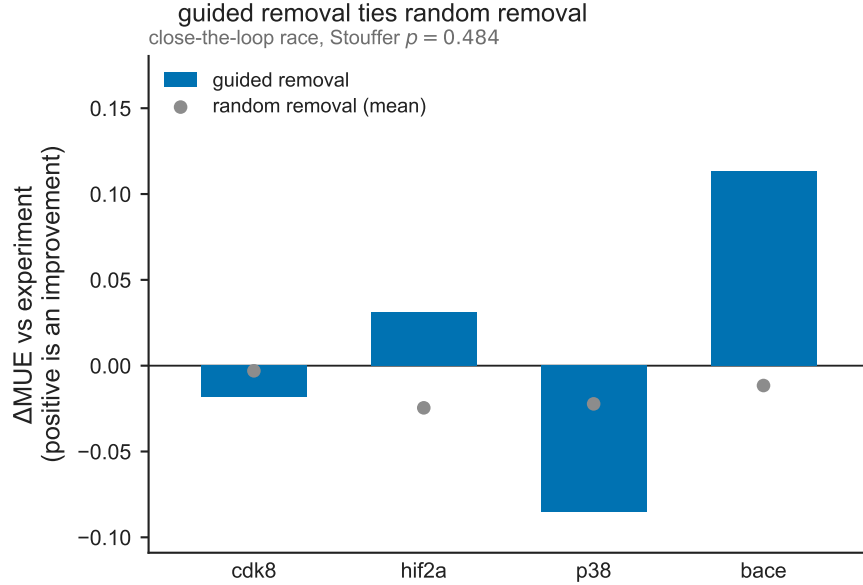


Figure S10: **Change in mean unsigned error against experimental affinity after guided versus random QC-edge removal.** Bars: change in mean unsigned error against experiment (ΔMUE) after guided removal of the QC-flagged high- $|z|$ edges, per system. Points: mean ΔMUE after a size-matched random removal, per system. Throughout, ΔMUE is $\text{MUE}(\text{full}) - \text{MUE}(\text{after})$, so a positive value is an improvement. Systems: cdk8, hif2a, p38, bace.

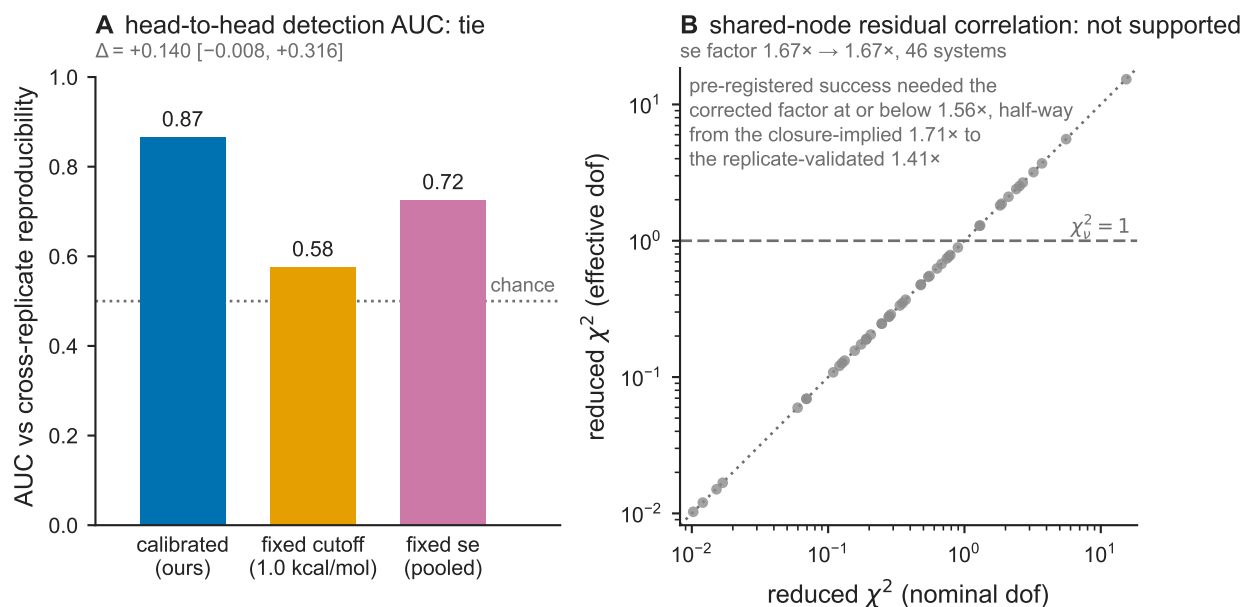


Figure S11: **Detection AUC of three flagging rules, and nominal versus correlation-corrected reduced χ^2 .** (A) Area under the ROC curve (AUC) scored against the per-system reproducible-systematic magnitude defined in Computational details, for the calibrated per-edge null, a fixed 1.0 kcal/mol hysteresis cutoff, and a fixed-standard-error χ^2 test, with the chance level (AUC = 0.5) marked. (B) Per-system reduced χ^2 at the nominal degrees of freedom versus at the shared-node-correlation-corrected effective degrees of freedom, with the $y=x$ line, the $\chi^2_v = 1$ reference, and the annotated inverted-median factor of 1.67 $\times$, the pre-registered functional evaluated over this panel's 46 systems.

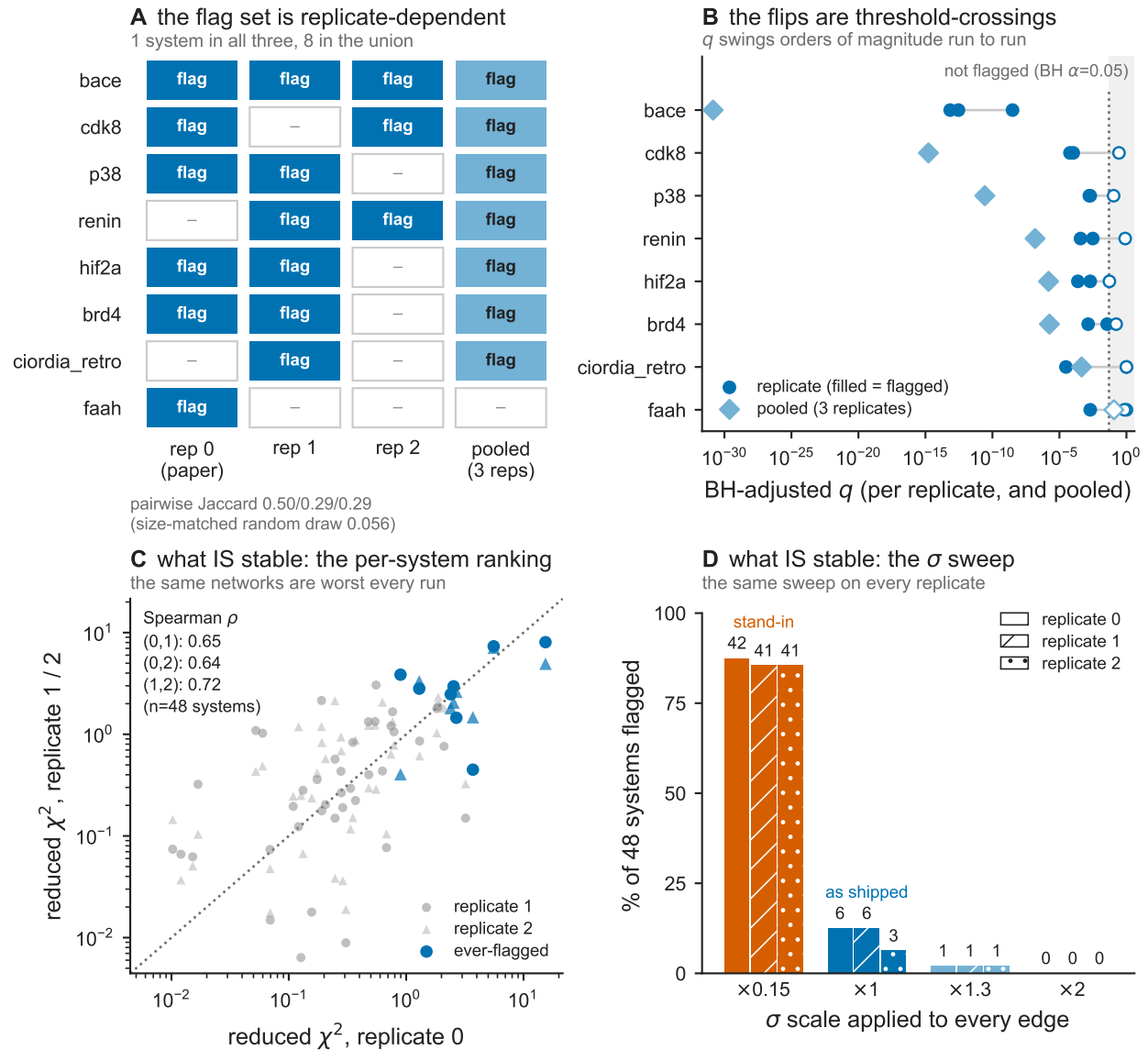


Figure S12: **Replicate stability of the flag set.** The Benjamini–Hochberg test at $\alpha = 0.05$ recomputed independently on each of the benchmark’s three protocol repeats. Per-system adjusted p -values, which systems are flagged on which replicate, the pairwise Jaccard overlap of the three flag sets against a random-draw reference, the flag set obtained from the three replicates pooled, and the selectivity of the test across a swept uniform σ scale on each replicate.

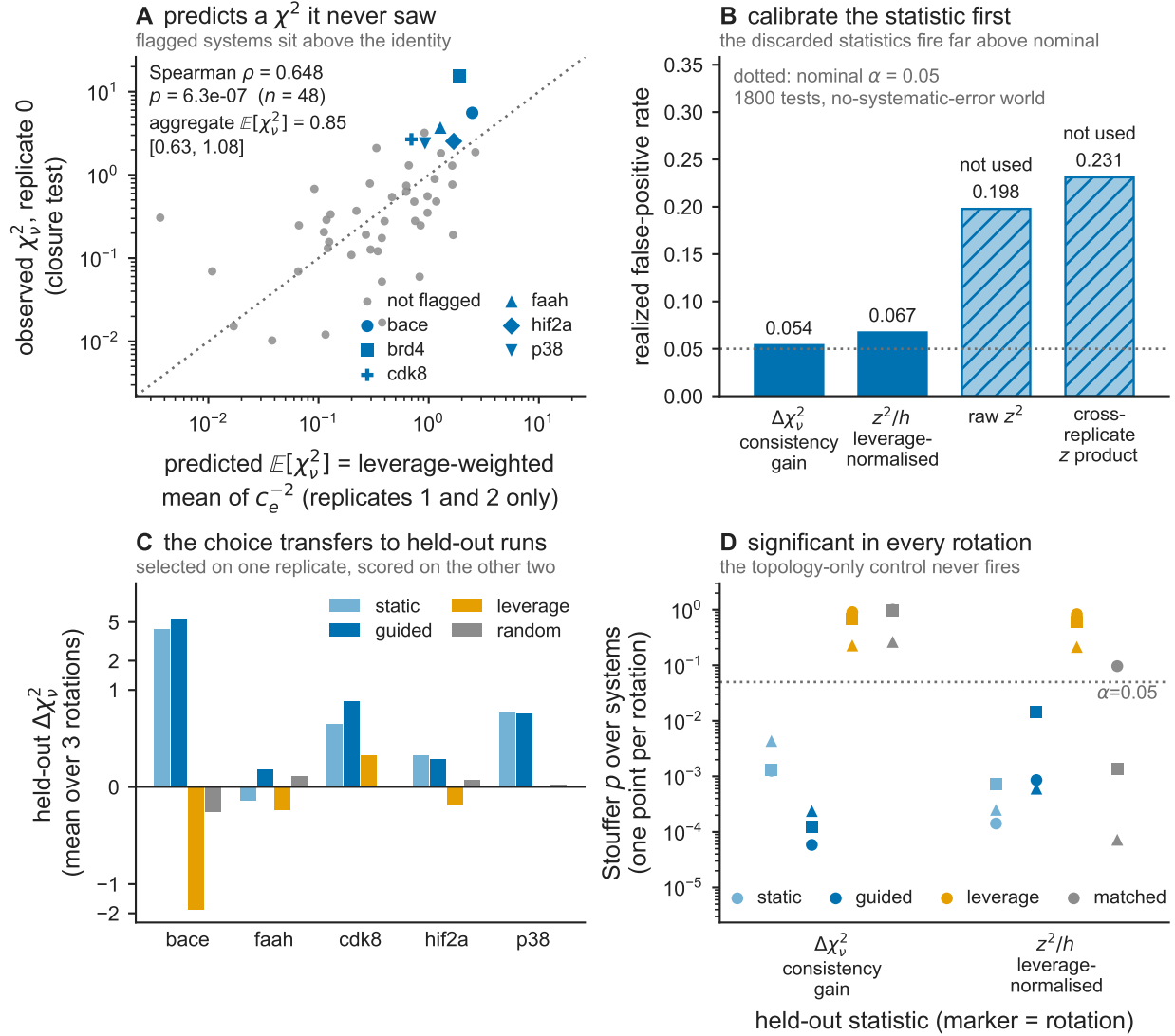


Figure S13: **Out-of-sample behaviour of the calibrated null.** (A) Per-system closure χ^2_v predicted from the per-edge miscalibration measured on the two replicates not being predicted, against the observed value, with the identity line and the flagged systems marked. (B) Realized false-positive rate of each candidate held-out statistic under a simulation with no systematic error anywhere, against the nominal level, with the two statistics that fire far above nominal marked. (C) Held-out change in reduced χ^2 on the two replicates not used for selection, after removing the edges selected on the third, averaged over the three rotations and shown per testable flagged system, for a static and a guided selection rule, a topology-only rule built from curl-leverage, and equal-size random removal. (D) Stouffer-combined p -value over systems, one point per rotation, for each of the two statistics that held their nominal level, under the same selection rules together with an $|z|$ -rank-matched control drawn from unflagged systems, with the $\alpha = 0.05$ line marked.

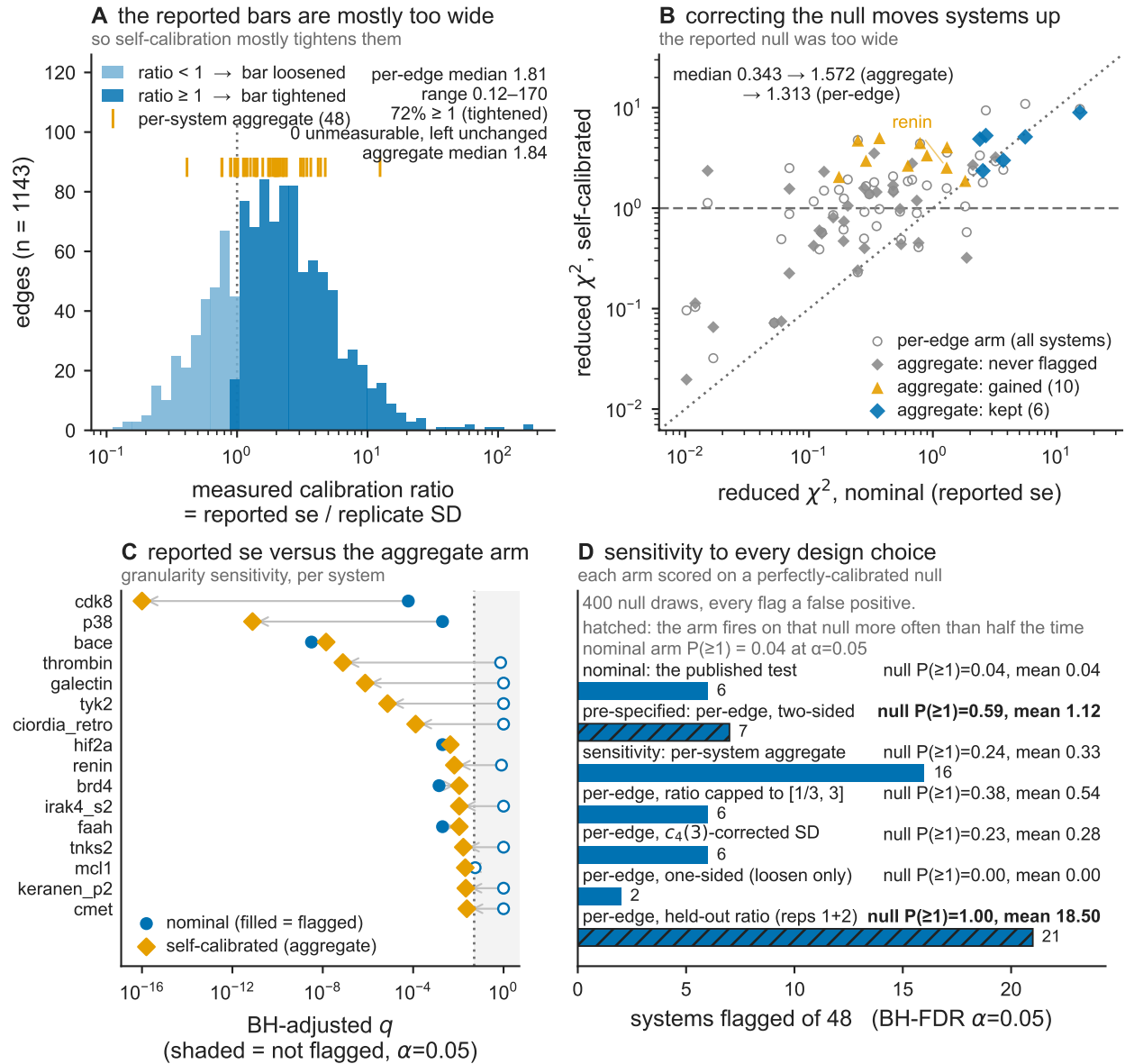


Figure S14: **Two-sided self-calibration of the per-edge error bar.** Per-system reduced χ^2 before and after replacing each system's reported standard error by its own replicate-measured calibration, which systems are gained and which are lost, and the realized probability of at least one false flag for each arm under a pure-null simulation, for the seven arms described in the text.

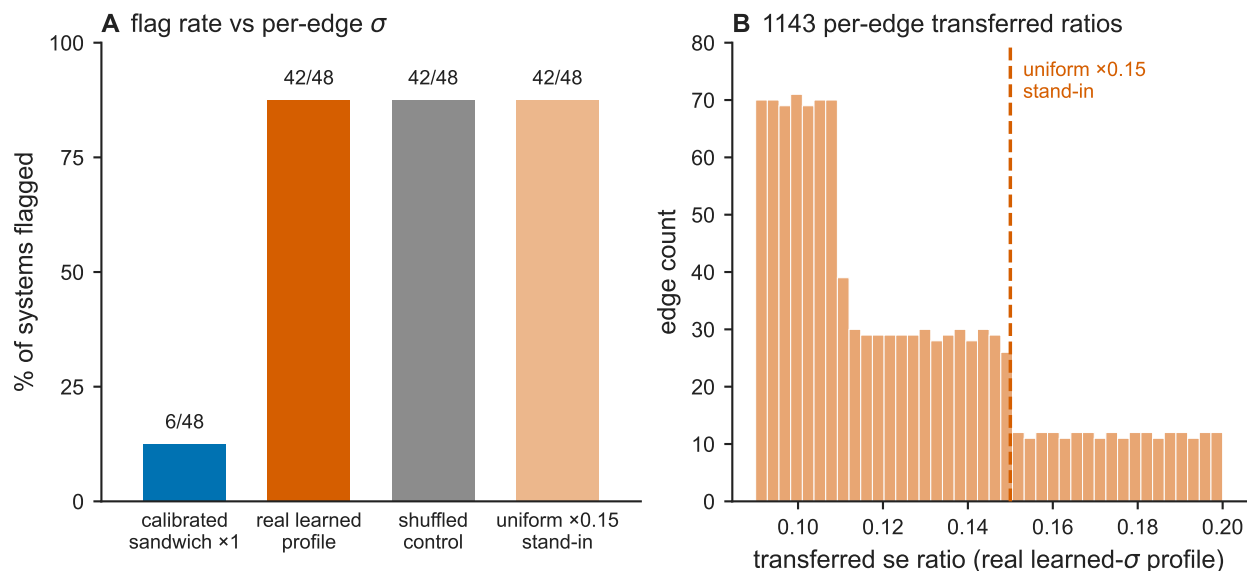


Figure S15: **Cycle-closure flagging under four per-edge σ assignments, and the distribution of the real-profile ratios.** (Left) Percentage of the 48-system OpenFE network flagged, for the calibrated sandwich, the learned head’s measured overlap-dependent miscalibration profile transferred by percentile rank onto each edge’s own overlap, a shuffled-profile control, and a uniform $\times 0.15$ shrink. (Right) Histogram of the 1143 per-edge transferred ratios from the real learned profile, with the uniform $\times 0.15$ stand-in marked.

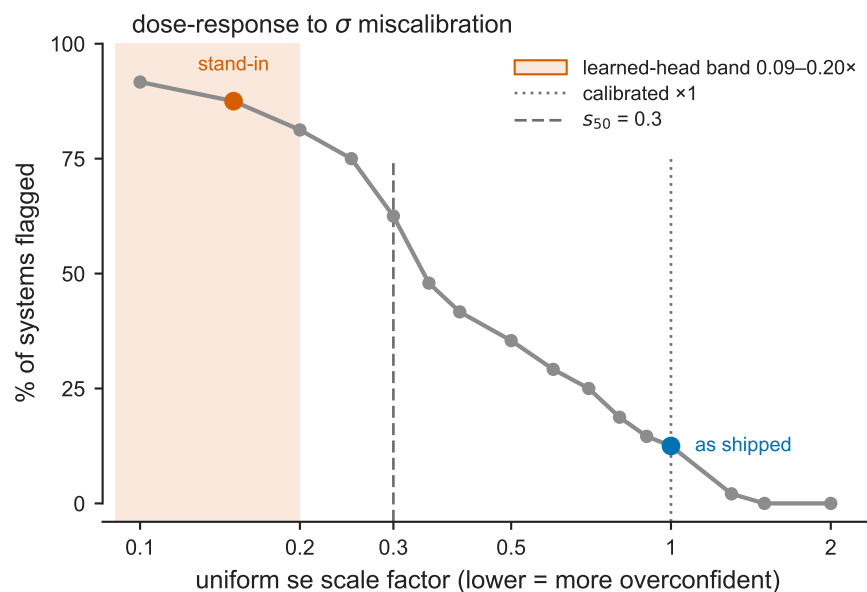


Figure S16: **Fraction of the OpenFE network flagged across a swept uniform σ scale factor.** Percentage of the 48-system network flagged by the cycle-closure test, across a frozen grid of 16 uniform standard-error scale factors from 2.0 $\times$ to 0.10 $\times$ (lower = more overconfident; 1.0 $\times$ is the calibrated sandwich). Vertical dotted line: the calibrated scale. Vertical dashed line: s_{50} , the largest scale at which at least half the systems are flagged. Shaded band: the learned head’s measured, rank-transferred miscalibration range.

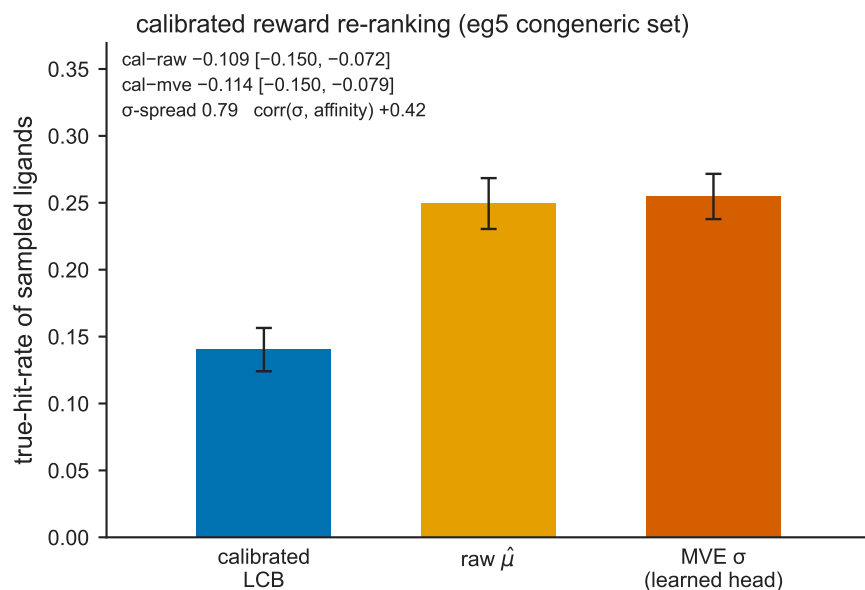


Figure S17: **True hit-rate of a trajectory-balance GFlowNet under three reward functions.** Mean true hit-rate of generated ligands, over 12 seeds with standard-error bars, for a calibrated lower-confidence-bound reward, a raw predicted-mean reward, and an overconfident MVE-uncertainty reward, on the **eg5** congeneric ligand series.

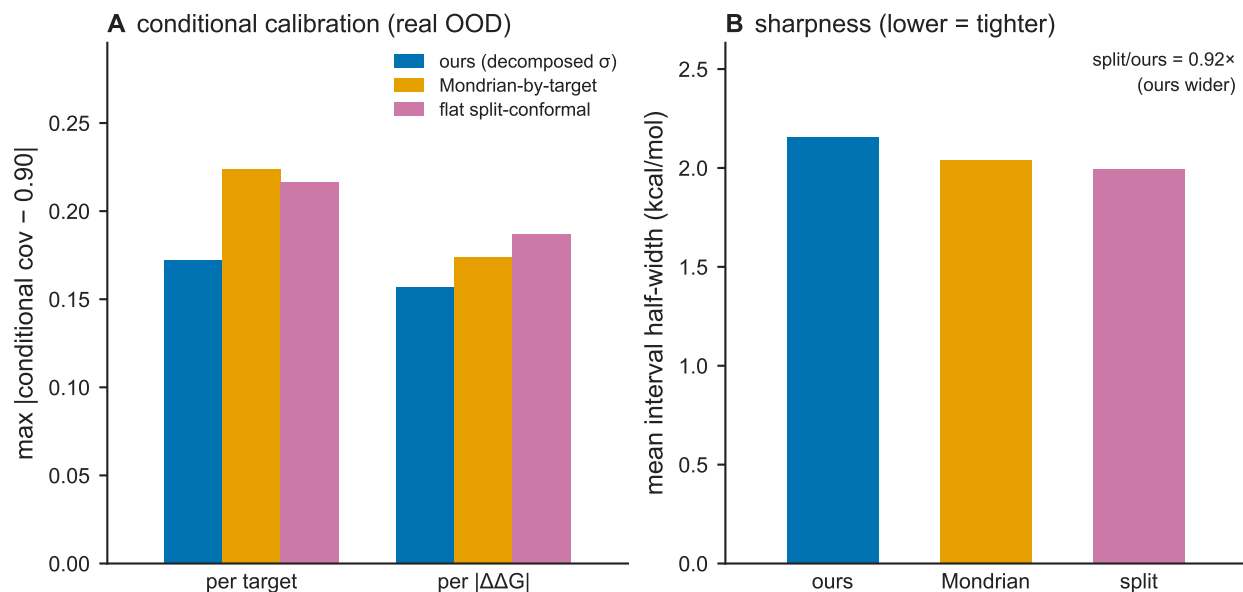


Figure S18: **Conditional calibration and interval sharpness on real out-of-distribution trunk residuals.** (A) Maximum conditional coverage error, computed per target and per $|\Delta\Delta G|$ bin, for the decomposed physics σ , a Mondrian-by-target conformal interval, and a flat split-conformal interval, each at approximately 90% marginal coverage. (B) Mean prediction-interval half-width for the same three methods.

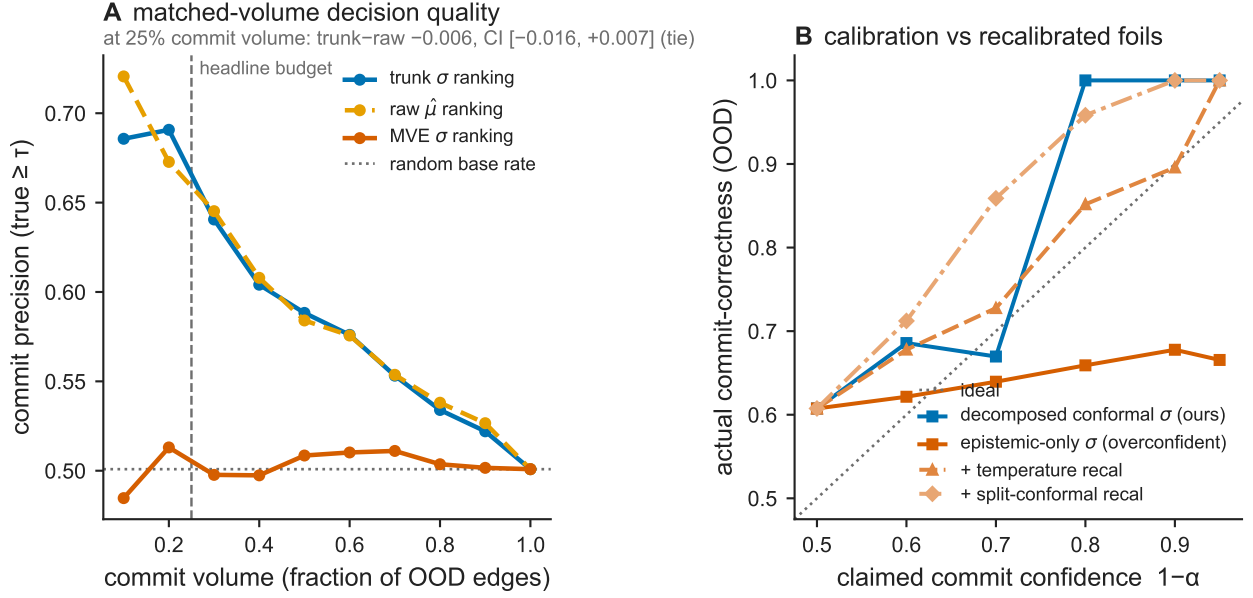


Figure S19: **Commit-to-synthesis ranking and calibration on real out-of-distribution edges.** (A) Commit precision (fraction of committed candidates with true value at or above a threshold τ) versus commit volume (fraction of edges committed), for candidates ranked by the trunk σ -aware margin, by the raw predicted mean, and by a learned MVE-head margin, with the random base rate marked. (B) Actual commit-correctness versus claimed commit confidence $1 - \alpha$, for the decomposed physics σ , an epistemic-only σ , and the same epistemic-only σ after temperature and after split-conformal recalibration, with the ideal (actual = claimed) line.

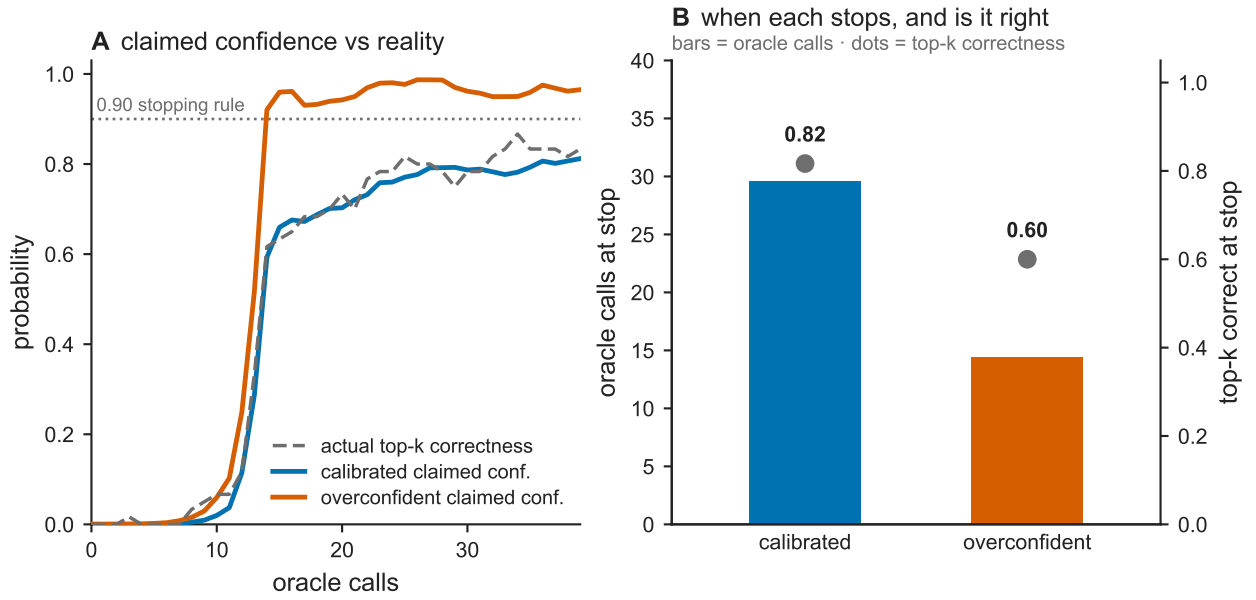


Figure S20: **Simulated active-learning stopping under calibrated versus counterfactually overconfident claimed uncertainty.** (A) Actual top- k correctness and claimed stopping confidence, averaged over 60 seeds, versus the number of oracle calls, for a calibrated learner and for a learner whose assumed posterior standard error is scaled by $0.15\times$. (B) Oracle calls at the stopping rule (bars) and top- k correctness at that stop (points), for the two learners.